Degree_project_ Tobias_Steinegger_2019

September 8, 2019

1 Investigating Environmental Hotspots of Swedish Household Consumption

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2 Impressum

This Jupyter notebook is the python-code for the Degree Project written by Tobias Steinegger to obtain his Degree in MSc. Sustainable Technology (*Industrial Ecology*). The project aims at estimating the environmental impacts of Swedish households by combining environmnetally extended multi-regional input-output tables (*EEMRIOT*), namely *EXIOBASE 3.4* and link them to existing Household Expenditure Surveys (*HBS*). Further description of the research methodology can be found in the thesis, linked underneath.

This approach lead to the research question of the degree project:

> *How expenditures and environmental impacts are distributed among different goods and services for different household groups and regions in Sweden?*

This research question is divided into two research objectives RO:
>**RO 1.** Estimate the consumption-based environmental impact of Sweden with the most recent available data;
>**RO 2.** Develop a methodological framework linking Swedish household expenditure surveys to environmentally extended multiregional input-output tables to investigate the environmental footprint and hotspots of a Swedish Household;

The aim of this project was therefore three-fold: (i) calculating the current consumption-based emissions of Sweden; (ii) developing a methodology to connect Swedish household consumption with environmentally-extended input-output tables; (iii) estimating the environmental impacts and hotspots of Swedish households.

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Program: MSc. Sustainable Technology
Institution: Royal Institute of Technology, KTH
Place: Stockholm

Contact: tobias.steinegger@bluewin.ch **Thesis available online:** https://kth.diva-portal.org/smash/get/diva2:1348995/FULLTEXT01.pdf **Date of submission:** eptember 2019 Remark:*** This is a student project and mark my first steps of programming in Python, therefore the author takes no liability over the data, the calculations or the emerging results! ***

3 RO 0.0 Prepartion of the Python Packages and the Database

First, all the necessary packages for this projects are imported:

```
In [1]: import pymrio
        import pandas as pd
        import numpy as np
        import seaborn as sb
        import matplotlib.pyplot as plt
        import country_converter as coco
        import sys as os
```

```
In [2]: print(" Pandas version: " + pd.__version__)
        print(" Numpy version: " + np.__version__)
        print(" Seaborn version: " + sb.__version__)
        #print(" Matplotlib version: " + plt.__version__)
        print(" Country converter version: " + coco.__version__)
```

```
 Pandas version: 0.24.2
 Numpy version: 1.16.2
 Seaborn version: 0.9.0
 Country converter version: 0.6.6
```

Needed for creating the table of content on top:

```
In [3]: %%javascript
        $.getScript('https://kmahelona.github.io/ipython_notebook_goodies/ipython_notebook_toc
```

```
<IPython.core.display.Javascript object>
```

The report is based on the environmental data provided in the *EXIOBASE 3.4 - IOT - 2011 - pxp* which is freely available on: https://exiobase.eu/index.php/data-download/exiobase3mon
For further information on the construction and underlying principle of the EXIOBASE 3, please see:
Wood, R., Stadler, K., Bulavskaya, T., Lutter, S., Giljum, S., de Koning, A., Kuenen, J., Schütz, H., Acosta-Fernández, J. and Usubiaga, A., 2015. Global sustainability accounting—developing EXIOBASE for multi-regional footprint analysis. Sustainability, 7(1), pp.138–163.

In a second step, the database Exiobase 3.4 is parsed and imported into the system as `exio3`

```
In [4]: exio3 = pymrio.parse_exiobase3(path='C:/Users/tobia/Desktop/Database_LCA/exiobase3.4_i
```

```
C:\Users\tobia\Anaconda3\envs\Pymrio\lib\site-packages\pymrio\core\fileio.py:288: FutureWarning
  header=_header))
```

In order to calculate all missing tables, using the `calc_all()` command.

```
In [5]: exio3.calc_all()
```

```
Out[5]: <pymrio.core.mriosystem.IOSystem at 0x1eefaa48160>
```

Plot the EXIO3 accounts for the environmental stressor *'CO2-combustion-air'* in a seperate window

```
In [6]: %matplotlib qt
        exio3.satellite.plot_account('CO2 - combustion - air')
        plt.show()
```

```
In [7]: #Total expenditure for Swedish households:
        exio3.Y.SE.sum()
```

```
Out[7]: category
        Final consumption expenditure by households
        Final consumption expenditure by non-profit organisations serving households (NPISH)
        Final consumption expenditure by government
        Gross fixed capital formation
        Changes in inventories
        Changes in valuables
        Exports: Total (fob)
        dtype: float64
```

This report is mainly interested in the consumption based accounts *'exio3.satellite.D_cba'*. Consumption-based accounts are defined as the environmental impacts that occur due to consumption, means: *Consumption = Production + Import -Export*

```
In [8]: exio3.satellite.D_cba
        exio3.satellite.D_cba_reg
```

```
Out[8]: region                                                        AT  \
        Taxes less subsidies on products purchased: Total   1.362251e+04
        Other net taxes on production                       6.103556e+03
        Compensation of employees; wages, salaries, & e...  1.061780e+04
        Compensation of employees; wages, salaries, & e...  6.942497e+04
        Compensation of employees; wages, salaries, & e...  7.836160e+04
        Operating surplus: Consumption of fixed capital     5.258742e+04
        Operating surplus: Rents on land                    0.000000e+00
        Operating surplus: Royalties on resources           0.000000e+00
        Operating surplus: Remaining net operating surplus  6.973212e+04
        Employment: Low-skilled male                        7.627747e+02
        Employment: Low-skilled female                      5.410121e+02
```

Employment: Medium-skilled male	2.709096e+03
Employment: Medium-skilled female	1.691989e+03
Employment: High-skilled male	1.078144e+03
Employment: High-skilled female	8.257957e+02
Employment hours: Low-skilled male	1.630001e+03
Employment hours: Low-skilled female	1.061174e+03
Employment hours: Medium-skilled male	5.441236e+03
Employment hours: Medium-skilled female	3.190786e+03
Employment hours: High-skilled male	1.934461e+03
Employment hours: High-skilled female	1.382385e+03
Employment: Vulnerable employment	2.164165e+03
Employment hours: Vulnerable employment	4.629404e+03
CO2 - combustion - air	9.176377e+10
CH4 - combustion - air	4.183363e+07
N2O - combustion - air	2.226504e+06
SOx - combustion - air	1.141587e+08
NOx - combustion - air	2.565178e+08
NH3 - combustion - air	2.568642e+07
CO - combustion - air	3.521075e+08
...	...
Water Withdrawal Blue - Manufacturing - Radio, ...	8.425529e+00
Water Withdrawal Blue - Manufacturing - Medical...	3.918157e+00
Water Withdrawal Blue - Manufacturing - Motor v...	5.919089e-01
Water Withdrawal Blue - Manufacturing - Other t...	3.425895e-01
Water Withdrawal Blue - Manufacturing - Furnitu...	1.761050e+01
Water Withdrawal Blue - Electricity - tower - E...	1.523300e+01
Water Withdrawal Blue - Electricity - tower - E...	1.179496e+01
Water Withdrawal Blue - Electricity - tower - E...	5.303830e+00
Water Withdrawal Blue - Electricity - tower - E...	0.000000e+00
Water Withdrawal Blue - Electricity - tower - E...	0.000000e+00
Water Withdrawal Blue - Electricity - tower - E...	5.985649e-02
Water Withdrawal Blue - Electricity - tower - E...	2.146999e-01
Water Withdrawal Blue - Electricity - tower - E...	0.000000e+00
Water Withdrawal Blue - Electricity - tower - E...	2.792530e-03
Water Withdrawal Blue - Electricity - tower - E...	0.000000e+00
Water Withdrawal Blue - Electricity - tower - E...	1.562018e-01
Water Withdrawal Blue - Electricity - tower - E...	0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...	8.775137e+02
Water Withdrawal Blue - Electricity - once-thro...	8.621259e+02
Water Withdrawal Blue - Electricity - once-thro...	1.820919e+02
Water Withdrawal Blue - Electricity - once-thro...	0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...	0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...	9.941109e+01
Water Withdrawal Blue - Electricity - once-thro...	4.706927e+00
Water Withdrawal Blue - Electricity - once-thro...	0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...	0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...	0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...	6.059659e-01

```
Water Withdrawal Blue - Electricity - once-thro...  0.000000e+00
Water Withdrawal Blue - Domestic - domestic Wat...  5.510130e+02

region                                                          BE  \
Taxes less subsidies on products purchased: Total   1.985186e+04
Other net taxes on production                       6.026566e+03
Compensation of employees; wages, salaries, & e...  1.397937e+04
Compensation of employees; wages, salaries, & e...  8.238805e+04
Compensation of employees; wages, salaries, & e...  1.025254e+05
Operating surplus: Consumption of fixed capital     6.457209e+04
Operating surplus: Rents on land                    0.000000e+00
Operating surplus: Royalties on resources           0.000000e+00
Operating surplus: Remaining net operating surplus  8.729508e+04
Employment: Low-skilled male                        1.609371e+03
Employment: Low-skilled female                      9.672420e+02
Employment: Medium-skilled male                     4.418317e+03
Employment: Medium-skilled female                   2.462307e+03
Employment: High-skilled male                       1.389545e+03
Employment: High-skilled female                     1.047464e+03
Employment hours: Low-skilled male                  3.556881e+03
Employment hours: Low-skilled female                2.019918e+03
Employment hours: Medium-skilled male               9.013137e+03
Employment hours: Medium-skilled female             4.763375e+03
Employment hours: High-skilled male                 2.519088e+03
Employment hours: High-skilled female               1.676831e+03
Employment: Vulnerable employment                   4.254463e+03
Employment hours: Vulnerable employment             8.982170e+03
CO2 - combustion - air                              1.346237e+11
CH4 - combustion - air                              3.929911e+07
N2O - combustion - air                              2.162601e+06
SOx - combustion - air                              2.064339e+08
NOx - combustion - air                              3.586554e+08
NH3 - combustion - air                              4.893308e+07
CO - combustion - air                               4.736812e+08
...                                                          ...
Water Withdrawal Blue - Manufacturing - Radio, ...  1.541913e+01
Water Withdrawal Blue - Manufacturing - Medical...  1.084419e+01
Water Withdrawal Blue - Manufacturing - Motor v...  2.086545e+00
Water Withdrawal Blue - Manufacturing - Other t...  7.881097e-01
Water Withdrawal Blue - Manufacturing - Furnitu...  3.370514e+01
Water Withdrawal Blue - Electricity - tower - E...  2.141398e+01
Water Withdrawal Blue - Electricity - tower - E...  1.066651e+01
Water Withdrawal Blue - Electricity - tower - E...  5.553049e+01
Water Withdrawal Blue - Electricity - tower - E...  0.000000e+00
Water Withdrawal Blue - Electricity - tower - E...  0.000000e+00
Water Withdrawal Blue - Electricity - tower - E...  9.242481e-02
Water Withdrawal Blue - Electricity - tower - E...  2.503380e+00
Water Withdrawal Blue - Electricity - tower - E...  0.000000e+00
```

Water Withdrawal Blue - Electricity - tower - E...	4.837791e-03
Water Withdrawal Blue - Electricity - tower - E...	0.000000e+00
Water Withdrawal Blue - Electricity - tower - E...	1.739495e-01
Water Withdrawal Blue - Electricity - tower - E...	0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...	1.731489e+03
Water Withdrawal Blue - Electricity - once-thro...	1.469286e+03
Water Withdrawal Blue - Electricity - once-thro...	1.751409e+02
Water Withdrawal Blue - Electricity - once-thro...	0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...	0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...	2.155024e+02
Water Withdrawal Blue - Electricity - once-thro...	5.029326e+00
Water Withdrawal Blue - Electricity - once-thro...	0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...	0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...	0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...	1.183512e+00
Water Withdrawal Blue - Electricity - once-thro...	0.000000e+00
Water Withdrawal Blue - Domestic - domestic Wat...	6.382930e+02

region	BG \
Taxes less subsidies on products purchased: Total	2.218660e+03
Other net taxes on production	5.462890e+02
Compensation of employees; wages, salaries, & e...	1.339204e+03
Compensation of employees; wages, salaries, & e...	8.730109e+03
Compensation of employees; wages, salaries, & e...	8.016956e+03
Operating surplus: Consumption of fixed capital	6.439380e+03
Operating surplus: Rents on land	0.000000e+00
Operating surplus: Royalties on resources	0.000000e+00
Operating surplus: Remaining net operating surplus	1.323182e+04
Employment: Low-skilled male	2.886025e+02
Employment: Low-skilled female	1.833455e+02
Employment: Medium-skilled male	1.056071e+03
Employment: Medium-skilled female	6.567788e+02
Employment: High-skilled male	3.905443e+02
Employment: High-skilled female	4.344605e+02
Employment hours: Low-skilled male	6.246711e+02
Employment hours: Low-skilled female	3.887402e+02
Employment hours: Medium-skilled male	2.223496e+03
Employment hours: Medium-skilled female	1.374990e+03
Employment hours: High-skilled male	8.098632e+02
Employment hours: High-skilled female	8.856906e+02
Employment: Vulnerable employment	5.776863e+02
Employment hours: Vulnerable employment	1.232450e+03
CO2 - combustion - air	5.484413e+10
CH4 - combustion - air	1.568348e+07
N2O - combustion - air	1.740141e+06
SOx - combustion - air	5.131265e+08
NOx - combustion - air	1.216228e+08
NH3 - combustion - air	4.127205e+06

```
CO - combustion - air                                  1.839602e+08
...                                                             ...
Water Withdrawal Blue - Manufacturing - Radio, ...  1.919775e+00
Water Withdrawal Blue - Manufacturing - Medical...  1.197699e+00
Water Withdrawal Blue - Manufacturing - Motor v...  1.034156e-01
Water Withdrawal Blue - Manufacturing - Other t...  1.012557e-01
Water Withdrawal Blue - Manufacturing - Furnitu...  6.055042e+00
Water Withdrawal Blue - Electricity - tower - E...  2.931250e+01
Water Withdrawal Blue - Electricity - tower - E...  1.289297e+00
Water Withdrawal Blue - Electricity - tower - E...  7.034228e-01
Water Withdrawal Blue - Electricity - tower - E...  0.000000e+00
Water Withdrawal Blue - Electricity - tower - E...  0.000000e+00
Water Withdrawal Blue - Electricity - tower - E...  1.864561e-02
Water Withdrawal Blue - Electricity - tower - E...  2.713030e-02
Water Withdrawal Blue - Electricity - tower - E...  0.000000e+00
Water Withdrawal Blue - Electricity - tower - E...  5.353939e-04
Water Withdrawal Blue - Electricity - tower - E...  0.000000e+00
Water Withdrawal Blue - Electricity - tower - E...  6.691421e-02
Water Withdrawal Blue - Electricity - tower - E...  0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...  1.745791e+03
Water Withdrawal Blue - Electricity - once-thro...  4.182108e+01
Water Withdrawal Blue - Electricity - once-thro...  2.246821e+03
Water Withdrawal Blue - Electricity - once-thro...  0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...  0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...  1.063810e+01
Water Withdrawal Blue - Electricity - once-thro...  3.896330e-01
Water Withdrawal Blue - Electricity - once-thro...  0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...  0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...  0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...  4.923729e-02
Water Withdrawal Blue - Electricity - once-thro...  0.000000e+00
Water Withdrawal Blue - Domestic - domestic Wat...  4.724050e+02

region                                                        CY  \
Taxes less subsidies on products purchased: Total   6.295381e+02
Other net taxes on production                       7.150655e+02
Compensation of employees; wages, salaries, & e...  7.977238e+02
Compensation of employees; wages, salaries, & e...  4.126077e+03
Compensation of employees; wages, salaries, & e...  4.473269e+03
Operating surplus: Consumption of fixed capital     2.647052e+03
Operating surplus: Rents on land                    0.000000e+00
Operating surplus: Royalties on resources           0.000000e+00
Operating surplus: Remaining net operating surplus  6.739052e+03
Employment: Low-skilled male                        6.911622e+01
Employment: Low-skilled female                      5.012146e+01
Employment: Medium-skilled male                     2.465020e+02
Employment: Medium-skilled female                   1.358268e+02
Employment: High-skilled male                       8.373947e+01
```

```
Employment: High-skilled female                          6.359718e+01
Employment hours: Low-skilled male                       1.508658e+02
Employment hours: Low-skilled female                     1.082663e+02
Employment hours: Medium-skilled male                    5.162262e+02
Employment hours: Medium-skilled female                  2.775036e+02
Employment hours: High-skilled male                      1.706327e+02
Employment hours: High-skilled female                    1.220984e+02
Employment: Vulnerable employment                        1.920886e+02
Employment hours: Vulnerable employment                  3.942167e+02
CO2 - combustion - air                                   1.355232e+10
CH4 - combustion - air                                   2.178159e+06
N2O - combustion - air                                   1.572908e+05
SOx - combustion - air                                   6.823941e+07
NOx - combustion - air                                   9.590995e+07
NH3 - combustion - air                                   2.137770e+06
CO - combustion - air                                    5.041962e+07
...                                                               ...
Water Withdrawal Blue - Manufacturing - Radio, ...       2.079286e-01
Water Withdrawal Blue - Manufacturing - Medical...       1.012964e-01
Water Withdrawal Blue - Manufacturing - Motor v...       2.572981e-02
Water Withdrawal Blue - Manufacturing - Other t...       4.623957e-01
Water Withdrawal Blue - Manufacturing - Furnitu...       1.217384e+00
Water Withdrawal Blue - Electricity - tower - E...       3.000253e+00
Water Withdrawal Blue - Electricity - tower - E...       3.369353e-01
Water Withdrawal Blue - Electricity - tower - E...       1.291071e-01
Water Withdrawal Blue - Electricity - tower - E...       0.000000e+00
Water Withdrawal Blue - Electricity - tower - E...       0.000000e+00
Water Withdrawal Blue - Electricity - tower - E...       6.139984e-03
Water Withdrawal Blue - Electricity - tower - E...       5.489776e-03
Water Withdrawal Blue - Electricity - tower - E...       0.000000e+00
Water Withdrawal Blue - Electricity - tower - E...       1.180265e-04
Water Withdrawal Blue - Electricity - tower - E...       0.000000e+00
Water Withdrawal Blue - Electricity - tower - E...       1.109688e-02
Water Withdrawal Blue - Electricity - tower - E...       0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...       8.430830e+00
Water Withdrawal Blue - Electricity - once-thro...       3.050523e+01
Water Withdrawal Blue - Electricity - once-thro...       6.281868e+00
Water Withdrawal Blue - Electricity - once-thro...       0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...       0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...       8.029524e+00
Water Withdrawal Blue - Electricity - once-thro...       7.535168e-02
Water Withdrawal Blue - Electricity - once-thro...       0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...       0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...       0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...       4.583864e-02
Water Withdrawal Blue - Electricity - once-thro...       0.000000e+00
Water Withdrawal Blue - Domestic - domestic Wat...       9.479360e+01
```

```
region                                                        CZ  \
Taxes less subsidies on products purchased: Total   8.103264e+03
Other net taxes on production                       1.070456e+03
Compensation of employees; wages, salaries, & e...  3.779616e+03
Compensation of employees; wages, salaries, & e...  2.988124e+04
Compensation of employees; wages, salaries, & e...  3.794607e+04
Operating surplus: Consumption of fixed capital     3.044064e+04
Operating surplus: Rents on land                    0.000000e+00
Operating surplus: Royalties on resources           0.000000e+00
Operating surplus: Remaining net operating surplus  4.562954e+04
Employment: Low-skilled male                        5.218465e+02
Employment: Low-skilled female                      3.716064e+02
Employment: Medium-skilled male                     2.252644e+03
Employment: Medium-skilled female                   1.254666e+03
Employment: High-skilled male                       9.904172e+02
Employment: High-skilled female                     9.308493e+02
Employment hours: Low-skilled male                  1.128018e+03
Employment hours: Low-skilled female                7.570142e+02
Employment hours: Medium-skilled male               4.512546e+03
Employment hours: Medium-skilled female             2.460733e+03
Employment hours: High-skilled male                 1.855846e+03
Employment hours: High-skilled female               1.674969e+03
Employment: Vulnerable employment                   1.772194e+03
Employment hours: Vulnerable employment             3.762902e+03
CO2 - combustion - air                              1.114138e+11
CH4 - combustion - air                              5.216886e+07
N2O - combustion - air                              2.985318e+06
SOx - combustion - air                              1.799166e+08
NOx - combustion - air                              2.295465e+08
NH3 - combustion - air                              1.757216e+07
CO - combustion - air                               4.546753e+08
...                                                          ...
Water Withdrawal Blue - Manufacturing - Radio, ...  9.763615e+00
Water Withdrawal Blue - Manufacturing - Medical...  4.359232e+00
Water Withdrawal Blue - Manufacturing - Motor v...  5.479398e-01
Water Withdrawal Blue - Manufacturing - Other t...  2.074322e-01
Water Withdrawal Blue - Manufacturing - Furnitu...  7.951827e+00
Water Withdrawal Blue - Electricity - tower - E...  1.023536e+01
Water Withdrawal Blue - Electricity - tower - E...  1.652158e+00
Water Withdrawal Blue - Electricity - tower - E...  2.978639e+00
Water Withdrawal Blue - Electricity - tower - E...  0.000000e+00
Water Withdrawal Blue - Electricity - tower - E...  0.000000e+00
Water Withdrawal Blue - Electricity - tower - E...  3.192144e-02
Water Withdrawal Blue - Electricity - tower - E...  1.245842e-01
Water Withdrawal Blue - Electricity - tower - E...  0.000000e+00
Water Withdrawal Blue - Electricity - tower - E...  1.853275e-03
Water Withdrawal Blue - Electricity - tower - E...  0.000000e+00
Water Withdrawal Blue - Electricity - tower - E...  7.759049e-02
```

```
Water Withdrawal Blue - Electricity - tower - E...  0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...  1.822194e+02
Water Withdrawal Blue - Electricity - once-thro...  1.186950e+02
Water Withdrawal Blue - Electricity - once-thro...  9.857418e+01
Water Withdrawal Blue - Electricity - once-thro...  0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...  0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...  3.430491e+01
Water Withdrawal Blue - Electricity - once-thro...  2.189812e+00
Water Withdrawal Blue - Electricity - once-thro...  0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...  0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...  0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...  3.096764e-01
Water Withdrawal Blue - Electricity - once-thro...  0.000000e+00
Water Withdrawal Blue - Domestic - domestic Wat...  7.302420e+02

region                                                        DE  \
Taxes less subsidies on products purchased: Total   1.180021e+05
Other net taxes on production                       1.984904e+04
Compensation of employees; wages, salaries, & e...  6.673981e+04
Compensation of employees; wages, salaries, & e...  5.641227e+05
Compensation of employees; wages, salaries, & e...  7.210208e+05
Operating surplus: Consumption of fixed capital     4.223236e+05
Operating surplus: Rents on land                    0.000000e+00
Operating surplus: Royalties on resources           0.000000e+00
Operating surplus: Remaining net operating surplus  6.554699e+05
Employment: Low-skilled male                        6.161907e+03
Employment: Low-skilled female                      4.154719e+03
Employment: Medium-skilled male                     2.323311e+04
Employment: Medium-skilled female                   1.470482e+04
Employment: High-skilled male                       9.703199e+03
Employment: High-skilled female                     8.854937e+03
Employment hours: Low-skilled male                  1.286064e+04
Employment hours: Low-skilled female                7.717936e+03
Employment hours: Medium-skilled male               4.487630e+04
Employment hours: Medium-skilled female             2.624602e+04
Employment hours: High-skilled male                 1.555775e+04
Employment hours: High-skilled female               1.295417e+04
Employment: Vulnerable employment                   1.964642e+04
Employment hours: Vulnerable employment             4.135181e+04
CO2 - combustion - air                              9.534470e+11
CH4 - combustion - air                              3.160890e+08
N2O - combustion - air                              2.375527e+07
SOx - combustion - air                              1.454505e+09
NOx - combustion - air                              2.627540e+09
NH3 - combustion - air                              2.800200e+08
CO - combustion - air                               3.570598e+09
...                                                          ...
Water Withdrawal Blue - Manufacturing - Radio, ...  1.156977e+02
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Water Withdrawal Blue - Manufacturing - Medical...	6.327725e+01
Water Withdrawal Blue - Manufacturing - Motor v...	7.858801e+00
Water Withdrawal Blue - Manufacturing - Other t...	1.047492e+01
Water Withdrawal Blue - Manufacturing - Furnitu...	1.566249e+02
Water Withdrawal Blue - Electricity - tower - E...	5.101949e+02
Water Withdrawal Blue - Electricity - tower - E...	5.606452e+01
Water Withdrawal Blue - Electricity - tower - E...	2.009605e+02
Water Withdrawal Blue - Electricity - tower - E...	0.000000e+00
Water Withdrawal Blue - Electricity - tower - E...	0.000000e+00
Water Withdrawal Blue - Electricity - tower - E...	1.232672e+00
Water Withdrawal Blue - Electricity - tower - E...	7.310849e+00
Water Withdrawal Blue - Electricity - tower - E...	0.000000e+00
Water Withdrawal Blue - Electricity - tower - E...	3.067771e-02
Water Withdrawal Blue - Electricity - tower - E...	0.000000e+00
Water Withdrawal Blue - Electricity - tower - E...	1.049131e+00
Water Withdrawal Blue - Electricity - tower - E...	0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...	7.117108e+03
Water Withdrawal Blue - Electricity - once-thro...	2.935697e+03
Water Withdrawal Blue - Electricity - once-thro...	4.924222e+03
Water Withdrawal Blue - Electricity - once-thro...	0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...	0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...	4.797078e+02
Water Withdrawal Blue - Electricity - once-thro...	2.452241e+02
Water Withdrawal Blue - Electricity - once-thro...	0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...	0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...	0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...	4.099801e+00
Water Withdrawal Blue - Electricity - once-thro...	0.000000e+00
Water Withdrawal Blue - Domestic - domestic Wat...	3.369450e+03

region	DK \
Taxes less subsidies on products purchased: Total	1.259920e+04
Other net taxes on production	2.971154e+03
Compensation of employees; wages, salaries, & e...	1.212026e+04
Compensation of employees; wages, salaries, & e...	5.906922e+04
Compensation of employees; wages, salaries, & e...	6.901867e+04
Operating surplus: Consumption of fixed capital	4.304456e+04
Operating surplus: Rents on land	0.000000e+00
Operating surplus: Royalties on resources	0.000000e+00
Operating surplus: Remaining net operating surplus	3.289242e+04
Employment: Low-skilled male	5.752234e+02
Employment: Low-skilled female	3.461894e+02
Employment: Medium-skilled male	1.844487e+03
Employment: Medium-skilled female	1.176033e+03
Employment: High-skilled male	7.622714e+02
Employment: High-skilled female	6.117144e+02
Employment hours: Low-skilled male	1.241258e+03
Employment hours: Low-skilled female	7.121857e+02

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Employment hours: Medium-skilled male                3.825879e+03
Employment hours: Medium-skilled female              2.303978e+03
Employment hours: High-skilled male                  1.477810e+03
Employment hours: High-skilled female                1.136849e+03
Employment: Vulnerable employment                    1.572395e+03
Employment hours: Vulnerable employment              3.038424e+03
CO2 - combustion - air                               7.034627e+10
CH4 - combustion - air                               2.386162e+07
N2O - combustion - air                               1.769006e+06
SOx - combustion - air                               2.493736e+08
NOx - combustion - air                               4.584186e+08
NH3 - combustion - air                               3.176564e+07
CO - combustion - air                                3.404348e+08
...                                                           ...
Water Withdrawal Blue - Manufacturing - Radio, ...   3.482780e+00
Water Withdrawal Blue - Manufacturing - Medical...   2.769366e+00
Water Withdrawal Blue - Manufacturing - Motor v...   6.155180e-01
Water Withdrawal Blue - Manufacturing - Other t...   5.575786e-01
Water Withdrawal Blue - Manufacturing - Furnitu...   2.614994e+01
Water Withdrawal Blue - Electricity - tower - E...   6.706169e+00
Water Withdrawal Blue - Electricity - tower - E...   1.643671e+00
Water Withdrawal Blue - Electricity - tower - E...   1.700204e+00
Water Withdrawal Blue - Electricity - tower - E...   0.000000e+00
Water Withdrawal Blue - Electricity - tower - E...   0.000000e+00
Water Withdrawal Blue - Electricity - tower - E...   2.804938e-02
Water Withdrawal Blue - Electricity - tower - E...   1.743876e+02
Water Withdrawal Blue - Electricity - tower - E...   0.000000e+00
Water Withdrawal Blue - Electricity - tower - E...   1.369022e-03
Water Withdrawal Blue - Electricity - tower - E...   0.000000e+00
Water Withdrawal Blue - Electricity - tower - E...   6.660749e-02
Water Withdrawal Blue - Electricity - tower - E...   0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...   1.237286e+02
Water Withdrawal Blue - Electricity - once-thro...   1.302786e+02
Water Withdrawal Blue - Electricity - once-thro...   6.795975e+01
Water Withdrawal Blue - Electricity - once-thro...   0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...   0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...   3.193765e+01
Water Withdrawal Blue - Electricity - once-thro...   2.698709e+00
Water Withdrawal Blue - Electricity - once-thro...   0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...   0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...   0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...   5.109842e-01
Water Withdrawal Blue - Electricity - once-thro...   0.000000e+00
Water Withdrawal Blue - Domestic - domestic Wat...   3.573010e+02

region                                                         EE  \
Taxes less subsidies on products purchased: Total   1.053775e+03
Other net taxes on production                       1.632947e+02
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Compensation of employees; wages, salaries, & e...  5.448535e+02
Compensation of employees; wages, salaries, & e...  3.304123e+03
Compensation of employees; wages, salaries, & e...  3.987665e+03
Operating surplus: Consumption of fixed capital     2.735572e+03
Operating surplus: Rents on land                    0.000000e+00
Operating surplus: Royalties on resources           0.000000e+00
Operating surplus: Remaining net operating surplus  3.902313e+03
Employment: Low-skilled male                        1.008421e+02
Employment: Low-skilled female                      7.148732e+01
Employment: Medium-skilled male                     2.882654e+02
Employment: Medium-skilled female                   1.517232e+02
Employment: High-skilled male                       9.716389e+01
Employment: High-skilled female                     1.263945e+02
Employment hours: Low-skilled male                  2.253055e+02
Employment hours: Low-skilled female                1.527721e+02
Employment hours: Medium-skilled male               6.090489e+02
Employment hours: Medium-skilled female             3.117819e+02
Employment hours: High-skilled male                 1.994933e+02
Employment hours: High-skilled female               2.488272e+02
Employment: Vulnerable employment                   1.834602e+02
Employment hours: Vulnerable employment             3.873478e+02
CO2 - combustion - air                              1.735572e+10
CH4 - combustion - air                              9.356093e+06
N2O - combustion - air                              6.245674e+05
SOx - combustion - air                              5.834907e+07
NOx - combustion - air                              3.496966e+07
NH3 - combustion - air                              2.227618e+06
CO - combustion - air                               8.453583e+07
...                                                          ...
Water Withdrawal Blue - Manufacturing - Radio, ...  8.691096e-01
Water Withdrawal Blue - Manufacturing - Medical...  3.995703e-01
Water Withdrawal Blue - Manufacturing - Motor v...  8.359101e-02
Water Withdrawal Blue - Manufacturing - Other t...  1.231763e-01
Water Withdrawal Blue - Manufacturing - Furnitu...  1.706609e+00
Water Withdrawal Blue - Electricity - tower - E...  1.113659e+00
Water Withdrawal Blue - Electricity - tower - E...  6.113282e+00
Water Withdrawal Blue - Electricity - tower - E...  2.653983e-01
Water Withdrawal Blue - Electricity - tower - E...  0.000000e+00
Water Withdrawal Blue - Electricity - tower - E...  0.000000e+00
Water Withdrawal Blue - Electricity - tower - E...  1.824580e-02
Water Withdrawal Blue - Electricity - tower - E...  1.437627e-02
Water Withdrawal Blue - Electricity - tower - E...  0.000000e+00
Water Withdrawal Blue - Electricity - tower - E...  1.340942e-04
Water Withdrawal Blue - Electricity - tower - E...  0.000000e+00
Water Withdrawal Blue - Electricity - tower - E...  2.198565e-02
Water Withdrawal Blue - Electricity - tower - E...  0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...  1.796111e+01
Water Withdrawal Blue - Electricity - once-thro...  2.674228e+01
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Water Withdrawal Blue - Electricity - once-thro...  1.311643e+01
Water Withdrawal Blue - Electricity - once-thro...  0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...  0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...  3.475645e+01
Water Withdrawal Blue - Electricity - once-thro...  6.389360e-01
Water Withdrawal Blue - Electricity - once-thro...  0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...  0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...  0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...  2.983251e-02
Water Withdrawal Blue - Electricity - once-thro...  0.000000e+00
Water Withdrawal Blue - Domestic - domestic Wat...  5.233010e+01

region                                                        ES  \
Taxes less subsidies on products purchased: Total   4.260830e+04
Other net taxes on production                       1.424046e+04
Compensation of employees; wages, salaries, & e...  4.810833e+04
Compensation of employees; wages, salaries, & e...  2.484917e+05
Compensation of employees; wages, salaries, & e...  2.522517e+05
Operating surplus: Consumption of fixed capital     1.725750e+05
Operating surplus: Rents on land                    0.000000e+00
Operating surplus: Royalties on resources           0.000000e+00
Operating surplus: Remaining net operating surplus  2.933309e+05
Employment: Low-skilled male                        3.756031e+03
Employment: Low-skilled female                      2.591410e+03
Employment: Medium-skilled male                     1.078246e+04
Employment: Medium-skilled female                   6.100886e+03
Employment: High-skilled male                       3.737733e+03
Employment: High-skilled female                     3.198043e+03
Employment hours: Low-skilled male                  8.320974e+03
Employment hours: Low-skilled female                5.403192e+03
Employment hours: Medium-skilled male               2.259001e+04
Employment hours: Medium-skilled female             1.237711e+04
Employment hours: High-skilled male                 7.462856e+03
Employment hours: High-skilled female               6.118161e+03
Employment: Vulnerable employment                   8.568683e+03
Employment hours: Vulnerable employment             1.401428e+04
CO2 - combustion - air                              3.289151e+11
CH4 - combustion - air                              1.096667e+08
N2O - combustion - air                              6.368892e+06
SOx - combustion - air                              4.596735e+08
NOx - combustion - air                              1.116275e+09
NH3 - combustion - air                              6.695715e+07
CO - combustion - air                               1.223493e+09
...                                                          ...
Water Withdrawal Blue - Manufacturing - Radio, ...  2.871080e+01
Water Withdrawal Blue - Manufacturing - Medical...  1.792360e+01
Water Withdrawal Blue - Manufacturing - Motor v...  1.630305e+00
Water Withdrawal Blue - Manufacturing - Other t...  2.068299e+00
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Water Withdrawal Blue - Manufacturing - Furnitu...	6.870458e+01
Water Withdrawal Blue - Electricity - tower - E...	5.321215e+01
Water Withdrawal Blue - Electricity - tower - E...	2.879413e+01
Water Withdrawal Blue - Electricity - tower - E...	3.035796e+01
Water Withdrawal Blue - Electricity - tower - E...	0.000000e+00
Water Withdrawal Blue - Electricity - tower - E...	0.000000e+00
Water Withdrawal Blue - Electricity - tower - E...	1.586025e-01
Water Withdrawal Blue - Electricity - tower - E...	1.148854e+00
Water Withdrawal Blue - Electricity - tower - E...	0.000000e+00
Water Withdrawal Blue - Electricity - tower - E...	9.674971e-01
Water Withdrawal Blue - Electricity - tower - E...	0.000000e+00
Water Withdrawal Blue - Electricity - tower - E...	3.904574e-01
Water Withdrawal Blue - Electricity - tower - E...	0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...	1.303500e+03
Water Withdrawal Blue - Electricity - once-thro...	8.156766e+02
Water Withdrawal Blue - Electricity - once-thro...	3.587750e+03
Water Withdrawal Blue - Electricity - once-thro...	0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...	0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...	2.902779e+02
Water Withdrawal Blue - Electricity - once-thro...	7.640021e+01
Water Withdrawal Blue - Electricity - once-thro...	0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...	0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...	0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...	2.638406e+00
Water Withdrawal Blue - Electricity - once-thro...	0.000000e+00
Water Withdrawal Blue - Domestic - domestic Wat...	2.861060e+03

region	FI	...	\
Taxes less subsidies on products purchased: Total	1.030063e+04	...	
Other net taxes on production	9.021262e+02	...	
Compensation of employees; wages, salaries, & e...	7.636838e+03	...	
Compensation of employees; wages, salaries, & e...	4.449555e+04	...	
Compensation of employees; wages, salaries, & e...	5.560204e+04	...	
Operating surplus: Consumption of fixed capital	3.527640e+04	...	
Operating surplus: Rents on land	0.000000e+00	...	
Operating surplus: Royalties on resources	0.000000e+00	...	
Operating surplus: Remaining net operating surplus	4.406422e+04	...	
Employment: Low-skilled male	4.132710e+02	...	
Employment: Low-skilled female	2.779811e+02	...	
Employment: Medium-skilled male	1.596426e+03	...	
Employment: Medium-skilled female	1.031233e+03	...	
Employment: High-skilled male	6.777373e+02	...	
Employment: High-skilled female	6.059195e+02	...	
Employment hours: Low-skilled male	8.998939e+02	...	
Employment hours: Low-skilled female	5.724383e+02	...	
Employment hours: Medium-skilled male	3.331826e+03	...	
Employment hours: Medium-skilled female	2.047771e+03	...	
Employment hours: High-skilled male	1.331222e+03	...	

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Employment hours: High-skilled female                1.136766e+03  ...
Employment: Vulnerable employment                    1.307165e+03  ...
Employment hours: Vulnerable employment              2.792002e+03  ...
CO2 - combustion - air                               7.798378e+10  ...
CH4 - combustion - air                               3.117240e+07  ...
N2O - combustion - air                               3.714788e+06  ...
SOx - combustion - air                               1.350690e+08  ...
NOx - combustion - air                               2.648465e+08  ...
NH3 - combustion - air                               1.730419e+07  ...
CO - combustion - air                                3.350213e+08  ...
...                                                           ...  ...
Water Withdrawal Blue - Manufacturing - Radio, ...   6.978133e+00  ...
Water Withdrawal Blue - Manufacturing - Medical...   2.203551e+00  ...
Water Withdrawal Blue - Manufacturing - Motor v...   4.765175e-01  ...
Water Withdrawal Blue - Manufacturing - Other t...   4.478914e-01  ...
Water Withdrawal Blue - Manufacturing - Furnitu...   9.145540e+00  ...
Water Withdrawal Blue - Electricity - tower - E...   6.754844e+00  ...
Water Withdrawal Blue - Electricity - tower - E...   3.155380e+00  ...
Water Withdrawal Blue - Electricity - tower - E...   1.835478e+00  ...
Water Withdrawal Blue - Electricity - tower - E...   0.000000e+00  ...
Water Withdrawal Blue - Electricity - tower - E...   0.000000e+00  ...
Water Withdrawal Blue - Electricity - tower - E...   2.761535e-02  ...
Water Withdrawal Blue - Electricity - tower - E...   1.821154e-01  ...
Water Withdrawal Blue - Electricity - tower - E...   0.000000e+00  ...
Water Withdrawal Blue - Electricity - tower - E...   1.459140e-03  ...
Water Withdrawal Blue - Electricity - tower - E...   0.000000e+00  ...
Water Withdrawal Blue - Electricity - tower - E...   6.806118e-02  ...
Water Withdrawal Blue - Electricity - tower - E...   0.000000e+00  ...
Water Withdrawal Blue - Electricity - once-thro...   4.177217e+02  ...
Water Withdrawal Blue - Electricity - once-thro...   2.174854e+02  ...
Water Withdrawal Blue - Electricity - once-thro...   8.101784e+01  ...
Water Withdrawal Blue - Electricity - once-thro...   0.000000e+00  ...
Water Withdrawal Blue - Electricity - once-thro...   0.000000e+00  ...
Water Withdrawal Blue - Electricity - once-thro...   3.286304e+01  ...
Water Withdrawal Blue - Electricity - once-thro...   5.948408e+02  ...
Water Withdrawal Blue - Electricity - once-thro...   0.000000e+00  ...
Water Withdrawal Blue - Electricity - once-thro...   0.000000e+00  ...
Water Withdrawal Blue - Electricity - once-thro...   0.000000e+00  ...
Water Withdrawal Blue - Electricity - once-thro...   3.142040e-01  ...
Water Withdrawal Blue - Electricity - once-thro...   0.000000e+00  ...
Water Withdrawal Blue - Domestic - domestic Wat...   3.585800e+02  ...

region                                                         TR  \
Taxes less subsidies on products purchased: Total   5.036719e+04
Other net taxes on production                       8.478190e+03
Compensation of employees; wages, salaries, & e...  2.049963e+04
Compensation of employees; wages, salaries, & e...  1.227783e+05
Compensation of employees; wages, salaries, & e...  9.904794e+04
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Operating surplus: Consumption of fixed capital	7.850444e+04
Operating surplus: Rents on land	0.000000e+00
Operating surplus: Royalties on resources	0.000000e+00
Operating surplus: Remaining net operating surplus	2.251767e+05
Employment: Low-skilled male	3.504514e+03
Employment: Low-skilled female	1.747052e+03
Employment: Medium-skilled male	1.327968e+04
Employment: Medium-skilled female	5.298452e+03
Employment: High-skilled male	4.006685e+03
Employment: High-skilled female	1.640590e+03
Employment hours: Low-skilled male	8.482730e+03
Employment hours: Low-skilled female	3.877001e+03
Employment hours: Medium-skilled male	3.180066e+04
Employment hours: Medium-skilled female	1.181053e+04
Employment hours: High-skilled male	9.865858e+03
Employment hours: High-skilled female	3.687502e+03
Employment: Vulnerable employment	1.152099e+04
Employment hours: Vulnerable employment	2.598081e+04
CO2 - combustion - air	3.612865e+11
CH4 - combustion - air	1.811200e+08
N2O - combustion - air	5.487008e+06
SOx - combustion - air	1.366697e+09
NOx - combustion - air	1.088364e+09
NH3 - combustion - air	5.553216e+07
CO - combustion - air	1.947259e+09
...	...
Water Withdrawal Blue - Manufacturing - Radio, ...	1.837190e+01
Water Withdrawal Blue - Manufacturing - Medical...	1.184129e+01
Water Withdrawal Blue - Manufacturing - Motor v...	2.188961e+00
Water Withdrawal Blue - Manufacturing - Other t...	1.832779e+00
Water Withdrawal Blue - Manufacturing - Furnitu...	1.591226e+01
Water Withdrawal Blue - Electricity - tower - E...	1.448965e+02
Water Withdrawal Blue - Electricity - tower - E...	2.833762e+01
Water Withdrawal Blue - Electricity - tower - E...	5.802068e+00
Water Withdrawal Blue - Electricity - tower - E...	0.000000e+00
Water Withdrawal Blue - Electricity - tower - E...	0.000000e+00
Water Withdrawal Blue - Electricity - tower - E...	1.311679e-01
Water Withdrawal Blue - Electricity - tower - E...	3.416572e-01
Water Withdrawal Blue - Electricity - tower - E...	0.000000e+00
Water Withdrawal Blue - Electricity - tower - E...	7.970594e-03
Water Withdrawal Blue - Electricity - tower - E...	0.000000e+00
Water Withdrawal Blue - Electricity - tower - E...	8.622961e-01
Water Withdrawal Blue - Electricity - tower - E...	0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...	5.446412e+02
Water Withdrawal Blue - Electricity - once-thro...	5.538887e+02
Water Withdrawal Blue - Electricity - once-thro...	2.983626e+02
Water Withdrawal Blue - Electricity - once-thro...	0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...	0.000000e+00

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Water Withdrawal Blue - Electricity - once-thro...  2.831543e+03
Water Withdrawal Blue - Electricity - once-thro...  4.294154e+00
Water Withdrawal Blue - Electricity - once-thro...  0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...  0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...  0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...  5.894416e-01
Water Withdrawal Blue - Electricity - once-thro...  0.000000e+00
Water Withdrawal Blue - Domestic - domestic Wat...  5.663050e+03

region                                                        TW  \
Taxes less subsidies on products purchased: Total   1.165991e+04
Other net taxes on production                       1.002672e+04
Compensation of employees; wages, salaries, & e...  1.694190e+04
Compensation of employees; wages, salaries, & e...  1.022466e+05
Compensation of employees; wages, salaries, & e...  4.975263e+04
Operating surplus: Consumption of fixed capital     4.942616e+04
Operating surplus: Rents on land                    0.000000e+00
Operating surplus: Royalties on resources           0.000000e+00
Operating surplus: Remaining net operating surplus  8.536556e+04
Employment: Low-skilled male                        1.586150e+03
Employment: Low-skilled female                      9.417579e+02
Employment: Medium-skilled male                     7.017019e+03
Employment: Medium-skilled female                   4.379147e+03
Employment: High-skilled male                       1.420937e+03
Employment: High-skilled female                     9.737073e+02
Employment hours: Low-skilled male                  3.390391e+03
Employment hours: Low-skilled female                1.986421e+03
Employment hours: Medium-skilled male               1.430201e+04
Employment hours: Medium-skilled female             8.839422e+03
Employment hours: High-skilled male                 3.024430e+03
Employment hours: High-skilled female               2.032577e+03
Employment: Vulnerable employment                   7.494697e+03
Employment hours: Vulnerable employment             1.482476e+04
CO2 - combustion - air                              2.050476e+11
CH4 - combustion - air                              3.803482e+07
N2O - combustion - air                              3.070765e+06
SOx - combustion - air                              3.252457e+08
NOx - combustion - air                              5.871927e+08
NH3 - combustion - air                              2.828222e+08
CO - combustion - air                               6.680881e+08
...                                                          ...
Water Withdrawal Blue - Manufacturing - Radio, ...  1.632154e+01
Water Withdrawal Blue - Manufacturing - Medical...  1.689581e+01
Water Withdrawal Blue - Manufacturing - Motor v...  1.969123e+02
Water Withdrawal Blue - Manufacturing - Other t...  3.194785e+01
Water Withdrawal Blue - Manufacturing - Furnitu...  1.486544e+01
Water Withdrawal Blue - Electricity - tower - E...  1.184496e+01
Water Withdrawal Blue - Electricity - tower - E...  3.234683e+00
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Water Withdrawal Blue - Electricity - tower - E...  3.227916e+00
Water Withdrawal Blue - Electricity - tower - E...  0.000000e+00
Water Withdrawal Blue - Electricity - tower - E...  0.000000e+00
Water Withdrawal Blue - Electricity - tower - E...  7.356887e-02
Water Withdrawal Blue - Electricity - tower - E...  1.670924e-01
Water Withdrawal Blue - Electricity - tower - E...  0.000000e+00
Water Withdrawal Blue - Electricity - tower - E...  2.710702e-03
Water Withdrawal Blue - Electricity - tower - E...  0.000000e+00
Water Withdrawal Blue - Electricity - tower - E...  1.419185e-01
Water Withdrawal Blue - Electricity - tower - E...  0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...  2.703587e+02
Water Withdrawal Blue - Electricity - once-thro...  2.718219e+02
Water Withdrawal Blue - Electricity - once-thro...  1.440440e+02
Water Withdrawal Blue - Electricity - once-thro...  0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...  0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...  1.762860e+02
Water Withdrawal Blue - Electricity - once-thro...  2.664086e+00
Water Withdrawal Blue - Electricity - once-thro...  0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...  0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...  0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...  8.247824e-01
Water Withdrawal Blue - Electricity - once-thro...  0.000000e+00
Water Withdrawal Blue - Domestic - domestic Wat...  3.424900e+03

region                                                        NO  \
Taxes less subsidies on products purchased: Total   1.541281e+04
Other net taxes on production                       1.790371e+03
Compensation of employees; wages, salaries, & e...  1.312564e+04
Compensation of employees; wages, salaries, & e...  7.620551e+04
Compensation of employees; wages, salaries, & e...  8.208182e+04
Operating surplus: Consumption of fixed capital     4.616871e+04
Operating surplus: Rents on land                    0.000000e+00
Operating surplus: Royalties on resources           0.000000e+00
Operating surplus: Remaining net operating surplus  7.712858e+04
Employment: Low-skilled male                        6.101026e+02
Employment: Low-skilled female                      3.668619e+02
Employment: Medium-skilled male                     2.426409e+03
Employment: Medium-skilled female                   1.465726e+03
Employment: High-skilled male                       8.441735e+02
Employment: High-skilled female                     6.876956e+02
Employment hours: Low-skilled male                  1.353115e+03
Employment hours: Low-skilled female                7.792391e+02
Employment hours: Medium-skilled male               5.023831e+03
Employment hours: Medium-skilled female             2.866233e+03
Employment hours: High-skilled male                 1.642989e+03
Employment hours: High-skilled female               1.247920e+03
Employment: Vulnerable employment                   1.950270e+03
Employment hours: Vulnerable employment             4.146107e+03
```

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CO2 - combustion - air                                  7.160644e+10
CH4 - combustion - air                                  1.618222e+07
N2O - combustion - air                                  1.444827e+06
SOx - combustion - air                                  2.623183e+08
NOx - combustion - air                                  4.572874e+08
NH3 - combustion - air                                  2.563692e+07
CO - combustion - air                                   2.687705e+08
...                                                              ...
Water Withdrawal Blue - Manufacturing - Radio, ...  1.069812e+01
Water Withdrawal Blue - Manufacturing - Medical...  6.348676e+00
Water Withdrawal Blue - Manufacturing - Motor v...  9.331555e-01
Water Withdrawal Blue - Manufacturing - Other t...  2.734062e+00
Water Withdrawal Blue - Manufacturing - Furnitu...  8.598012e+01
Water Withdrawal Blue - Electricity - tower - E...  1.172896e+01
Water Withdrawal Blue - Electricity - tower - E...  2.267021e+00
Water Withdrawal Blue - Electricity - tower - E...  2.551123e+00
Water Withdrawal Blue - Electricity - tower - E...  0.000000e+00
Water Withdrawal Blue - Electricity - tower - E...  0.000000e+00
Water Withdrawal Blue - Electricity - tower - E...  4.458976e-02
Water Withdrawal Blue - Electricity - tower - E...  3.794810e-01
Water Withdrawal Blue - Electricity - tower - E...  0.000000e+00
Water Withdrawal Blue - Electricity - tower - E...  2.995815e-03
Water Withdrawal Blue - Electricity - tower - E...  0.000000e+00
Water Withdrawal Blue - Electricity - tower - E...  1.013269e-01
Water Withdrawal Blue - Electricity - tower - E...  0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...  2.333656e+02
Water Withdrawal Blue - Electricity - once-thro...  1.548900e+02
Water Withdrawal Blue - Electricity - once-thro...  1.166955e+02
Water Withdrawal Blue - Electricity - once-thro...  0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...  0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...  4.957668e+01
Water Withdrawal Blue - Electricity - once-thro...  3.290840e+00
Water Withdrawal Blue - Electricity - once-thro...  0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...  0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...  0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...  4.274919e-01
Water Withdrawal Blue - Electricity - once-thro...  0.000000e+00
Water Withdrawal Blue - Domestic - domestic Wat...  7.543170e+02

region                                                      ID  \
Taxes less subsidies on products purchased: Total   3.282434e+04
Other net taxes on production                       1.673609e+04
Compensation of employees; wages, salaries, & e...  2.470071e+04
Compensation of employees; wages, salaries, & e...  1.346768e+05
Compensation of employees; wages, salaries, & e...  5.049470e+04
Operating surplus: Consumption of fixed capital     7.065823e+04
Operating surplus: Rents on land                    0.000000e+00
Operating surplus: Royalties on resources           0.000000e+00
```

Operating surplus: Remaining net operating surplus	2.955254e+05
Employment: Low-skilled male	1.213243e+04
Employment: Low-skilled female	4.752781e+03
Employment: Medium-skilled male	4.439541e+04
Employment: Medium-skilled female	2.412643e+04
Employment: High-skilled male	5.830617e+03
Employment: High-skilled female	3.130031e+03
Employment hours: Low-skilled male	2.705972e+04
Employment hours: Low-skilled female	1.046199e+04
Employment hours: Medium-skilled male	9.158495e+04
Employment hours: Medium-skilled female	4.909153e+04
Employment hours: High-skilled male	1.230545e+04
Employment hours: High-skilled female	6.008021e+03
Employment: Vulnerable employment	5.286797e+04
Employment hours: Vulnerable employment	1.065753e+05
CO2 - combustion - air	4.207423e+11
CH4 - combustion - air	5.753229e+08
N2O - combustion - air	1.368899e+07
SOx - combustion - air	1.076933e+09
NOx - combustion - air	1.299285e+09
NH3 - combustion - air	1.561020e+08
CO - combustion - air	1.205931e+10
...	...
Water Withdrawal Blue - Manufacturing - Radio, ...	3.279274e+01
Water Withdrawal Blue - Manufacturing - Medical...	1.661048e+01
Water Withdrawal Blue - Manufacturing - Motor v...	3.750754e+00
Water Withdrawal Blue - Manufacturing - Other t...	2.229895e+00
Water Withdrawal Blue - Manufacturing - Furnitu...	2.353067e+01
Water Withdrawal Blue - Electricity - tower - E...	2.181919e+01
Water Withdrawal Blue - Electricity - tower - E...	2.654471e+00
Water Withdrawal Blue - Electricity - tower - E...	2.067576e+00
Water Withdrawal Blue - Electricity - tower - E...	0.000000e+00
Water Withdrawal Blue - Electricity - tower - E...	0.000000e+00
Water Withdrawal Blue - Electricity - tower - E...	7.953156e-02
Water Withdrawal Blue - Electricity - tower - E...	2.034251e-01
Water Withdrawal Blue - Electricity - tower - E...	0.000000e+00
Water Withdrawal Blue - Electricity - tower - E...	1.735953e-03
Water Withdrawal Blue - Electricity - tower - E...	0.000000e+00
Water Withdrawal Blue - Electricity - tower - E...	2.374340e+00
Water Withdrawal Blue - Electricity - tower - E...	0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...	3.976928e+02
Water Withdrawal Blue - Electricity - once-thro...	9.761132e+03
Water Withdrawal Blue - Electricity - once-thro...	1.238135e+02
Water Withdrawal Blue - Electricity - once-thro...	0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...	0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...	1.830260e+02
Water Withdrawal Blue - Electricity - once-thro...	1.385897e+00
Water Withdrawal Blue - Electricity - once-thro...	0.000000e+00

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Water Withdrawal Blue - Electricity - once-thro...  0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...  0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...  2.174445e+00
Water Withdrawal Blue - Electricity - once-thro...  0.000000e+00
Water Withdrawal Blue - Domestic - domestic Wat...  1.054960e+04

region                                                        ZA  \
Taxes less subsidies on products purchased: Total   2.067937e+04
Other net taxes on production                       6.952306e+03
Compensation of employees; wages, salaries, & e...  1.246853e+04
Compensation of employees; wages, salaries, & e...  4.888275e+04
Compensation of employees; wages, salaries, & e...  6.873167e+04
Operating surplus: Consumption of fixed capital     7.357428e+04
Operating surplus: Rents on land                    0.000000e+00
Operating surplus: Royalties on resources           0.000000e+00
Operating surplus: Remaining net operating surplus  6.537635e+04
Employment: Low-skilled male                        2.273744e+03
Employment: Low-skilled female                      2.008837e+03
Employment: Medium-skilled male                     5.206959e+03
Employment: Medium-skilled female                   2.602516e+03
Employment: High-skilled male                       1.852499e+03
Employment: High-skilled female                     1.230903e+03
Employment hours: Low-skilled male                  5.118908e+03
Employment hours: Low-skilled female                4.419096e+03
Employment hours: Medium-skilled male               1.164176e+04
Employment hours: Medium-skilled female             5.745241e+03
Employment hours: High-skilled male                 4.147249e+03
Employment hours: High-skilled female               2.687040e+03
Employment: Vulnerable employment                   5.548696e+03
Employment hours: Vulnerable employment             1.235098e+04
CO2 - combustion - air                              2.802167e+11
CH4 - combustion - air                              9.812903e+07
N2O - combustion - air                              5.805085e+06
SOx - combustion - air                              9.267547e+08
NOx - combustion - air                              6.888304e+08
NH3 - combustion - air                              3.831204e+07
CO - combustion - air                               2.187387e+09
...                                                          ...
Water Withdrawal Blue - Manufacturing - Radio, ...  1.845405e+01
Water Withdrawal Blue - Manufacturing - Medical...  6.586184e+00
Water Withdrawal Blue - Manufacturing - Motor v...  2.313082e+00
Water Withdrawal Blue - Manufacturing - Other t...  4.356376e-01
Water Withdrawal Blue - Manufacturing - Furnitu...  2.550134e+01
Water Withdrawal Blue - Electricity - tower - E...  2.068538e+02
Water Withdrawal Blue - Electricity - tower - E...  1.156433e+00
Water Withdrawal Blue - Electricity - tower - E...  1.128334e+00
Water Withdrawal Blue - Electricity - tower - E...  0.000000e+00
Water Withdrawal Blue - Electricity - tower - E...  0.000000e+00
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Water Withdrawal Blue - Electricity - tower - E...	4.362584e-02
Water Withdrawal Blue - Electricity - tower - E...	7.691986e-02
Water Withdrawal Blue - Electricity - tower - E...	0.000000e+00
Water Withdrawal Blue - Electricity - tower - E...	1.108583e-03
Water Withdrawal Blue - Electricity - tower - E...	0.000000e+00
Water Withdrawal Blue - Electricity - tower - E...	7.420643e-02
Water Withdrawal Blue - Electricity - tower - E...	0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...	1.291210e+02
Water Withdrawal Blue - Electricity - once-thro...	2.046795e+02
Water Withdrawal Blue - Electricity - once-thro...	5.066108e+01
Water Withdrawal Blue - Electricity - once-thro...	0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...	0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...	6.327482e+01
Water Withdrawal Blue - Electricity - once-thro...	9.401259e-01
Water Withdrawal Blue - Electricity - once-thro...	0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...	0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...	0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...	1.333958e+00
Water Withdrawal Blue - Electricity - once-thro...	0.000000e+00
Water Withdrawal Blue - Domestic - domestic Wat...	3.315080e+03

region	WA \
Taxes less subsidies on products purchased: Total	2.238648e+05
Other net taxes on production	6.121378e+04
Compensation of employees; wages, salaries, & e...	1.079187e+05
Compensation of employees; wages, salaries, & e...	6.677931e+05
Compensation of employees; wages, salaries, & e...	3.325094e+05
Operating surplus: Consumption of fixed capital	2.283011e+05
Operating surplus: Rents on land	0.000000e+00
Operating surplus: Royalties on resources	0.000000e+00
Operating surplus: Remaining net operating surplus	3.891866e+05
Employment: Low-skilled male	4.249360e+04
Employment: Low-skilled female	1.568796e+04
Employment: Medium-skilled male	1.572065e+05
Employment: Medium-skilled female	8.641473e+04
Employment: High-skilled male	2.200772e+04
Employment: High-skilled female	1.230325e+04
Employment hours: Low-skilled male	9.177467e+04
Employment hours: Low-skilled female	3.263834e+04
Employment hours: Medium-skilled male	3.206715e+05
Employment hours: Medium-skilled female	1.734801e+05
Employment hours: High-skilled male	4.998389e+04
Employment hours: High-skilled female	2.789085e+04
Employment: Vulnerable employment	1.893901e+05
Employment hours: Vulnerable employment	3.747105e+05
CO2 - combustion - air	1.763511e+12
CH4 - combustion - air	1.243645e+09
N2O - combustion - air	3.941151e+07

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SOx - combustion - air                                  4.175343e+09
NOx - combustion - air                                  4.214161e+09
NH3 - combustion - air                                  1.228480e+10
CO - combustion - air                                   1.442947e+10
...                                                              ...
Water Withdrawal Blue - Manufacturing - Radio, ...      6.069030e+02
Water Withdrawal Blue - Manufacturing - Medical...      3.061935e+02
Water Withdrawal Blue - Manufacturing - Motor v...      3.861821e+01
Water Withdrawal Blue - Manufacturing - Other t...      2.354902e+01
Water Withdrawal Blue - Manufacturing - Furnitu...      2.952842e+03
Water Withdrawal Blue - Electricity - tower - E...      2.656774e+02
Water Withdrawal Blue - Electricity - tower - E...      1.094754e+02
Water Withdrawal Blue - Electricity - tower - E...      2.655587e+01
Water Withdrawal Blue - Electricity - tower - E...      0.000000e+00
Water Withdrawal Blue - Electricity - tower - E...      0.000000e+00
Water Withdrawal Blue - Electricity - tower - E...      4.668001e+00
Water Withdrawal Blue - Electricity - tower - E...      3.220025e+00
Water Withdrawal Blue - Electricity - tower - E...      0.000000e+00
Water Withdrawal Blue - Electricity - tower - E...      1.629405e-02
Water Withdrawal Blue - Electricity - tower - E...      0.000000e+00
Water Withdrawal Blue - Electricity - tower - E...      1.594066e+01
Water Withdrawal Blue - Electricity - tower - E...      0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...      1.060270e+04
Water Withdrawal Blue - Electricity - once-thro...      3.063506e+04
Water Withdrawal Blue - Electricity - once-thro...      2.377705e+03
Water Withdrawal Blue - Electricity - once-thro...      0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...      0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...      6.520032e+03
Water Withdrawal Blue - Electricity - once-thro...      1.925127e+01
Water Withdrawal Blue - Electricity - once-thro...      0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...      0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...      0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...      6.051377e+02
Water Withdrawal Blue - Electricity - once-thro...      0.000000e+00
Water Withdrawal Blue - Domestic - domestic Wat...      4.960021e+04

region                                                            WL  \
Taxes less subsidies on products purchased: Total       2.287465e+05
Other net taxes on production                           6.899727e+04
Compensation of employees; wages, salaries, & e...      5.755465e+04
Compensation of employees; wages, salaries, & e...      4.852468e+05
Compensation of employees; wages, salaries, & e...      3.131235e+05
Operating surplus: Consumption of fixed capital         2.062088e+05
Operating surplus: Rents on land                        0.000000e+00
Operating surplus: Royalties on resources               0.000000e+00
Operating surplus: Remaining net operating surplus      3.279145e+05
Employment: Low-skilled male                            1.203375e+04
Employment: Low-skilled female                          5.281257e+03
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Employment: Medium-skilled male	5.175541e+04
Employment: Medium-skilled female	2.146227e+04
Employment: High-skilled male	1.604123e+04
Employment: High-skilled female	1.064722e+04
Employment hours: Low-skilled male	2.853242e+04
Employment hours: Low-skilled female	1.245087e+04
Employment hours: Medium-skilled male	1.184261e+05
Employment hours: Medium-skilled female	4.993405e+04
Employment hours: High-skilled male	3.804239e+04
Employment hours: High-skilled female	2.530596e+04
Employment: Vulnerable employment	4.120442e+04
Employment hours: Vulnerable employment	9.445070e+04
CO2 - combustion - air	8.744709e+11
CH4 - combustion - air	4.313833e+08
N2O - combustion - air	1.668775e+07
SOx - combustion - air	1.601665e+09
NOx - combustion - air	2.361402e+09
NH3 - combustion - air	1.977369e+08
CO - combustion - air	6.409022e+09
...	...
Water Withdrawal Blue - Manufacturing - Radio, ...	1.170600e+02
Water Withdrawal Blue - Manufacturing - Medical...	8.051360e+01
Water Withdrawal Blue - Manufacturing - Motor v...	3.295063e+01
Water Withdrawal Blue - Manufacturing - Other t...	1.943982e+01
Water Withdrawal Blue - Manufacturing - Furnitu...	4.783243e+02
Water Withdrawal Blue - Electricity - tower - E...	1.054297e+02
Water Withdrawal Blue - Electricity - tower - E...	1.858300e+01
Water Withdrawal Blue - Electricity - tower - E...	2.361481e+01
Water Withdrawal Blue - Electricity - tower - E...	0.000000e+00
Water Withdrawal Blue - Electricity - tower - E...	0.000000e+00
Water Withdrawal Blue - Electricity - tower - E...	2.692745e+00
Water Withdrawal Blue - Electricity - tower - E...	1.313581e+00
Water Withdrawal Blue - Electricity - tower - E...	0.000000e+00
Water Withdrawal Blue - Electricity - tower - E...	2.115586e-02
Water Withdrawal Blue - Electricity - tower - E...	0.000000e+00
Water Withdrawal Blue - Electricity - tower - E...	3.013388e+00
Water Withdrawal Blue - Electricity - tower - E...	0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...	5.304854e+03
Water Withdrawal Blue - Electricity - once-thro...	1.352025e+04
Water Withdrawal Blue - Electricity - once-thro...	7.256615e+03
Water Withdrawal Blue - Electricity - once-thro...	0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...	0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...	1.634215e+03
Water Withdrawal Blue - Electricity - once-thro...	8.549371e+01
Water Withdrawal Blue - Electricity - once-thro...	0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...	0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...	0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...	2.485698e+00

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Water Withdrawal Blue - Electricity - once-thro...  0.000000e+00
Water Withdrawal Blue - Domestic - domestic Wat...  2.873534e+04

region                                                          WE  \
Taxes less subsidies on products purchased: Total   3.114963e+04
Other net taxes on production                       9.978492e+03
Compensation of employees; wages, salaries, & e...  1.672947e+04
Compensation of employees; wages, salaries, & e...  8.662124e+04
Compensation of employees; wages, salaries, & e...  6.926108e+04
Operating surplus: Consumption of fixed capital     3.515773e+04
Operating surplus: Rents on land                    0.000000e+00
Operating surplus: Royalties on resources           0.000000e+00
Operating surplus: Remaining net operating surplus  5.834717e+04
Employment: Low-skilled male                        1.220815e+04
Employment: Low-skilled female                      7.857200e+03
Employment: Medium-skilled male                     4.753357e+04
Employment: Medium-skilled female                   2.163811e+04
Employment: High-skilled male                       1.899807e+04
Employment: High-skilled female                     1.641271e+04
Employment hours: Low-skilled male                  2.562845e+04
Employment hours: Low-skilled female                1.604547e+04
Employment hours: Medium-skilled male               9.641374e+04
Employment hours: Medium-skilled female             4.279110e+04
Employment hours: High-skilled male                 3.748559e+04
Employment hours: High-skilled female               3.188465e+04
Employment: Vulnerable employment                   2.580715e+04
Employment hours: Vulnerable employment             5.307907e+04
CO2 - combustion - air                              3.900407e+11
CH4 - combustion - air                              3.514788e+08
N2O - combustion - air                              4.516904e+06
SOx - combustion - air                              1.730444e+09
NOx - combustion - air                              8.688573e+08
NH3 - combustion - air                              3.079799e+07
CO - combustion - air                               1.176236e+09
...                                                          ...
Water Withdrawal Blue - Manufacturing - Radio, ...  3.624107e+01
Water Withdrawal Blue - Manufacturing - Medical...  2.490514e+01
Water Withdrawal Blue - Manufacturing - Motor v...  1.904817e+00
Water Withdrawal Blue - Manufacturing - Other t...  5.591797e-01
Water Withdrawal Blue - Manufacturing - Furnitu...  5.338343e+01
Water Withdrawal Blue - Electricity - tower - E...  2.244045e+02
Water Withdrawal Blue - Electricity - tower - E...  4.365372e+01
Water Withdrawal Blue - Electricity - tower - E...  6.484692e+01
Water Withdrawal Blue - Electricity - tower - E...  0.000000e+00
Water Withdrawal Blue - Electricity - tower - E...  0.000000e+00
Water Withdrawal Blue - Electricity - tower - E...  2.097626e+00
Water Withdrawal Blue - Electricity - tower - E...  1.966629e-01
Water Withdrawal Blue - Electricity - tower - E...  0.000000e+00
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Water Withdrawal Blue - Electricity - tower - E...	2.913893e-03
Water Withdrawal Blue - Electricity - tower - E...	0.000000e+00
Water Withdrawal Blue - Electricity - tower - E...	1.635785e+01
Water Withdrawal Blue - Electricity - tower - E...	0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...	2.972414e+03
Water Withdrawal Blue - Electricity - once-thro...	1.674657e+03
Water Withdrawal Blue - Electricity - once-thro...	2.503779e+03
Water Withdrawal Blue - Electricity - once-thro...	0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...	0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...	8.191719e+01
Water Withdrawal Blue - Electricity - once-thro...	2.548704e+00
Water Withdrawal Blue - Electricity - once-thro...	0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...	0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...	0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...	1.327595e+00
Water Withdrawal Blue - Electricity - once-thro...	0.000000e+00
Water Withdrawal Blue - Domestic - domestic Wat...	8.287718e+03

region	WF	WM
Taxes less subsidies on products purchased: Total	1.586049e+05	2.513780e+05
Other net taxes on production	4.371538e+04	6.946837e+04
Compensation of employees; wages, salaries, & e...	8.415286e+04	7.952591e+04
Compensation of employees; wages, salaries, & e...	2.352330e+05	4.444607e+05
Compensation of employees; wages, salaries, & e...	2.262130e+05	3.825625e+05
Operating surplus: Consumption of fixed capital	1.286965e+05	2.127556e+05
Operating surplus: Rents on land	0.000000e+00	0.000000e+00
Operating surplus: Royalties on resources	0.000000e+00	0.000000e+00
Operating surplus: Remaining net operating surplus	2.189974e+05	3.521483e+05
Employment: Low-skilled male	8.235181e+04	9.944481e+03
Employment: Low-skilled female	5.209717e+04	3.848001e+03
Employment: Medium-skilled male	7.832752e+04	3.471130e+04
Employment: Medium-skilled female	2.855333e+04	1.309351e+04
Employment: High-skilled male	2.430436e+04	8.453866e+03
Employment: High-skilled female	1.350477e+04	3.478473e+03
Employment hours: Low-skilled male	1.900017e+05	2.395763e+04
Employment hours: Low-skilled female	1.187287e+05	8.761828e+03
Employment hours: Medium-skilled male	1.729137e+05	8.193434e+04
Employment hours: Medium-skilled female	6.206372e+04	2.902995e+04
Employment hours: High-skilled male	5.285178e+04	2.154652e+04
Employment hours: High-skilled female	2.885405e+04	8.809134e+03
Employment: Vulnerable employment	6.159106e+04	3.200682e+04
Employment hours: Vulnerable employment	1.332365e+05	7.248616e+04
CO2 - combustion - air	6.444419e+11	1.621372e+12
CH4 - combustion - air	2.310766e+09	5.794814e+08
N2O - combustion - air	5.232415e+07	1.978707e+07
SOx - combustion - air	1.504555e+09	2.736370e+09
NOx - combustion - air	2.180307e+09	3.865177e+09
NH3 - combustion - air	2.110186e+08	4.350334e+08

```
CO - combustion - air                                3.227115e+10  6.021470e+09
...                                                           ...           ...
Water Withdrawal Blue - Manufacturing - Radio, ...   5.771092e+01  8.889573e+01
Water Withdrawal Blue - Manufacturing - Medical...   5.340814e+01  4.863421e+01
Water Withdrawal Blue - Manufacturing - Motor v...   2.390375e+01  2.332213e+01
Water Withdrawal Blue - Manufacturing - Other t...   1.492215e+01  7.556941e+00
Water Withdrawal Blue - Manufacturing - Furnitu...   1.888092e+03  2.575762e+02
Water Withdrawal Blue - Electricity - tower - E...   9.530241e+01  1.408189e+02
Water Withdrawal Blue - Electricity - tower - E...   1.240819e+01  4.573095e+01
Water Withdrawal Blue - Electricity - tower - E...   8.590237e+00  1.958943e+01
Water Withdrawal Blue - Electricity - tower - E...   0.000000e+00  0.000000e+00
Water Withdrawal Blue - Electricity - tower - E...   0.000000e+00  0.000000e+00
Water Withdrawal Blue - Electricity - tower - E...   6.273516e-01  4.926269e+00
Water Withdrawal Blue - Electricity - tower - E...   7.837026e-01  1.382612e+00
Water Withdrawal Blue - Electricity - tower - E...   0.000000e+00  0.000000e+00
Water Withdrawal Blue - Electricity - tower - E...   2.042506e-02  2.188442e-02
Water Withdrawal Blue - Electricity - tower - E...   0.000000e+00  0.000000e+00
Water Withdrawal Blue - Electricity - tower - E...   1.461663e+00  1.049639e+00
Water Withdrawal Blue - Electricity - tower - E...   0.000000e+00  0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...   1.142738e+03  2.546115e+03
Water Withdrawal Blue - Electricity - once-thro...   1.284487e+03  3.545118e+04
Water Withdrawal Blue - Electricity - once-thro...   3.888284e+02  8.139868e+02
Water Withdrawal Blue - Electricity - once-thro...   0.000000e+00  0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...   0.000000e+00  0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...   4.376361e+02  1.066696e+04
Water Withdrawal Blue - Electricity - once-thro...   8.034716e+00  1.407639e+01
Water Withdrawal Blue - Electricity - once-thro...   0.000000e+00  0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...   0.000000e+00  0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...   0.000000e+00  0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...   2.486178e+00  4.133830e+00
Water Withdrawal Blue - Electricity - once-thro...   0.000000e+00  0.000000e+00
Water Withdrawal Blue - Domestic - domestic Wat...   1.844529e+04  7.123540e+03

[1104 rows x 49 columns]
```

The foundation of the calculation is now created and ready for further calculations in RO 1.0.
There are certain important commands that help navigating through the panda frames:

- prints the shape of the data frame, in forms of rows x columns (120x1104): `df.shape()`
- returns a list of all the indices of the dataframe (index = rows): `df.index()`
- returns a list of all the columns of the dataframe (columns = columns): `df.columns()`
- command for display all rows and columns, with no 'gap' in between:`with pd.option_context('display.max_rows', None, 'display.max_columns', None): display(df)`
- To call only one specific index with a label: `df.loc['label']`
- To call a specific position in the index, only taking integers: `df.iloc[0,:]`
- Call one specific row: `df.xs('Ris', level=3)`
- To show output in a seperate window: `%matplotlib qt`
- To show output inline: `%matplotlib inline`

4 RO 1.0 Estimate the consumption-based environmental impacts of Sweden with the most recent data available

4.1 RO 1.1 Aggregation

This report aggregates the different countries and region into three main categories. Due to the focus on the report the categories are: *'SE', 'EU' and 'RoW'*

> **SE(1):** Sweden **EU(26):** Austria, Italy, Belgium, Latvia, Bulgaria, Lithuania, Croatia, Luxembourg, Cyprus, Malta, Czechia, Netherlands, Denmark, Poland, Estonia, Portugal, Finland, Romania, France, Slovakia, Germany, Slovenia, Greece, Spain, Hungary, Ireland, United Kingdom **RoW(21):** Rest of Countries

In [9]:
```
# A list with all the regions from exio3 is created and then the ones chosen according
regions = list(exio3.get_regions())
EU = regions[0:24]+ regions[25:28]
SE = regions[24]
RoW = regions[28:]

regions
```

Out[9]:
```
['AT',
 'BE',
 'BG',
 'CY',
 'CZ',
 'DE',
 'DK',
 'EE',
 'ES',
 'FI',
 'FR',
 'GR',
 'HR',
 'HU',
 'IE',
 'IT',
 'LT',
 'LU',
 'LV',
 'MT',
 'NL',
 'PL',
 'PT',
 'RO',
 'SE',
 'SI',
 'SK',
 'GB',
```

```
'US',
'JP',
'CN',
'CA',
'KR',
'BR',
'IN',
'MX',
'RU',
'AU',
'CH',
'TR',
'TW',
'NO',
'ID',
'ZA',
'WA',
'WL',
'WE',
'WF',
'WM']
```

Region aggreg matrix: 3 regions, row_SE = SE(24), row_EU = EU excl. SE, row_NONEU = Row, therefore we create the countrymatrix cm

```
In [10]: # The variable cm is constructed by filling the three different np.arrays with a 1 in
         # Example: the row_SE is a np.array with all 0 and a length of the variable regions. 
         # therefore on place 24 a 1 instead of 0 is paced. The same with the other two rows. 
         row_SE = np.zeros((1,len(regions)))
         row_EU = np.zeros((1,len(regions)))
         row_RoW = np.zeros((1,len(regions)))
         cm = np.concatenate((row_SE,row_EU,row_RoW))
         cm[0,24] = 1
         cm[1,0:24] = 1
         cm[1,25:28] = 1
         cm[2,28:] = 1

         cm
```

```
Out[10]: array([[0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0.,
                 0., 0., 0., 0., 0., 0., 0., 0., 1., 0., 0., 0., 0., 0., 0., 0.,
                 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0.,
                 0.],
                [1., 1., 1., 1., 1., 1., 1., 1., 1., 1., 1., 1., 1., 1., 1., 1.,
                 1., 1., 1., 1., 1., 1., 1., 1., 0., 1., 1., 1., 0., 0., 0., 0.,
                 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0.,
                 0.],
                [0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0.,
```

```
                0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 1., 1., 1., 1.,
                1., 1., 1., 1., 1., 1., 1., 1., 1., 1., 1., 1., 1., 1., 1., 1.,
                1.]])
```

In a next step we need to create the sector aggregation matrix: no aggregation is needed, therefore identity matrix of dimension 200x200 according to all products.

```
In [11]: # Defining the production matrix
         pro = list(exio3.get_sectors())
         pm = np.identity(len(pro))

         pm
```

```
Out[11]: array([[1., 0., 0., ..., 0., 0., 0.],
                [0., 1., 0., ..., 0., 0., 0.],
                [0., 0., 1., ..., 0., 0., 0.],
                ...,
                [0., 0., 0., ..., 1., 0., 0.],
                [0., 0., 0., ..., 0., 1., 0.],
                [0., 0., 0., ..., 0., 0., 1.]])
```

Now we have the two different aggregation matrices (country matrix and product matrix) which can now be multiplied.

```
In [12]: # Elias made that, therefore we do to.
         # This is not making a copy! Just same pointer!
         SE_aggr = exio3.copy('SE_aggr')
         SE_aggr
```

```
Out[12]: <pymrio.core.mriosystem.IOSystem at 0x1eefaa487b8>
```

Important note from Elia: Aggregate method bug: run calc_all first, and convert to dataframe the x vector, before running aggregation https://github.com/konstantinstadler/pymrio/issues/21

```
In [13]: SE_aggr.x = pd.DataFrame(SE_aggr.x)

         #The actual multiplication / aggregation
         SE_aggr.aggregate(region_agg=cm, region_names=['SE','EU','RoW'], sector_agg=pm, secto
```

```
Out[13]: <pymrio.core.mriosystem.IOSystem at 0x1eefaa487b8>
```

```
In [14]: SE_aggr.get_sectors()
         SE_aggr.get_regions()
```

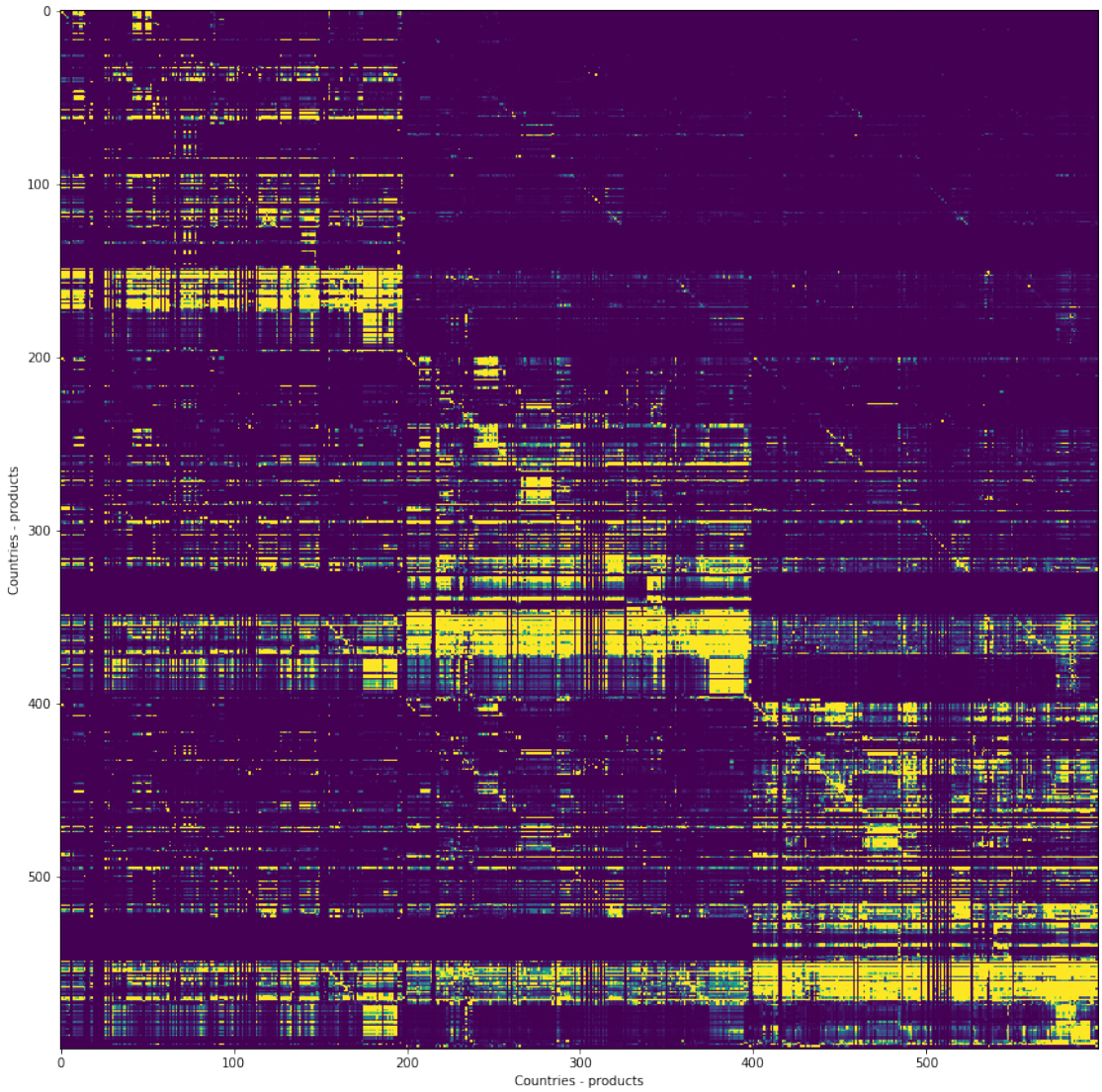
```
Out[14]: Index(['SE', 'EU', 'RoW'], dtype='object', name='region')
```

Recalculate all the matrices belonging to the system.

```
In [15]: SE_aggr.calc_all()
```

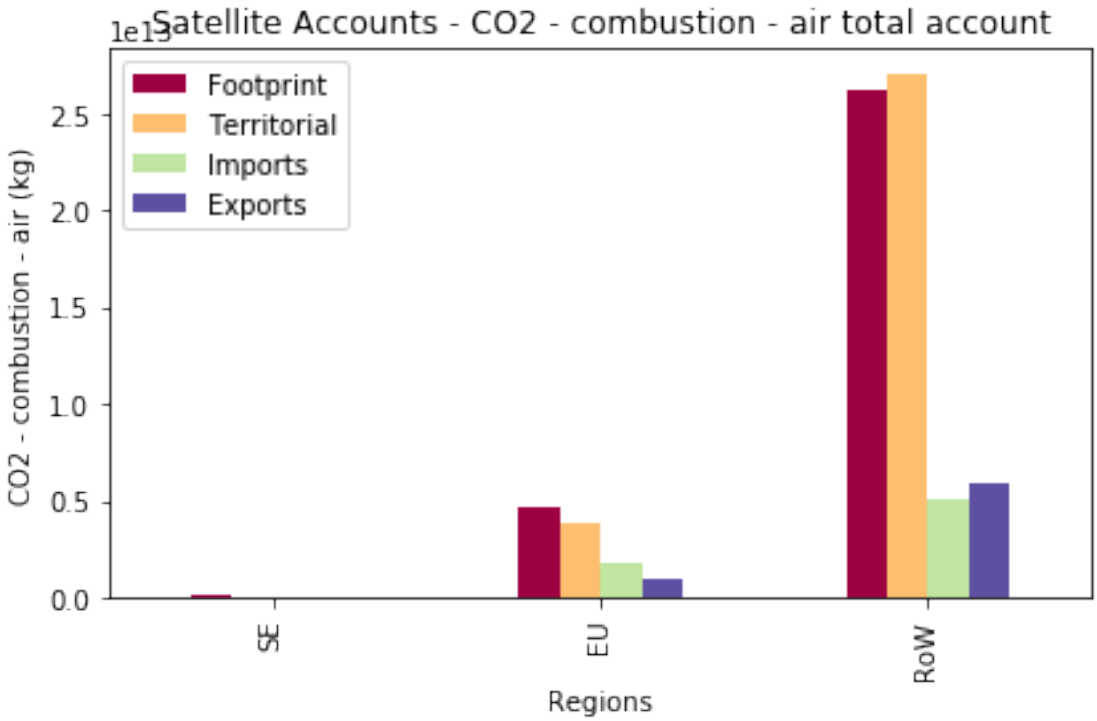
Out[15]: <pymrio.core.mriosystem.IOSystem at 0x1eefaa487b8>

In [16]:
```
# This plot gives a overview of the interaction between the different products of dif
# It shows that the calculation are conducted correctly.
%matplotlib inline
plt.figure(figsize=(15,15))
plt.imshow(SE_aggr.A, vmax=1E-3)
plt.xlabel('Countries - products')
plt.ylabel('Countries - products')
plt.show()
```



In [17]:
```
# This plot shows one of the 1104 stressors ('CO2 - combustion - air') and how they a
# Footprint is the consumption based account. Means: Footprint = Territorial + Import
```

```
SE_aggr.satellite.plot_account('CO2 - combustion - air', per_capita=False, file_name=
plt.show()
```




```
In [18]: # This are different commands to show the dataframe, comprised in the aggregated data
         # The meaning of the different varibles can be seen in the report or the Pymrio manua
         SE_aggr.Y.SE
         SE_aggr.satellite.M
         SE_aggr.satellite.D_imp.SE
```

```
Out[18]: sector                                                  Paddy rice  \
         Taxes less subsidies on products purchased: Total   1.995815e-03
         Other net taxes on production                       7.564636e-03
         Compensation of employees; wages, salaries, & e...  1.933978e-03
         Compensation of employees; wages, salaries, & e...  1.839900e-02
         Compensation of employees; wages, salaries, & e...  1.159709e-02
         Operating surplus: Consumption of fixed capital     2.745377e-02
         Operating surplus: Rents on land                    0.000000e+00
         Operating surplus: Royalties on resources           0.000000e+00
         Operating surplus: Remaining net operating surplus  7.345348e-02
         Employment: Low-skilled male                        6.490581e-04
         Employment: Low-skilled female                      3.481853e-04
         Employment: Medium-skilled male                     1.427404e-03
         Employment: Medium-skilled female                   5.228215e-04
         Employment: High-skilled male                       3.810989e-04
```

```
Employment: High-skilled female                      1.665364e-04
Employment hours: Low-skilled male                   1.408526e-03
Employment hours: Low-skilled female                 7.549724e-04
Employment hours: Medium-skilled male                3.083570e-03
Employment hours: Medium-skilled female              1.124722e-03
Employment hours: High-skilled male                  8.177132e-04
Employment hours: High-skilled female                3.552615e-04
Employment: Vulnerable employment                    1.734940e-03
Employment hours: Vulnerable employment              4.065431e-03
CO2 - combustion - air                               8.834109e+04
CH4 - combustion - air                               7.425526e+00
N2O - combustion - air                               2.241121e+00
SOx - combustion - air                               2.669222e+01
NOx - combustion - air                               5.182672e+02
NH3 - combustion - air                               3.604994e+00
CO - combustion - air                                1.039923e+03
...                                                           ...
Water Withdrawal Blue - Manufacturing - Radio, ...   1.618188e-07
Water Withdrawal Blue - Manufacturing - Medical...   6.073193e-08
Water Withdrawal Blue - Manufacturing - Motor v...   1.159998e-08
Water Withdrawal Blue - Manufacturing - Other t...   1.067016e-08
Water Withdrawal Blue - Manufacturing - Furnitu...   4.674555e-07
Water Withdrawal Blue - Electricity - tower - E...   3.838980e-07
Water Withdrawal Blue - Electricity - tower - E...   1.781204e-07
Water Withdrawal Blue - Electricity - tower - E...   9.219179e-08
Water Withdrawal Blue - Electricity - tower - E...   0.000000e+00
Water Withdrawal Blue - Electricity - tower - E...   0.000000e+00
Water Withdrawal Blue - Electricity - tower - E...   6.691294e-09
Water Withdrawal Blue - Electricity - tower - E...   4.340356e-08
Water Withdrawal Blue - Electricity - tower - E...   0.000000e+00
Water Withdrawal Blue - Electricity - tower - E...   1.037880e-10
Water Withdrawal Blue - Electricity - tower - E...   0.000000e+00
Water Withdrawal Blue - Electricity - tower - E...   5.366217e-08
Water Withdrawal Blue - Electricity - tower - E...   0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...   1.115904e-05
Water Withdrawal Blue - Electricity - once-thro...   2.544224e-05
Water Withdrawal Blue - Electricity - once-thro...   4.467644e-06
Water Withdrawal Blue - Electricity - once-thro...   0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...   0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...   7.718592e-06
Water Withdrawal Blue - Electricity - once-thro...   6.526192e-08
Water Withdrawal Blue - Electricity - once-thro...   0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...   0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...   0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...   3.441407e-08
Water Withdrawal Blue - Electricity - once-thro...   0.000000e+00
Water Withdrawal Blue - Domestic - domestic Wat...   0.000000e+00
```

```
sector                                                        Wheat  \
Taxes less subsidies on products purchased: Total   6.195243e+00
Other net taxes on production                       1.147986e-01
Compensation of employees; wages, salaries, & e...  2.678154e+00
Compensation of employees; wages, salaries, & e...  1.916290e+01
Compensation of employees; wages, salaries, & e...  1.323501e+01
Operating surplus: Consumption of fixed capital     1.544153e+01
Operating surplus: Rents on land                    0.000000e+00
Operating surplus: Royalties on resources           0.000000e+00
Operating surplus: Remaining net operating surplus  2.749943e+01
Employment: Low-skilled male                        7.816798e-01
Employment: Low-skilled female                      4.276864e-01
Employment: Medium-skilled male                     3.925184e+00
Employment: Medium-skilled female                   2.317917e+00
Employment: High-skilled male                       3.553866e-01
Employment: High-skilled female                     1.963737e-01
Employment hours: Low-skilled male                  1.474875e+00
Employment hours: Low-skilled female                7.928498e-01
Employment hours: Medium-skilled male               6.913929e+00
Employment hours: Medium-skilled female             4.004163e+00
Employment hours: High-skilled male                 7.147138e-01
Employment hours: High-skilled female               3.869395e-01
Employment: Vulnerable employment                   5.049251e+00
Employment hours: Vulnerable employment             8.603704e+00
CO2 - combustion - air                              5.811302e+07
CH4 - combustion - air                              1.992948e+04
N2O - combustion - air                              9.594828e+02
SOx - combustion - air                              1.183445e+05
NOx - combustion - air                              2.105038e+05
NH3 - combustion - air                              6.739445e+03
CO - combustion - air                               1.780141e+05
...                                                          ...
Water Withdrawal Blue - Manufacturing - Radio, ...  2.015575e-03
Water Withdrawal Blue - Manufacturing - Medical...  6.579551e-04
Water Withdrawal Blue - Manufacturing - Motor v...  8.676144e-05
Water Withdrawal Blue - Manufacturing - Other t...  8.279024e-05
Water Withdrawal Blue - Manufacturing - Furnitu...  1.269563e-03
Water Withdrawal Blue - Electricity - tower - E...  7.093707e-03
Water Withdrawal Blue - Electricity - tower - E...  3.363574e-03
Water Withdrawal Blue - Electricity - tower - E...  2.248335e-03
Water Withdrawal Blue - Electricity - tower - E...  0.000000e+00
Water Withdrawal Blue - Electricity - tower - E...  0.000000e+00
Water Withdrawal Blue - Electricity - tower - E...  3.040912e-05
Water Withdrawal Blue - Electricity - tower - E...  9.439349e-05
Water Withdrawal Blue - Electricity - tower - E...  0.000000e+00
Water Withdrawal Blue - Electricity - tower - E...  1.193462e-06
Water Withdrawal Blue - Electricity - tower - E...  0.000000e+00
Water Withdrawal Blue - Electricity - tower - E...  1.248675e-04
```

```
Water Withdrawal Blue - Electricity - tower - E...  0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...  1.479273e-01
Water Withdrawal Blue - Electricity - once-thro...  1.646703e-01
Water Withdrawal Blue - Electricity - once-thro...  8.798736e-02
Water Withdrawal Blue - Electricity - once-thro...  0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...  0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...  2.752417e-02
Water Withdrawal Blue - Electricity - once-thro...  5.467505e-02
Water Withdrawal Blue - Electricity - once-thro...  0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...  0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...  0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...  3.476000e-04
Water Withdrawal Blue - Electricity - once-thro...  0.000000e+00
Water Withdrawal Blue - Domestic - domestic Wat...  0.000000e+00

sector                                              Cereal grains nec  \
Taxes less subsidies on products purchased: Total        7.060391e-01
Other net taxes on production                            1.456833e-02
Compensation of employees; wages, salaries, & e...       3.174429e-01
Compensation of employees; wages, salaries, & e...       2.070926e+00
Compensation of employees; wages, salaries, & e...       1.716176e+00
Operating surplus: Consumption of fixed capital          1.962293e+00
Operating surplus: Rents on land                         0.000000e+00
Operating surplus: Royalties on resources                0.000000e+00
Operating surplus: Remaining net operating surplus       3.331118e+00
Employment: Low-skilled male                             4.506090e-02
Employment: Low-skilled female                           2.490076e-02
Employment: Medium-skilled male                          1.596334e-01
Employment: Medium-skilled female                        7.944501e-02
Employment: High-skilled male                            4.336145e-02
Employment: High-skilled female                          2.436005e-02
Employment hours: Low-skilled male                       9.990229e-02
Employment hours: Low-skilled female                     5.352854e-02
Employment hours: Medium-skilled male                    3.364192e-01
Employment hours: Medium-skilled female                  1.634231e-01
Employment hours: High-skilled male                      8.872090e-02
Employment hours: High-skilled female                    4.834673e-02
Employment: Vulnerable employment                        1.311809e-01
Employment hours: Vulnerable employment                  2.781731e-01
CO2 - combustion - air                                   7.346027e+06
CH4 - combustion - air                                   2.649818e+03
N2O - combustion - air                                   1.225698e+02
SOx - combustion - air                                   1.505149e+04
NOx - combustion - air                                   2.468600e+04
NH3 - combustion - air                                   8.045226e+02
CO - combustion - air                                    2.223387e+04
...                                                               ...
Water Withdrawal Blue - Manufacturing - Radio, ...       3.036648e-04
```

Water Withdrawal Blue - Manufacturing - Medical...	7.771980e-05
Water Withdrawal Blue - Manufacturing - Motor v...	1.054119e-05
Water Withdrawal Blue - Manufacturing - Other t...	9.695030e-06
Water Withdrawal Blue - Manufacturing - Furnitu...	1.469020e-04
Water Withdrawal Blue - Electricity - tower - E...	9.030442e-04
Water Withdrawal Blue - Electricity - tower - E...	4.528516e-04
Water Withdrawal Blue - Electricity - tower - E...	2.831037e-04
Water Withdrawal Blue - Electricity - tower - E...	0.000000e+00
Water Withdrawal Blue - Electricity - tower - E...	0.000000e+00
Water Withdrawal Blue - Electricity - tower - E...	3.675639e-06
Water Withdrawal Blue - Electricity - tower - E...	1.116069e-05
Water Withdrawal Blue - Electricity - tower - E...	0.000000e+00
Water Withdrawal Blue - Electricity - tower - E...	1.420385e-07
Water Withdrawal Blue - Electricity - tower - E...	0.000000e+00
Water Withdrawal Blue - Electricity - tower - E...	1.538957e-05
Water Withdrawal Blue - Electricity - tower - E...	0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...	1.991341e-02
Water Withdrawal Blue - Electricity - once-thro...	2.169583e-02
Water Withdrawal Blue - Electricity - once-thro...	1.113913e-02
Water Withdrawal Blue - Electricity - once-thro...	0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...	0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...	3.311239e-03
Water Withdrawal Blue - Electricity - once-thro...	7.665813e-03
Water Withdrawal Blue - Electricity - once-thro...	0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...	0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...	0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...	4.280259e-05
Water Withdrawal Blue - Electricity - once-thro...	0.000000e+00
Water Withdrawal Blue - Domestic - domestic Wat...	0.000000e+00

sector	Vegetables, fruit, nuts \
Taxes less subsidies on products purchased: Total	7.386633e+01
Other net taxes on production	5.854344e+00
Compensation of employees; wages, salaries, & e...	5.158708e+01
Compensation of employees; wages, salaries, & e...	2.612128e+02
Compensation of employees; wages, salaries, & e...	1.221060e+02
Operating surplus: Consumption of fixed capital	2.399582e+02
Operating surplus: Rents on land	0.000000e+00
Operating surplus: Royalties on resources	0.000000e+00
Operating surplus: Remaining net operating surplus	4.108412e+02
Employment: Low-skilled male	3.719312e+01
Employment: Low-skilled female	2.121934e+01
Employment: Medium-skilled male	7.604862e+01
Employment: Medium-skilled female	3.203757e+01
Employment: High-skilled male	5.772712e+00
Employment: High-skilled female	2.166199e+00
Employment hours: Low-skilled male	8.360722e+01
Employment hours: Low-skilled female	4.749438e+01

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Employment hours: Medium-skilled male                          1.481360e+02
Employment hours: Medium-skilled female                        5.942873e+01
Employment hours: High-skilled male                            1.249986e+01
Employment hours: High-skilled female                          4.575633e+00
Employment: Vulnerable employment                              8.416285e+01
Employment hours: Vulnerable employment                        1.553392e+02
CO2 - combustion - air                                         4.545078e+08
CH4 - combustion - air                                         1.254390e+05
N2O - combustion - air                                         8.269531e+03
SOx - combustion - air                                         6.706034e+05
NOx - combustion - air                                         1.868070e+06
NH3 - combustion - air                                         1.169731e+05
CO - combustion - air                                          2.980671e+06
...                                                                     ...
Water Withdrawal Blue - Manufacturing - Radio, ...             9.670470e-03
Water Withdrawal Blue - Manufacturing - Medical...             5.699775e-03
Water Withdrawal Blue - Manufacturing - Motor v...             1.240125e-03
Water Withdrawal Blue - Manufacturing - Other t...             4.472604e-04
Water Withdrawal Blue - Manufacturing - Furnitu...             3.386147e-02
Water Withdrawal Blue - Electricity - tower - E...             1.191600e-01
Water Withdrawal Blue - Electricity - tower - E...             3.106863e-02
Water Withdrawal Blue - Electricity - tower - E...             2.913722e-02
Water Withdrawal Blue - Electricity - tower - E...             0.000000e+00
Water Withdrawal Blue - Electricity - tower - E...             0.000000e+00
Water Withdrawal Blue - Electricity - tower - E...             9.389527e-04
Water Withdrawal Blue - Electricity - tower - E...             1.658729e-03
Water Withdrawal Blue - Electricity - tower - E...             0.000000e+00
Water Withdrawal Blue - Electricity - tower - E...             9.188127e-05
Water Withdrawal Blue - Electricity - tower - E...             0.000000e+00
Water Withdrawal Blue - Electricity - tower - E...             1.293873e-03
Water Withdrawal Blue - Electricity - tower - E...             0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...             4.799597e+00
Water Withdrawal Blue - Electricity - once-thro...             9.183548e+00
Water Withdrawal Blue - Electricity - once-thro...             1.434772e+00
Water Withdrawal Blue - Electricity - once-thro...             0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...             0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...             1.447528e+00
Water Withdrawal Blue - Electricity - once-thro...             5.274569e-02
Water Withdrawal Blue - Electricity - once-thro...             0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...             0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...             0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...             6.659013e-03
Water Withdrawal Blue - Electricity - once-thro...             0.000000e+00
Water Withdrawal Blue - Domestic - domestic Wat...             0.000000e+00

sector                                                      Oil seeds  \
Taxes less subsidies on products purchased: Total   7.774982e+00
Other net taxes on production                      -5.538774e+00
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Compensation of employees; wages, salaries, & e...  4.576988e+00
Compensation of employees; wages, salaries, & e...  2.966931e+01
Compensation of employees; wages, salaries, & e...  1.381682e+01
Operating surplus: Consumption of fixed capital     2.338704e+01
Operating surplus: Rents on land                    0.000000e+00
Operating surplus: Royalties on resources           0.000000e+00
Operating surplus: Remaining net operating surplus  2.747949e+01
Employment: Low-skilled male                        8.847696e-01
Employment: Low-skilled female                      5.042516e-01
Employment: Medium-skilled male                     4.905386e+00
Employment: Medium-skilled female                   2.007103e+00
Employment: High-skilled male                       3.888898e-01
Employment: High-skilled female                     1.846507e-01
Employment hours: Low-skilled male                  1.865991e+00
Employment hours: Low-skilled female                1.027143e+00
Employment hours: Medium-skilled male               9.709213e+00
Employment hours: Medium-skilled female             3.835762e+00
Employment hours: High-skilled male                 7.977439e-01
Employment hours: High-skilled female               3.621691e-01
Employment: Vulnerable employment                   5.462279e+00
Employment hours: Vulnerable employment             1.084818e+01
CO2 - combustion - air                              4.724899e+07
CH4 - combustion - air                              9.737961e+03
N2O - combustion - air                              9.486374e+02
SOx - combustion - air                              7.611466e+04
NOx - combustion - air                              2.843720e+05
NH3 - combustion - air                              4.374118e+03
CO - combustion - air                               3.884165e+05
...                                                          ...
Water Withdrawal Blue - Manufacturing - Radio, ...  8.139225e-04
Water Withdrawal Blue - Manufacturing - Medical...  3.204507e-04
Water Withdrawal Blue - Manufacturing - Motor v...  7.089712e-05
Water Withdrawal Blue - Manufacturing - Other t...  4.458226e-05
Water Withdrawal Blue - Manufacturing - Furnitu...  7.500858e-04
Water Withdrawal Blue - Electricity - tower - E...  5.424767e-03
Water Withdrawal Blue - Electricity - tower - E...  1.401728e-03
Water Withdrawal Blue - Electricity - tower - E...  1.825393e-03
Water Withdrawal Blue - Electricity - tower - E...  0.000000e+00
Water Withdrawal Blue - Electricity - tower - E...  0.000000e+00
Water Withdrawal Blue - Electricity - tower - E...  4.099224e-05
Water Withdrawal Blue - Electricity - tower - E...  8.362895e-05
Water Withdrawal Blue - Electricity - tower - E...  0.000000e+00
Water Withdrawal Blue - Electricity - tower - E...  1.209899e-06
Water Withdrawal Blue - Electricity - tower - E...  0.000000e+00
Water Withdrawal Blue - Electricity - tower - E...  8.943617e-05
Water Withdrawal Blue - Electricity - tower - E...  0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...  1.063291e-01
Water Withdrawal Blue - Electricity - once-thro...  2.230099e-01
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Water Withdrawal Blue - Electricity - once-thro...  8.779901e-02
Water Withdrawal Blue - Electricity - once-thro...  0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...  0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...  5.708136e-02
Water Withdrawal Blue - Electricity - once-thro...  8.538240e-03
Water Withdrawal Blue - Electricity - once-thro...  0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...  0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...  0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...  2.219714e-04
Water Withdrawal Blue - Electricity - once-thro...  0.000000e+00
Water Withdrawal Blue - Domestic - domestic Wat...  0.000000e+00

sector                                              Sugar cane, sugar beet  \
Taxes less subsidies on products purchased: Total             1.569654e-03
Other net taxes on production                                 5.415737e-04
Compensation of employees; wages, salaries, & e...            9.871885e-04
Compensation of employees; wages, salaries, & e...            6.275376e-03
Compensation of employees; wages, salaries, & e...            5.158509e-03
Operating surplus: Consumption of fixed capital               4.247795e-03
Operating surplus: Rents on land                              0.000000e+00
Operating surplus: Royalties on resources                     0.000000e+00
Operating surplus: Remaining net operating surplus            7.700056e-03
Employment: Low-skilled male                                  1.941835e-04
Employment: Low-skilled female                                1.164131e-04
Employment: Medium-skilled male                               5.572787e-04
Employment: Medium-skilled female                             3.273454e-04
Employment: High-skilled male                                 1.507554e-04
Employment: High-skilled female                               9.126861e-05
Employment hours: Low-skilled male                            4.336865e-04
Employment hours: Low-skilled female                          2.532237e-04
Employment hours: Medium-skilled male                         1.172172e-03
Employment hours: Medium-skilled female                       6.759670e-04
Employment hours: High-skilled male                           3.097580e-04
Employment hours: High-skilled female                         1.824641e-04
Employment: Vulnerable employment                             5.006029e-04
Employment hours: Vulnerable employment                       1.069269e-03
CO2 - combustion - air                                        1.567972e+04
CH4 - combustion - air                                        5.780716e+00
N2O - combustion - air                                        2.441945e-01
SOx - combustion - air                                        3.361586e+01
NOx - combustion - air                                        4.216088e+01
NH3 - combustion - air                                        2.747108e+00
CO - combustion - air                                         3.417990e+01
...                                                                     ...
Water Withdrawal Blue - Manufacturing - Radio, ...            6.855177e-07
Water Withdrawal Blue - Manufacturing - Medical...            2.262137e-07
Water Withdrawal Blue - Manufacturing - Motor v...            2.693095e-08
Water Withdrawal Blue - Manufacturing - Other t...            3.118278e-08
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Water Withdrawal Blue - Manufacturing - Furnitu...	5.069537e-07
Water Withdrawal Blue - Electricity - tower - E...	2.409449e-06
Water Withdrawal Blue - Electricity - tower - E...	7.670642e-07
Water Withdrawal Blue - Electricity - tower - E...	7.541019e-07
Water Withdrawal Blue - Electricity - tower - E...	0.000000e+00
Water Withdrawal Blue - Electricity - tower - E...	0.000000e+00
Water Withdrawal Blue - Electricity - tower - E...	1.381300e-08
Water Withdrawal Blue - Electricity - tower - E...	3.375394e-08
Water Withdrawal Blue - Electricity - tower - E...	0.000000e+00
Water Withdrawal Blue - Electricity - tower - E...	4.747822e-10
Water Withdrawal Blue - Electricity - tower - E...	0.000000e+00
Water Withdrawal Blue - Electricity - tower - E...	6.088619e-08
Water Withdrawal Blue - Electricity - tower - E...	0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...	4.713419e-05
Water Withdrawal Blue - Electricity - once-thro...	5.121912e-05
Water Withdrawal Blue - Electricity - once-thro...	2.871055e-05
Water Withdrawal Blue - Electricity - once-thro...	0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...	0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...	1.184282e-05
Water Withdrawal Blue - Electricity - once-thro...	8.510013e-06
Water Withdrawal Blue - Electricity - once-thro...	0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...	0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...	0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...	1.969818e-07
Water Withdrawal Blue - Electricity - once-thro...	0.000000e+00
Water Withdrawal Blue - Domestic - domestic Wat...	0.000000e+00

sector	Plant-based fibers \
Taxes less subsidies on products purchased: Total	2.902851e-05
Other net taxes on production	5.191635e-06
Compensation of employees; wages, salaries, & e...	1.976357e-05
Compensation of employees; wages, salaries, & e...	1.134300e-04
Compensation of employees; wages, salaries, & e...	2.527940e-05
Operating surplus: Consumption of fixed capital	7.134037e-05
Operating surplus: Rents on land	0.000000e+00
Operating surplus: Royalties on resources	0.000000e+00
Operating surplus: Remaining net operating surplus	3.281405e-04
Employment: Low-skilled male	1.239370e-05
Employment: Low-skilled female	8.264344e-06
Employment: Medium-skilled male	4.932782e-05
Employment: Medium-skilled female	3.039128e-05
Employment: High-skilled male	1.295416e-06
Employment: High-skilled female	4.702445e-07
Employment hours: Low-skilled male	2.580782e-05
Employment hours: Low-skilled female	1.707927e-05
Employment hours: Medium-skilled male	9.420567e-05
Employment hours: Medium-skilled female	5.747639e-05
Employment hours: High-skilled male	2.737558e-06

Employment hours: High-skilled female	9.369482e-07
Employment: Vulnerable employment	8.045810e-05
Employment hours: Vulnerable employment	1.528367e-04
CO2 - combustion - air	3.516314e+02
CH4 - combustion - air	6.864867e-02
N2O - combustion - air	7.288729e-03
SOx - combustion - air	5.954905e-01
NOx - combustion - air	2.940720e+00
NH3 - combustion - air	2.545962e-02
CO - combustion - air	1.367837e+01
...	...
Water Withdrawal Blue - Manufacturing - Radio, ...	1.741005e-09
Water Withdrawal Blue - Manufacturing - Medical...	1.802926e-09
Water Withdrawal Blue - Manufacturing - Motor v...	3.075973e-10
Water Withdrawal Blue - Manufacturing - Other t...	2.985610e-10
Water Withdrawal Blue - Manufacturing - Furnitu...	2.626817e-09
Water Withdrawal Blue - Electricity - tower - E...	4.130667e-08
Water Withdrawal Blue - Electricity - tower - E...	2.448568e-09
Water Withdrawal Blue - Electricity - tower - E...	3.535351e-09
Water Withdrawal Blue - Electricity - tower - E...	0.000000e+00
Water Withdrawal Blue - Electricity - tower - E...	0.000000e+00
Water Withdrawal Blue - Electricity - tower - E...	1.353694e-10
Water Withdrawal Blue - Electricity - tower - E...	1.000448e-09
Water Withdrawal Blue - Electricity - tower - E...	0.000000e+00
Water Withdrawal Blue - Electricity - tower - E...	9.165920e-11
Water Withdrawal Blue - Electricity - tower - E...	0.000000e+00
Water Withdrawal Blue - Electricity - tower - E...	5.254287e-10
Water Withdrawal Blue - Electricity - tower - E...	0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...	8.019848e-07
Water Withdrawal Blue - Electricity - once-thro...	1.561007e-07
Water Withdrawal Blue - Electricity - once-thro...	2.815524e-07
Water Withdrawal Blue - Electricity - once-thro...	0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...	0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...	9.111375e-08
Water Withdrawal Blue - Electricity - once-thro...	7.265064e-09
Water Withdrawal Blue - Electricity - once-thro...	0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...	0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...	0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...	2.141898e-10
Water Withdrawal Blue - Electricity - once-thro...	0.000000e+00
Water Withdrawal Blue - Domestic - domestic Wat...	0.000000e+00

sector	Crops nec \
Taxes less subsidies on products purchased: Total	1.568651e+01
Other net taxes on production	4.943935e+00
Compensation of employees; wages, salaries, & e...	1.099460e+01
Compensation of employees; wages, salaries, & e...	5.945653e+01
Compensation of employees; wages, salaries, & e...	1.456970e+01

Operating surplus: Consumption of fixed capital	2.629629e+01
Operating surplus: Rents on land	0.000000e+00
Operating surplus: Royalties on resources	0.000000e+00
Operating surplus: Remaining net operating surplus	6.021834e+01
Employment: Low-skilled male	1.365118e+01
Employment: Low-skilled female	7.629567e+00
Employment: Medium-skilled male	2.626594e+01
Employment: Medium-skilled female	9.812880e+00
Employment: High-skilled male	1.584214e+00
Employment: High-skilled female	5.522403e-01
Employment hours: Low-skilled male	3.139161e+01
Employment hours: Low-skilled female	1.754071e+01
Employment hours: Medium-skilled male	5.419823e+01
Employment hours: Medium-skilled female	1.968874e+01
Employment hours: High-skilled male	3.624221e+00
Employment hours: High-skilled female	1.258049e+00
Employment: Vulnerable employment	2.622562e+01
Employment hours: Vulnerable employment	5.258993e+01
CO2 - combustion - air	8.818821e+07
CH4 - combustion - air	1.720421e+04
N2O - combustion - air	2.016045e+03
SOx - combustion - air	1.611325e+05
NOx - combustion - air	6.524806e+05
NH3 - combustion - air	9.273636e+04
CO - combustion - air	1.282386e+06
...	...
Water Withdrawal Blue - Manufacturing - Radio, ...	1.335357e-03
Water Withdrawal Blue - Manufacturing - Medical...	7.389307e-04
Water Withdrawal Blue - Manufacturing - Motor v...	3.419766e-04
Water Withdrawal Blue - Manufacturing - Other t...	1.412840e-04
Water Withdrawal Blue - Manufacturing - Furnitu...	2.167677e-03
Water Withdrawal Blue - Electricity - tower - E...	1.370171e-02
Water Withdrawal Blue - Electricity - tower - E...	3.135623e-03
Water Withdrawal Blue - Electricity - tower - E...	2.204912e-03
Water Withdrawal Blue - Electricity - tower - E...	0.000000e+00
Water Withdrawal Blue - Electricity - tower - E...	0.000000e+00
Water Withdrawal Blue - Electricity - tower - E...	7.105090e-04
Water Withdrawal Blue - Electricity - tower - E...	1.082091e-04
Water Withdrawal Blue - Electricity - tower - E...	0.000000e+00
Water Withdrawal Blue - Electricity - tower - E...	1.156875e-06
Water Withdrawal Blue - Electricity - tower - E...	0.000000e+00
Water Withdrawal Blue - Electricity - tower - E...	3.299667e-04
Water Withdrawal Blue - Electricity - tower - E...	0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...	3.959624e-01
Water Withdrawal Blue - Electricity - once-thro...	2.281089e-01
Water Withdrawal Blue - Electricity - once-thro...	1.907830e-01
Water Withdrawal Blue - Electricity - once-thro...	0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...	0.000000e+00

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Water Withdrawal Blue - Electricity - once-thro...  5.687061e-02
Water Withdrawal Blue - Electricity - once-thro...  7.713630e-03
Water Withdrawal Blue - Electricity - once-thro...  0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...  0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...  0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...  2.263897e-04
Water Withdrawal Blue - Electricity - once-thro...  0.000000e+00
Water Withdrawal Blue - Domestic - domestic Wat...  0.000000e+00

sector                                                    Cattle  \
Taxes less subsidies on products purchased: Total   1.945545e-01
Other net taxes on production                       5.799399e-02
Compensation of employees; wages, salaries, & e...  1.083481e-01
Compensation of employees; wages, salaries, & e...  6.924271e-01
Compensation of employees; wages, salaries, & e...  4.600488e-01
Operating surplus: Consumption of fixed capital     4.389482e-01
Operating surplus: Rents on land                    0.000000e+00
Operating surplus: Royalties on resources           0.000000e+00
Operating surplus: Remaining net operating surplus  7.832816e-01
Employment: Low-skilled male                        5.244610e-02
Employment: Low-skilled female                      2.914694e-02
Employment: Medium-skilled male                     1.196594e-01
Employment: Medium-skilled female                   5.435387e-02
Employment: High-skilled male                       1.574430e-02
Employment: High-skilled female                     7.511429e-03
Employment hours: Low-skilled male                  1.193219e-01
Employment hours: Low-skilled female                6.583195e-02
Employment hours: Medium-skilled male               2.477515e-01
Employment hours: Medium-skilled female             1.099636e-01
Employment hours: High-skilled male                 3.353095e-02
Employment hours: High-skilled female               1.541716e-02
Employment: Vulnerable employment                   1.197577e-01
Employment hours: Vulnerable employment             2.439618e-01
CO2 - combustion - air                              1.428263e+06
CH4 - combustion - air                              4.326255e+02
N2O - combustion - air                              2.719374e+01
SOx - combustion - air                              3.136703e+03
NOx - combustion - air                              5.907605e+03
NH3 - combustion - air                              5.690145e+02
CO - combustion - air                               6.918646e+03
...                                                          ...
Water Withdrawal Blue - Manufacturing - Radio, ...  1.094893e-04
Water Withdrawal Blue - Manufacturing - Medical...  2.481800e-05
Water Withdrawal Blue - Manufacturing - Motor v...  3.853361e-06
Water Withdrawal Blue - Manufacturing - Other t...  4.171943e-06
Water Withdrawal Blue - Manufacturing - Furnitu...  5.829440e-05
Water Withdrawal Blue - Electricity - tower - E...  2.461743e-04
Water Withdrawal Blue - Electricity - tower - E...  8.588476e-05
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Water Withdrawal Blue - Electricity - tower - E...  7.340447e-05
Water Withdrawal Blue - Electricity - tower - E...  0.000000e+00
Water Withdrawal Blue - Electricity - tower - E...  0.000000e+00
Water Withdrawal Blue - Electricity - tower - E...  2.915001e-06
Water Withdrawal Blue - Electricity - tower - E...  3.592250e-06
Water Withdrawal Blue - Electricity - tower - E...  0.000000e+00
Water Withdrawal Blue - Electricity - tower - E...  6.702962e-08
Water Withdrawal Blue - Electricity - tower - E...  0.000000e+00
Water Withdrawal Blue - Electricity - tower - E...  3.896350e-06
Water Withdrawal Blue - Electricity - tower - E...  0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...  5.528853e-03
Water Withdrawal Blue - Electricity - once-thro...  6.010562e-03
Water Withdrawal Blue - Electricity - once-thro...  3.084065e-03
Water Withdrawal Blue - Electricity - once-thro...  0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...  0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...  1.372984e-03
Water Withdrawal Blue - Electricity - once-thro...  8.124988e-04
Water Withdrawal Blue - Electricity - once-thro...  0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...  0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...  0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...  1.905788e-05
Water Withdrawal Blue - Electricity - once-thro...  0.000000e+00
Water Withdrawal Blue - Domestic - domestic Wat...  0.000000e+00

sector                                                      Pigs  ...  \
Taxes less subsidies on products purchased: Total   6.861986e-01  ...
Other net taxes on production                       2.228731e-01  ...
Compensation of employees; wages, salaries, & e...  3.453445e-01  ...
Compensation of employees; wages, salaries, & e...  2.361186e+00  ...
Compensation of employees; wages, salaries, & e...  1.756361e+00  ...
Operating surplus: Consumption of fixed capital     1.545611e+00  ...
Operating surplus: Rents on land                    0.000000e+00  ...
Operating surplus: Royalties on resources           0.000000e+00  ...
Operating surplus: Remaining net operating surplus  2.644914e+00  ...
Employment: Low-skilled male                        9.629277e-02  ...
Employment: Low-skilled female                      5.296800e-02  ...
Employment: Medium-skilled male                     2.572525e-01  ...
Employment: Medium-skilled female                   1.325581e-01  ...
Employment: High-skilled male                       4.951766e-02  ...
Employment: High-skilled female                     2.479638e-02  ...
Employment hours: Low-skilled male                  2.171010e-01  ...
Employment hours: Low-skilled female                1.175321e-01  ...
Employment hours: Medium-skilled male               5.383141e-01  ...
Employment hours: Medium-skilled female             2.715911e-01  ...
Employment hours: High-skilled male                 1.032716e-01  ...
Employment hours: High-skilled female               4.973251e-02  ...
Employment: Vulnerable employment                   2.530041e-01  ...
Employment hours: Vulnerable employment             5.278613e-01  ...
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CO2 - combustion - air	4.848272e+06	...
CH4 - combustion - air	1.492953e+03	...
N2O - combustion - air	9.091435e+01	...
SOx - combustion - air	1.199928e+04	...
NOx - combustion - air	1.957674e+04	...
NH3 - combustion - air	1.967191e+03	...
CO - combustion - air	1.650286e+04	...
...	...	...
Water Withdrawal Blue - Manufacturing - Radio, ...	4.271691e-04	...
Water Withdrawal Blue - Manufacturing - Medical...	9.214064e-05	...
Water Withdrawal Blue - Manufacturing - Motor v...	1.285250e-05	...
Water Withdrawal Blue - Manufacturing - Other t...	1.736377e-05	...
Water Withdrawal Blue - Manufacturing - Furnitu...	2.240737e-04	...
Water Withdrawal Blue - Electricity - tower - E...	9.919292e-04	...
Water Withdrawal Blue - Electricity - tower - E...	3.222336e-04	...
Water Withdrawal Blue - Electricity - tower - E...	2.904473e-04	...
Water Withdrawal Blue - Electricity - tower - E...	0.000000e+00	...
Water Withdrawal Blue - Electricity - tower - E...	0.000000e+00	...
Water Withdrawal Blue - Electricity - tower - E...	5.740860e-06	...
Water Withdrawal Blue - Electricity - tower - E...	1.464228e-05	...
Water Withdrawal Blue - Electricity - tower - E...	0.000000e+00	...
Water Withdrawal Blue - Electricity - tower - E...	2.993925e-07	...
Water Withdrawal Blue - Electricity - tower - E...	0.000000e+00	...
Water Withdrawal Blue - Electricity - tower - E...	1.199360e-05	...
Water Withdrawal Blue - Electricity - tower - E...	0.000000e+00	...
Water Withdrawal Blue - Electricity - once-thro...	1.928416e-02	...
Water Withdrawal Blue - Electricity - once-thro...	2.204305e-02	...
Water Withdrawal Blue - Electricity - once-thro...	1.123196e-02	...
Water Withdrawal Blue - Electricity - once-thro...	0.000000e+00	...
Water Withdrawal Blue - Electricity - once-thro...	0.000000e+00	...
Water Withdrawal Blue - Electricity - once-thro...	4.924568e-03	...
Water Withdrawal Blue - Electricity - once-thro...	2.670092e-03	...
Water Withdrawal Blue - Electricity - once-thro...	0.000000e+00	...
Water Withdrawal Blue - Electricity - once-thro...	0.000000e+00	...
Water Withdrawal Blue - Electricity - once-thro...	0.000000e+00	...
Water Withdrawal Blue - Electricity - once-thro...	7.630408e-05	...
Water Withdrawal Blue - Electricity - once-thro...	0.000000e+00	...
Water Withdrawal Blue - Domestic - domestic Wat...	0.000000e+00	...

sector	Paper for treatment: landfill	\
Taxes less subsidies on products purchased: Total	0.0	
Other net taxes on production	0.0	
Compensation of employees; wages, salaries, & e...	0.0	
Compensation of employees; wages, salaries, & e...	0.0	
Compensation of employees; wages, salaries, & e...	0.0	
Operating surplus: Consumption of fixed capital	0.0	
Operating surplus: Rents on land	0.0	
Operating surplus: Royalties on resources	0.0	

Operating surplus: Remaining net operating surplus	0.0
Employment: Low-skilled male	0.0
Employment: Low-skilled female	0.0
Employment: Medium-skilled male	0.0
Employment: Medium-skilled female	0.0
Employment: High-skilled male	0.0
Employment: High-skilled female	0.0
Employment hours: Low-skilled male	0.0
Employment hours: Low-skilled female	0.0
Employment hours: Medium-skilled male	0.0
Employment hours: Medium-skilled female	0.0
Employment hours: High-skilled male	0.0
Employment hours: High-skilled female	0.0
Employment: Vulnerable employment	0.0
Employment hours: Vulnerable employment	0.0
CO2 - combustion - air	0.0
CH4 - combustion - air	0.0
N2O - combustion - air	0.0
SOx - combustion - air	0.0
NOx - combustion - air	0.0
NH3 - combustion - air	0.0
CO - combustion - air	0.0
...	...
Water Withdrawal Blue - Manufacturing - Radio, ...	0.0
Water Withdrawal Blue - Manufacturing - Medical...	0.0
Water Withdrawal Blue - Manufacturing - Motor v...	0.0
Water Withdrawal Blue - Manufacturing - Other t...	0.0
Water Withdrawal Blue - Manufacturing - Furnitu...	0.0
Water Withdrawal Blue - Electricity - tower - E...	0.0
Water Withdrawal Blue - Electricity - tower - E...	0.0
Water Withdrawal Blue - Electricity - tower - E...	0.0
Water Withdrawal Blue - Electricity - tower - E...	0.0
Water Withdrawal Blue - Electricity - tower - E...	0.0
Water Withdrawal Blue - Electricity - tower - E...	0.0
Water Withdrawal Blue - Electricity - tower - E...	0.0
Water Withdrawal Blue - Electricity - tower - E...	0.0
Water Withdrawal Blue - Electricity - tower - E...	0.0
Water Withdrawal Blue - Electricity - tower - E...	0.0
Water Withdrawal Blue - Electricity - tower - E...	0.0
Water Withdrawal Blue - Electricity - once-thro...	0.0
Water Withdrawal Blue - Electricity - once-thro...	0.0
Water Withdrawal Blue - Electricity - once-thro...	0.0
Water Withdrawal Blue - Electricity - once-thro...	0.0
Water Withdrawal Blue - Electricity - once-thro...	0.0
Water Withdrawal Blue - Electricity - once-thro...	0.0
Water Withdrawal Blue - Electricity - once-thro...	0.0
Water Withdrawal Blue - Electricity - once-thro...	0.0

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Water Withdrawal Blue - Electricity - once-thro...                          0.0
Water Withdrawal Blue - Electricity - once-thro...                          0.0
Water Withdrawal Blue - Electricity - once-thro...                          0.0
Water Withdrawal Blue - Electricity - once-thro...                          0.0
Water Withdrawal Blue - Domestic - domestic Wat...                          0.0

sector                                                Plastic waste for treatment: landf
Taxes less subsidies on products purchased: Total
Other net taxes on production
Compensation of employees; wages, salaries, & e...
Compensation of employees; wages, salaries, & e...
Compensation of employees; wages, salaries, & e...
Operating surplus: Consumption of fixed capital
Operating surplus: Rents on land
Operating surplus: Royalties on resources
Operating surplus: Remaining net operating surplus
Employment: Low-skilled male
Employment: Low-skilled female
Employment: Medium-skilled male
Employment: Medium-skilled female
Employment: High-skilled male
Employment: High-skilled female
Employment hours: Low-skilled male
Employment hours: Low-skilled female
Employment hours: Medium-skilled male
Employment hours: Medium-skilled female
Employment hours: High-skilled male
Employment hours: High-skilled female
Employment: Vulnerable employment
Employment hours: Vulnerable employment
CO2 - combustion - air
CH4 - combustion - air
N2O - combustion - air
SOx - combustion - air
NOx - combustion - air
NH3 - combustion - air
CO - combustion - air
...
Water Withdrawal Blue - Manufacturing - Radio, ...
Water Withdrawal Blue - Manufacturing - Medical...
Water Withdrawal Blue - Manufacturing - Motor v...
Water Withdrawal Blue - Manufacturing - Other t...
Water Withdrawal Blue - Manufacturing - Furnitu...
Water Withdrawal Blue - Electricity - tower - E...
Water Withdrawal Blue - Electricity - tower - E...
Water Withdrawal Blue - Electricity - tower - E...
Water Withdrawal Blue - Electricity - tower - E...
Water Withdrawal Blue - Electricity - tower - E...
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Water Withdrawal Blue - Electricity - tower - E...
Water Withdrawal Blue - Electricity - tower - E...
Water Withdrawal Blue - Electricity - tower - E...
Water Withdrawal Blue - Electricity - tower - E...
Water Withdrawal Blue - Electricity - tower - E...
Water Withdrawal Blue - Electricity - tower - E...
Water Withdrawal Blue - Electricity - tower - E...
Water Withdrawal Blue - Electricity - once-thro...
Water Withdrawal Blue - Electricity - once-thro...
Water Withdrawal Blue - Electricity - once-thro...
Water Withdrawal Blue - Electricity - once-thro...
Water Withdrawal Blue - Electricity - once-thro...
Water Withdrawal Blue - Electricity - once-thro...
Water Withdrawal Blue - Electricity - once-thro...
Water Withdrawal Blue - Electricity - once-thro...
Water Withdrawal Blue - Electricity - once-thro...
Water Withdrawal Blue - Electricity - once-thro...
Water Withdrawal Blue - Electricity - once-thro...
Water Withdrawal Blue - Electricity - once-thro...
Water Withdrawal Blue - Domestic - domestic Wat...

sector                                                Inert/metal/hazardous waste for t
Taxes less subsidies on products purchased: Total
Other net taxes on production
Compensation of employees; wages, salaries, & e...
Compensation of employees; wages, salaries, & e...
Compensation of employees; wages, salaries, & e...
Operating surplus: Consumption of fixed capital
Operating surplus: Rents on land
Operating surplus: Royalties on resources
Operating surplus: Remaining net operating surplus
Employment: Low-skilled male
Employment: Low-skilled female
Employment: Medium-skilled male
Employment: Medium-skilled female
Employment: High-skilled male
Employment: High-skilled female
Employment hours: Low-skilled male
Employment hours: Low-skilled female
Employment hours: Medium-skilled male
Employment hours: Medium-skilled female
Employment hours: High-skilled male
Employment hours: High-skilled female
Employment: Vulnerable employment
Employment hours: Vulnerable employment
CO2 - combustion - air
CH4 - combustion - air
N2O - combustion - air
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SOx - combustion - air
NOx - combustion - air
NH3 - combustion - air
CO - combustion - air
...
Water Withdrawal Blue - Manufacturing - Radio, ...
Water Withdrawal Blue - Manufacturing - Medical...
Water Withdrawal Blue - Manufacturing - Motor v...
Water Withdrawal Blue - Manufacturing - Other t...
Water Withdrawal Blue - Manufacturing - Furnitu...
Water Withdrawal Blue - Electricity - tower - E...
Water Withdrawal Blue - Electricity - tower - E...
Water Withdrawal Blue - Electricity - tower - E...
Water Withdrawal Blue - Electricity - tower - E...
Water Withdrawal Blue - Electricity - tower - E...
Water Withdrawal Blue - Electricity - tower - E...
Water Withdrawal Blue - Electricity - tower - E...
Water Withdrawal Blue - Electricity - tower - E...
Water Withdrawal Blue - Electricity - tower - E...
Water Withdrawal Blue - Electricity - tower - E...
Water Withdrawal Blue - Electricity - tower - E...
Water Withdrawal Blue - Electricity - once-thro...
Water Withdrawal Blue - Electricity - once-thro...
Water Withdrawal Blue - Electricity - once-thro...
Water Withdrawal Blue - Electricity - once-thro...
Water Withdrawal Blue - Electricity - once-thro...
Water Withdrawal Blue - Electricity - once-thro...
Water Withdrawal Blue - Electricity - once-thro...
Water Withdrawal Blue - Electricity - once-thro...
Water Withdrawal Blue - Electricity - once-thro...
Water Withdrawal Blue - Electricity - once-thro...
Water Withdrawal Blue - Electricity - once-thro...
Water Withdrawal Blue - Domestic - domestic Wat...

sector                                                Textiles waste for treatment: lan
Taxes less subsidies on products purchased: Total
Other net taxes on production
Compensation of employees; wages, salaries, & e...
Compensation of employees; wages, salaries, & e...
Compensation of employees; wages, salaries, & e...
Operating surplus: Consumption of fixed capital
Operating surplus: Rents on land
Operating surplus: Royalties on resources
Operating surplus: Remaining net operating surplus
Employment: Low-skilled male
Employment: Low-skilled female
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Employment: Medium-skilled male
Employment: Medium-skilled female
Employment: High-skilled male
Employment: High-skilled female
Employment hours: Low-skilled male
Employment hours: Low-skilled female
Employment hours: Medium-skilled male
Employment hours: Medium-skilled female
Employment hours: High-skilled male
Employment hours: High-skilled female
Employment: Vulnerable employment
Employment hours: Vulnerable employment
CO2 - combustion - air
CH4 - combustion - air
N2O - combustion - air
SOx - combustion - air
NOx - combustion - air
NH3 - combustion - air
CO - combustion - air
...
Water Withdrawal Blue - Manufacturing - Radio, ...
Water Withdrawal Blue - Manufacturing - Medical...
Water Withdrawal Blue - Manufacturing - Motor v...
Water Withdrawal Blue - Manufacturing - Other t...
Water Withdrawal Blue - Manufacturing - Furnitu...
Water Withdrawal Blue - Electricity - tower - E...
Water Withdrawal Blue - Electricity - tower - E...
Water Withdrawal Blue - Electricity - tower - E...
Water Withdrawal Blue - Electricity - tower - E...
Water Withdrawal Blue - Electricity - tower - E...
Water Withdrawal Blue - Electricity - tower - E...
Water Withdrawal Blue - Electricity - tower - E...
Water Withdrawal Blue - Electricity - tower - E...
Water Withdrawal Blue - Electricity - tower - E...
Water Withdrawal Blue - Electricity - tower - E...
Water Withdrawal Blue - Electricity - tower - E...
Water Withdrawal Blue - Electricity - tower - E...
Water Withdrawal Blue - Electricity - once-thro...
Water Withdrawal Blue - Electricity - once-thro...
Water Withdrawal Blue - Electricity - once-thro...
Water Withdrawal Blue - Electricity - once-thro...
Water Withdrawal Blue - Electricity - once-thro...
Water Withdrawal Blue - Electricity - once-thro...
Water Withdrawal Blue - Electricity - once-thro...
Water Withdrawal Blue - Electricity - once-thro...
Water Withdrawal Blue - Electricity - once-thro...
Water Withdrawal Blue - Electricity - once-thro...
Water Withdrawal Blue - Electricity - once-thro...
```

sector	Wood waste for treatment: landfil
Water Withdrawal Blue - Electricity - once-thro...	
Water Withdrawal Blue - Domestic - domestic Wat...	
Taxes less subsidies on products purchased: Total	0.0
Other net taxes on production	0.0
Compensation of employees; wages, salaries, & e...	0.0
Compensation of employees; wages, salaries, & e...	0.0
Compensation of employees; wages, salaries, & e...	0.0
Operating surplus: Consumption of fixed capital	0.0
Operating surplus: Rents on land	0.0
Operating surplus: Royalties on resources	0.0
Operating surplus: Remaining net operating surplus	0.0
Employment: Low-skilled male	0.0
Employment: Low-skilled female	0.0
Employment: Medium-skilled male	0.0
Employment: Medium-skilled female	0.0
Employment: High-skilled male	0.0
Employment: High-skilled female	0.0
Employment hours: Low-skilled male	0.0
Employment hours: Low-skilled female	0.0
Employment hours: Medium-skilled male	0.0
Employment hours: Medium-skilled female	0.0
Employment hours: High-skilled male	0.0
Employment hours: High-skilled female	0.0
Employment: Vulnerable employment	0.0
Employment hours: Vulnerable employment	0.0
CO2 - combustion - air	0.0
CH4 - combustion - air	0.0
N2O - combustion - air	0.0
SOx - combustion - air	0.0
NOx - combustion - air	0.0
NH3 - combustion - air	0.0
CO - combustion - air	0.0
...	..
Water Withdrawal Blue - Manufacturing - Radio, ...	0.0
Water Withdrawal Blue - Manufacturing - Medical...	0.0
Water Withdrawal Blue - Manufacturing - Motor v...	0.0
Water Withdrawal Blue - Manufacturing - Other t...	0.0
Water Withdrawal Blue - Manufacturing - Furnitu...	0.0
Water Withdrawal Blue - Electricity - tower - E...	0.0
Water Withdrawal Blue - Electricity - tower - E...	0.0
Water Withdrawal Blue - Electricity - tower - E...	0.0
Water Withdrawal Blue - Electricity - tower - E...	0.0
Water Withdrawal Blue - Electricity - tower - E...	0.0
Water Withdrawal Blue - Electricity - tower - E...	0.0
Water Withdrawal Blue - Electricity - tower - E...	0.0
Water Withdrawal Blue - Electricity - tower - E...	0.0

Water Withdrawal Blue - Electricity - tower - E...	0.0
Water Withdrawal Blue - Electricity - tower - E...	0.0
Water Withdrawal Blue - Electricity - tower - E...	0.0
Water Withdrawal Blue - Electricity - tower - E...	0.0
Water Withdrawal Blue - Electricity - once-thro...	0.0
Water Withdrawal Blue - Electricity - once-thro...	0.0
Water Withdrawal Blue - Electricity - once-thro...	0.0
Water Withdrawal Blue - Electricity - once-thro...	0.0
Water Withdrawal Blue - Electricity - once-thro...	0.0
Water Withdrawal Blue - Electricity - once-thro...	0.0
Water Withdrawal Blue - Electricity - once-thro...	0.0
Water Withdrawal Blue - Electricity - once-thro...	0.0
Water Withdrawal Blue - Electricity - once-thro...	0.0
Water Withdrawal Blue - Electricity - once-thro...	0.0
Water Withdrawal Blue - Electricity - once-thro...	0.0
Water Withdrawal Blue - Electricity - once-thro...	0.0
Water Withdrawal Blue - Domestic - domestic Wat...	0.0

sector	Membership organisation services r
Taxes less subsidies on products purchased: Total	1
Other net taxes on production	5
Compensation of employees; wages, salaries, & e...	8
Compensation of employees; wages, salaries, & e...	5
Compensation of employees; wages, salaries, & e...	4
Operating surplus: Consumption of fixed capital	3
Operating surplus: Rents on land	0
Operating surplus: Royalties on resources	0
Operating surplus: Remaining net operating surplus	5
Employment: Low-skilled male	1
Employment: Low-skilled female	9
Employment: Medium-skilled male	5
Employment: Medium-skilled female	2
Employment: High-skilled male	1
Employment: High-skilled female	6
Employment hours: Low-skilled male	4
Employment hours: Low-skilled female	2
Employment hours: Medium-skilled male	1
Employment hours: Medium-skilled female	5
Employment hours: High-skilled male	2
Employment hours: High-skilled female	1
Employment: Vulnerable employment	5
Employment hours: Vulnerable employment	1
CO2 - combustion - air	1
CH4 - combustion - air	2
N2O - combustion - air	2
SOx - combustion - air	2
NOx - combustion - air	3
NH3 - combustion - air	6

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CO - combustion - air                                                    3
...
Water Withdrawal Blue - Manufacturing - Radio, ...                       9
Water Withdrawal Blue - Manufacturing - Medical...                       3
Water Withdrawal Blue - Manufacturing - Motor v...                       3
Water Withdrawal Blue - Manufacturing - Other t...                       7
Water Withdrawal Blue - Manufacturing - Furnitu...                       3
Water Withdrawal Blue - Electricity - tower - E...                       2
Water Withdrawal Blue - Electricity - tower - E...                       6
Water Withdrawal Blue - Electricity - tower - E...                       5
Water Withdrawal Blue - Electricity - tower - E...                       0
Water Withdrawal Blue - Electricity - tower - E...                       0
Water Withdrawal Blue - Electricity - tower - E...                       1
Water Withdrawal Blue - Electricity - tower - E...                       2
Water Withdrawal Blue - Electricity - tower - E...                       0
Water Withdrawal Blue - Electricity - tower - E...                       5
Water Withdrawal Blue - Electricity - tower - E...                       0
Water Withdrawal Blue - Electricity - tower - E...                       2
Water Withdrawal Blue - Electricity - tower - E...                       0
Water Withdrawal Blue - Electricity - once-thro...                       4
Water Withdrawal Blue - Electricity - once-thro...                       5
Water Withdrawal Blue - Electricity - once-thro...                       2
Water Withdrawal Blue - Electricity - once-thro...                       0
Water Withdrawal Blue - Electricity - once-thro...                       0
Water Withdrawal Blue - Electricity - once-thro...                       1
Water Withdrawal Blue - Electricity - once-thro...                       1
Water Withdrawal Blue - Electricity - once-thro...                       0
Water Withdrawal Blue - Electricity - once-thro...                       0
Water Withdrawal Blue - Electricity - once-thro...                       0
Water Withdrawal Blue - Electricity - once-thro...                       2
Water Withdrawal Blue - Electricity - once-thro...                       0
Water Withdrawal Blue - Domestic - domestic Wat...                       0

sector                                                  Recreational, cultural and sportin
Taxes less subsidies on products purchased: Total
Other net taxes on production
Compensation of employees; wages, salaries, & e...
Compensation of employees; wages, salaries, & e...
Compensation of employees; wages, salaries, & e...
Operating surplus: Consumption of fixed capital
Operating surplus: Rents on land
Operating surplus: Royalties on resources
Operating surplus: Remaining net operating surplus
Employment: Low-skilled male
Employment: Low-skilled female
Employment: Medium-skilled male
Employment: Medium-skilled female
Employment: High-skilled male
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Employment: High-skilled female
Employment hours: Low-skilled male
Employment hours: Low-skilled female
Employment hours: Medium-skilled male
Employment hours: Medium-skilled female
Employment hours: High-skilled male
Employment hours: High-skilled female
Employment: Vulnerable employment
Employment hours: Vulnerable employment
CO2 - combustion - air
CH4 - combustion - air
N2O - combustion - air
SOx - combustion - air
NOx - combustion - air
NH3 - combustion - air
CO - combustion - air
...
Water Withdrawal Blue - Manufacturing - Radio, ...
Water Withdrawal Blue - Manufacturing - Medical...
Water Withdrawal Blue - Manufacturing - Motor v...
Water Withdrawal Blue - Manufacturing - Other t...
Water Withdrawal Blue - Manufacturing - Furnitu...
Water Withdrawal Blue - Electricity - tower - E...
Water Withdrawal Blue - Electricity - tower - E...
Water Withdrawal Blue - Electricity - tower - E...
Water Withdrawal Blue - Electricity - tower - E...
Water Withdrawal Blue - Electricity - tower - E...
Water Withdrawal Blue - Electricity - tower - E...
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Water Withdrawal Blue - Electricity - tower - E...
Water Withdrawal Blue - Electricity - tower - E...
Water Withdrawal Blue - Electricity - tower - E...
Water Withdrawal Blue - Electricity - tower - E...
Water Withdrawal Blue - Electricity - once-thro...
Water Withdrawal Blue - Electricity - once-thro...
Water Withdrawal Blue - Electricity - once-thro...
Water Withdrawal Blue - Electricity - once-thro...
Water Withdrawal Blue - Electricity - once-thro...
Water Withdrawal Blue - Electricity - once-thro...
Water Withdrawal Blue - Electricity - once-thro...
Water Withdrawal Blue - Electricity - once-thro...
Water Withdrawal Blue - Electricity - once-thro...
Water Withdrawal Blue - Electricity - once-thro...
Water Withdrawal Blue - Electricity - once-thro...
Water Withdrawal Blue - Electricity - once-thro...
Water Withdrawal Blue - Domestic - domestic Wat...
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sector	Other services (93) \
Taxes less subsidies on products purchased: Total	2.060625e+01
Other net taxes on production	7.235202e+00
Compensation of employees; wages, salaries, & e...	1.191397e+01
Compensation of employees; wages, salaries, & e...	7.832720e+01
Compensation of employees; wages, salaries, & e...	6.303171e+01
Operating surplus: Consumption of fixed capital	4.612496e+01
Operating surplus: Rents on land	0.000000e+00
Operating surplus: Royalties on resources	0.000000e+00
Operating surplus: Remaining net operating surplus	7.928833e+01
Employment: Low-skilled male	2.464839e+00
Employment: Low-skilled female	1.326323e+00
Employment: Medium-skilled male	6.980127e+00
Employment: Medium-skilled female	3.866667e+00
Employment: High-skilled male	1.707918e+00
Employment: High-skilled female	8.624937e-01
Employment hours: Low-skilled male	5.546914e+00
Employment hours: Low-skilled female	2.907072e+00
Employment hours: Medium-skilled male	1.484088e+01
Employment hours: Medium-skilled female	8.055547e+00
Employment hours: High-skilled male	3.564956e+00
Employment hours: High-skilled female	1.730584e+00
Employment: Vulnerable employment	6.865002e+00
Employment hours: Vulnerable employment	1.464848e+01
CO2 - combustion - air	1.447428e+08
CH4 - combustion - air	4.373819e+04
N2O - combustion - air	2.920695e+03
SOx - combustion - air	3.418009e+05
NOx - combustion - air	4.922347e+05
NH3 - combustion - air	1.269446e+05
CO - combustion - air	4.223046e+05
...	...
Water Withdrawal Blue - Manufacturing - Radio, ...	4.291276e-02
Water Withdrawal Blue - Manufacturing - Medical...	1.752856e-02
Water Withdrawal Blue - Manufacturing - Motor v...	4.121262e-04
Water Withdrawal Blue - Manufacturing - Other t...	8.276037e-04
Water Withdrawal Blue - Manufacturing - Furnitu...	9.556752e-03
Water Withdrawal Blue - Electricity - tower - E...	3.290488e-02
Water Withdrawal Blue - Electricity - tower - E...	9.477832e-03
Water Withdrawal Blue - Electricity - tower - E...	9.725356e-03
Water Withdrawal Blue - Electricity - tower - E...	0.000000e+00
Water Withdrawal Blue - Electricity - tower - E...	0.000000e+00
Water Withdrawal Blue - Electricity - tower - E...	1.541307e-04
Water Withdrawal Blue - Electricity - tower - E...	4.767884e-04
Water Withdrawal Blue - Electricity - tower - E...	0.000000e+00
Water Withdrawal Blue - Electricity - tower - E...	8.973588e-06
Water Withdrawal Blue - Electricity - tower - E...	0.000000e+00
Water Withdrawal Blue - Electricity - tower - E...	4.151054e-04

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Water Withdrawal Blue - Electricity - tower - E...         0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...         6.432351e-01
Water Withdrawal Blue - Electricity - once-thro...         6.945115e-01
Water Withdrawal Blue - Electricity - once-thro...         3.739655e-01
Water Withdrawal Blue - Electricity - once-thro...         0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...         0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...         1.951586e-01
Water Withdrawal Blue - Electricity - once-thro...         3.695945e-02
Water Withdrawal Blue - Electricity - once-thro...         0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...         0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...         0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...         3.192070e-03
Water Withdrawal Blue - Electricity - once-thro...         0.000000e+00
Water Withdrawal Blue - Domestic - domestic Wat...         0.000000e+00

sector                                                  Private households with employed p
Taxes less subsidies on products purchased: Total                                          1
Other net taxes on production                                                              3
Compensation of employees; wages, salaries, & e...                                         7
Compensation of employees; wages, salaries, & e...                                         3
Compensation of employees; wages, salaries, & e...                                         3
Operating surplus: Consumption of fixed capital                                            2
Operating surplus: Rents on land                                                           0
Operating surplus: Royalties on resources                                                  0
Operating surplus: Remaining net operating surplus                                         2
Employment: Low-skilled male                                                               1
Employment: Low-skilled female                                                             5
Employment: Medium-skilled male                                                            3
Employment: Medium-skilled female                                                          1
Employment: High-skilled male                                                              1
Employment: High-skilled female                                                            5
Employment hours: Low-skilled male                                                         3
Employment hours: Low-skilled female                                                       1
Employment hours: Medium-skilled male                                                      8
Employment hours: Medium-skilled female                                                    3
Employment hours: High-skilled male                                                        2
Employment hours: High-skilled female                                                      1
Employment: Vulnerable employment                                                          4
Employment hours: Vulnerable employment                                                    8
CO2 - combustion - air                                                                     6
CH4 - combustion - air                                                                     1
N2O - combustion - air                                                                     1
SOx - combustion - air                                                                     1
NOx - combustion - air                                                                     1
NH3 - combustion - air                                                                     3
CO - combustion - air                                                                      2
...
Water Withdrawal Blue - Manufacturing - Radio, ...                                         2
```

```
Water Withdrawal Blue - Manufacturing - Medical...
Water Withdrawal Blue - Manufacturing - Motor v...
Water Withdrawal Blue - Manufacturing - Other t...
Water Withdrawal Blue - Manufacturing - Furnitu...
Water Withdrawal Blue - Electricity - tower - E...
Water Withdrawal Blue - Electricity - tower - E...
Water Withdrawal Blue - Electricity - tower - E...
Water Withdrawal Blue - Electricity - tower - E...
Water Withdrawal Blue - Electricity - tower - E...
Water Withdrawal Blue - Electricity - tower - E...
Water Withdrawal Blue - Electricity - tower - E...
Water Withdrawal Blue - Electricity - tower - E...
Water Withdrawal Blue - Electricity - tower - E...
Water Withdrawal Blue - Electricity - tower - E...
Water Withdrawal Blue - Electricity - tower - E...
Water Withdrawal Blue - Electricity - tower - E...
Water Withdrawal Blue - Electricity - once-thro...
Water Withdrawal Blue - Electricity - once-thro...
Water Withdrawal Blue - Electricity - once-thro...
Water Withdrawal Blue - Electricity - once-thro...
Water Withdrawal Blue - Electricity - once-thro...
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Water Withdrawal Blue - Electricity - once-thro...
Water Withdrawal Blue - Electricity - once-thro...
Water Withdrawal Blue - Electricity - once-thro...
Water Withdrawal Blue - Electricity - once-thro...
Water Withdrawal Blue - Electricity - once-thro...
Water Withdrawal Blue - Domestic - domestic Wat...

sector                                                Extra-territorial organizations ar
Taxes less subsidies on products purchased: Total
Other net taxes on production
Compensation of employees; wages, salaries, & e...
Compensation of employees; wages, salaries, & e...
Compensation of employees; wages, salaries, & e...
Operating surplus: Consumption of fixed capital
Operating surplus: Rents on land
Operating surplus: Royalties on resources
Operating surplus: Remaining net operating surplus
Employment: Low-skilled male
Employment: Low-skilled female
Employment: Medium-skilled male
Employment: Medium-skilled female
Employment: High-skilled male
Employment: High-skilled female
Employment hours: Low-skilled male
Employment hours: Low-skilled female
```

```
Employment hours: Medium-skilled male
Employment hours: Medium-skilled female
Employment hours: High-skilled male
Employment hours: High-skilled female
Employment: Vulnerable employment
Employment hours: Vulnerable employment
CO2 - combustion - air
CH4 - combustion - air
N2O - combustion - air
SOx - combustion - air
NOx - combustion - air
NH3 - combustion - air
CO - combustion - air
...
Water Withdrawal Blue - Manufacturing - Radio, ...
Water Withdrawal Blue - Manufacturing - Medical...
Water Withdrawal Blue - Manufacturing - Motor v...
Water Withdrawal Blue - Manufacturing - Other t...
Water Withdrawal Blue - Manufacturing - Furnitu...
Water Withdrawal Blue - Electricity - tower - E...
Water Withdrawal Blue - Electricity - tower - E...
Water Withdrawal Blue - Electricity - tower - E...
Water Withdrawal Blue - Electricity - tower - E...
Water Withdrawal Blue - Electricity - tower - E...
Water Withdrawal Blue - Electricity - tower - E...
Water Withdrawal Blue - Electricity - tower - E...
Water Withdrawal Blue - Electricity - tower - E...
Water Withdrawal Blue - Electricity - tower - E...
Water Withdrawal Blue - Electricity - tower - E...
Water Withdrawal Blue - Electricity - tower - E...
Water Withdrawal Blue - Electricity - tower - E...
Water Withdrawal Blue - Electricity - once-thro...
Water Withdrawal Blue - Electricity - once-thro...
Water Withdrawal Blue - Electricity - once-thro...
Water Withdrawal Blue - Electricity - once-thro...
Water Withdrawal Blue - Electricity - once-thro...
Water Withdrawal Blue - Electricity - once-thro...
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Water Withdrawal Blue - Electricity - once-thro...
Water Withdrawal Blue - Electricity - once-thro...
Water Withdrawal Blue - Electricity - once-thro...
Water Withdrawal Blue - Electricity - once-thro...
Water Withdrawal Blue - Domestic - domestic Wat...

[1104 rows x 200 columns]
```

As a final step, the whole database `SE_aggr` was saved to a folder on the desktop.

```
In [20]: save_folder_full = 'C:/Users/tobia/Desktop/Database_LCA/SE_aggr'
         SE_aggr.save_all('C:/Users/tobia/Desktop/Database_LCA/SE_aggr')
```

```
Out[20]: <pymrio.core.mriosystem.IOSystem at 0x1eefaa487b8>
```

4.2 RO 1.2 Characterisation

Now we are importing the correspondence file for the characterisation of the environmental stressors into the four footprint (Carbon, Land, Material, Blue Water) according to Steinmann et al. (2018)

```
In [43]: characterisation = pd.read_csv('C:/Users/tobia/Dropbox/KTH Sustainable Technology/4_A
         characterisation
```

```
Out[43]:                                Taxes less subsidies on products purchased: Total  \
         Carbon Footprint [t CO2-eq.]                                                     0
         Land footprint [km2]                                                             0
         Material footprint [kt]                                                          0
         Water footprint [Mm3]                                                            0

                                        Other net taxes on production  \
         Carbon Footprint [t CO2-eq.]                               0
         Land footprint [km2]                                       0
         Material footprint [kt]                                    0
         Water footprint [Mm3]                                      0

                                        Compensation of employees; wages, salaries, & employers
         Carbon Footprint [t CO2-eq.]                                                     0
         Land footprint [km2]                                                             0
         Material footprint [kt]                                                          0
         Water footprint [Mm3]                                                            0

                                        Compensation of employees; wages, salaries, & employers
         Carbon Footprint [t CO2-eq.]                                                     0
         Land footprint [km2]                                                             0
         Material footprint [kt]                                                          0
         Water footprint [Mm3]                                                            0

                                        Compensation of employees; wages, salaries, & employers
         Carbon Footprint [t CO2-eq.]                                                     0
         Land footprint [km2]                                                             0
         Material footprint [kt]                                                          0
         Water footprint [Mm3]                                                            0

                                        Operating surplus: Consumption of fixed capital  \
         Carbon Footprint [t CO2-eq.]                                                   0
         Land footprint [km2]                                                           0
         Material footprint [kt]                                                        0
         Water footprint [Mm3]                                                          0
```

```
                                Operating surplus: Rents on land  \
Carbon Footprint [t CO2-eq.]                                    0
Land footprint [km2]                                            0
Material footprint [kt]                                         0
Water footprint [Mm3]                                           0

                                Operating surplus: Royalties on resources  \
Carbon Footprint [t CO2-eq.]                                            0
Land footprint [km2]                                                    0
Material footprint [kt]                                                 0
Water footprint [Mm3]                                                   0

                                Operating surplus: Remaining net operating surplus  \
Carbon Footprint [t CO2-eq.]                                                      0
Land footprint [km2]                                                              0
Material footprint [kt]                                                           0
Water footprint [Mm3]                                                             0

                                Employment: Low-skilled male  ...  \
Carbon Footprint [t CO2-eq.]                              0  ...
Land footprint [km2]                                      0  ...
Material footprint [kt]                                   0  ...
Water footprint [Mm3]                                     0  ...

                                Water Withdrawal Blue - Electricity - once-through - Ele
Carbon Footprint [t CO2-eq.]                                                      0
Land footprint [km2]                                                              0
Material footprint [kt]                                                           0
Water footprint [Mm3]                                                             0

                                Water Withdrawal Blue - Electricity - once-through - Ele
Carbon Footprint [t CO2-eq.]                                                      0
Land footprint [km2]                                                              0
Material footprint [kt]                                                           0
Water footprint [Mm3]                                                             0

                                Water Withdrawal Blue - Electricity - once-through - Ele
Carbon Footprint [t CO2-eq.]                                                      0
Land footprint [km2]                                                              0
Material footprint [kt]                                                           0
Water footprint [Mm3]                                                             0

                                Water Withdrawal Blue - Electricity - once-through - Ele
Carbon Footprint [t CO2-eq.]                                                      0
Land footprint [km2]                                                              0
Material footprint [kt]                                                           0
Water footprint [Mm3]                                                             0
```

```
                                Water Withdrawal Blue - Electricity - once-through - Ele
Carbon Footprint [t CO2-eq.]                                                          0
Land footprint [km2]                                                                  0
Material footprint [kt]                                                               0
Water footprint [Mm3]                                                                 0

                                Water Withdrawal Blue - Electricity - once-through - Ele
Carbon Footprint [t CO2-eq.]                                                          0
Land footprint [km2]                                                                  0
Material footprint [kt]                                                               0
Water footprint [Mm3]                                                                 0

                                Water Withdrawal Blue - Electricity - once-through - Ele
Carbon Footprint [t CO2-eq.]                                                          0
Land footprint [km2]                                                                  0
Material footprint [kt]                                                               0
Water footprint [Mm3]                                                                 0

                                Water Withdrawal Blue - Electricity - once-through - Ele
Carbon Footprint [t CO2-eq.]                                                          0
Land footprint [km2]                                                                  0
Material footprint [kt]                                                               0
Water footprint [Mm3]                                                                 0

                                Water Withdrawal Blue - Electricity - once-through - Ele
Carbon Footprint [t CO2-eq.]                                                          0
Land footprint [km2]                                                                  0
Material footprint [kt]                                                               0
Water footprint [Mm3]                                                                 0

                                Water Withdrawal Blue - Electricity - once-through - Ele
Carbon Footprint [t CO2-eq.]                                                          0
Land footprint [km2]                                                                  0
Material footprint [kt]                                                               0
Water footprint [Mm3]                                                                 0

                                Water Withdrawal Blue - Domestic - domestic Water Withdr
Carbon Footprint [t CO2-eq.]                                                          0
Land footprint [km2]                                                                  0
Material footprint [kt]                                                               0
Water footprint [Mm3]                                                                 0

[4 rows x 1104 columns]
```

After creating the characterisation matrix, we need to multiply the consumption based account `D_cba` with the characterisation matrix `characterisation`.

```
In [44]: stressor_aggr_indicator = SE_aggr.satellite.D_cba.T.dot(characterisation.T)
         stressor_aggr_indicator
```

```
Out[44]:                                                    Carbon Footprint [t CO2-eq
         region sector
         SE     Paddy rice                                                   4.300361e+0
```

	Wheat	2.708281e+0
	Cereal grains nec	3.951627e+0
	Vegetables, fruit, nuts	7.062354e+0
	Oil seeds	1.847766e+0
	Sugar cane, sugar beet	8.788258e+0
	Plant-based fibers	4.240767e-0
	Crops nec	3.812132e+0
	Cattle	1.352361e+0
	Pigs	3.745667e+0
	Poultry	1.681942e+0
	Meat animals nec	3.446644e+0
	Animal products nec	3.198690e+0
	Raw milk	6.566744e+0
	Wool, silk-worm cocoons	1.707951e+0
	Manure (conventional treatment)	0.000000e+0
	Manure (biogas treatment)	0.000000e+0
	Products of forestry, logging and related servi...	1.768469e+0
	Fish and other fishing products; services incid...	8.836144e+0
	Anthracite	6.103460e+0
	Coking Coal	7.129399e+0
	Other Bituminous Coal	7.552722e+0
	Sub-Bituminous Coal	1.891207e+0
	Patent Fuel	4.032836e-0
	Lignite/Brown Coal	2.357234e-0
	BKB/Peat Briquettes	9.876062e-0
	Peat	4.506679e+0
	Crude petroleum and services related to crude o...	0.000000e+0
	Natural gas and services related to natural gas...	1.221635e+0
	Natural Gas Liquids	0.000000e+0
...		.
RoW	Research and development services (73)	2.915531e+0
	Other business services (74)	1.383797e+0
	Public administration and defence services; com...	2.417794e+0
	Education services (80)	6.501118e+0
	Health and social work services (85)	1.401999e+0
	Food waste for treatment: incineration	7.628030e+0
	Paper waste for treatment: incineration	7.827239e+0
	Plastic waste for treatment: incineration	1.302653e+0
	Intert/metal waste for treatment: incineration	1.198959e+0
	Textiles waste for treatment: incineration	1.147540e+0
	Wood waste for treatment: incineration	2.619927e+0
	Oil/hazardous waste for treatment: incineration	1.214962e+0
	Food waste for treatment: biogasification and l...	1.751310e+0
	Paper waste for treatment: biogasification and ...	1.595978e+0
	Sewage sludge for treatment: biogasification an...	2.195699e+0
	Food waste for treatment: composting and land a...	4.516161e+0
	Paper and wood waste for treatment: composting ...	9.544979e+0
	Food waste for treatment: waste water treatment	4.156771e+0

Other waste for treatment: waste water treatment	5.195173e+0
Food waste for treatment: landfill	1.248464e+0
Paper for treatment: landfill	3.786981e+0
Plastic waste for treatment: landfill	1.278173e+0
Inert/metal/hazardous waste for treatment: land...	1.416448e+0
Textiles waste for treatment: landfill	1.052631e+0
Wood waste for treatment: landfill	9.548759e+0
Membership organisation services n.e.c. (91)	1.706119e+0
Recreational, cultural and sporting services (92)	2.637563e+0
Other services (93)	1.959581e+0
Private households with employed persons (95)	4.393735e+0
Extra-territorial organizations and bodies	0.000000e+0

region	sector	Land footprint [km2] \
SE	Paddy rice	4.059979e-01
	Wheat	4.528841e+03
	Cereal grains nec	7.696287e+02
	Vegetables, fruit, nuts	3.492989e+03
	Oil seeds	3.182918e+03
	Sugar cane, sugar beet	1.347821e+00
	Plant-based fibers	2.246884e-03
	Crops nec	1.616982e+03
	Cattle	1.556224e+03
	Pigs	8.311629e+01
	Poultry	4.835901e+02
	Meat animals nec	1.756265e+03
	Animal products nec	1.682377e+02
	Raw milk	3.126695e+01
	Wool, silk-worm cocoons	3.055607e-04
	Manure (conventional treatment)	0.000000e+00
	Manure (biogas treatment)	0.000000e+00
	Products of forestry, logging and related servi...	8.427092e+04
	Fish and other fishing products; services incid...	3.863772e+02
	Anthracite	1.078962e-04
	Coking Coal	4.419943e-01
	Other Bituminous Coal	4.130683e+00
	Sub-Bituminous Coal	6.719669e-04
	Patent Fuel	3.521535e-08
	Lignite/Brown Coal	2.283058e-05
	BKB/Peat Briquettes	2.207442e-07
	Peat	3.809134e+00
	Crude petroleum and services related to crude o...	0.000000e+00
	Natural gas and services related to natural gas...	2.923347e+01
	Natural Gas Liquids	0.000000e+00
...		...
RoW	Research and development services (73)	3.196585e+05
	Other business services (74)	1.644698e+05

Public administration and defence services; com...	2.336304e+06
Education services (80)	6.545770e+05
Health and social work services (85)	1.983176e+06
Food waste for treatment: incineration	3.323382e+03
Paper waste for treatment: incineration	3.593115e+03
Plastic waste for treatment: incineration	1.631881e+03
Intert/metal waste for treatment: incineration	2.771507e+03
Textiles waste for treatment: incineration	9.684105e+02
Wood waste for treatment: incineration	1.151531e+03
Oil/hazardous waste for treatment: incineration	1.820596e+03
Food waste for treatment: biogasification and l...	2.185901e+02
Paper waste for treatment: biogasification and ...	1.886180e+01
Sewage sludge for treatment: biogasification an...	1.094397e+03
Food waste for treatment: composting and land a...	2.637375e+03
Paper and wood waste for treatment: composting ...	3.283087e+02
Food waste for treatment: waste water treatment	1.529476e+04
Other waste for treatment: waste water treatment	5.002475e+04
Food waste for treatment: landfill	1.795624e+04
Paper for treatment: landfill	8.782914e+03
Plastic waste for treatment: landfill	4.716502e+03
Inert/metal/hazardous waste for treatment: land...	1.459872e+04
Textiles waste for treatment: landfill	2.260474e+03
Wood waste for treatment: landfill	2.965684e+03
Membership organisation services n.e.c. (91)	2.612347e+05
Recreational, cultural and sporting services (92)	4.344801e+05
Other services (93)	2.896393e+05
Private households with employed persons (95)	9.212777e+04
Extra-territorial organizations and bodies	0.000000e+00

region	sector	Material footprint [kt] \
SE	Paddy rice	2.334721e-01
	Wheat	1.234295e+03
	Cereal grains nec	1.719641e+02
	Vegetables, fruit, nuts	4.115869e+03
	Oil seeds	5.764728e+02
	Sugar cane, sugar beet	3.899792e+00
	Plant-based fibers	6.882919e-04
	Crops nec	4.346631e+02
	Cattle	6.266705e+02
	Pigs	1.156021e+02
	Poultry	7.598228e+02
	Meat animals nec	1.241676e+03
	Animal products nec	5.992581e+01
	Raw milk	7.154615e+00
	Wool, silk-worm cocoons	5.517894e-04
	Manure (conventional treatment)	0.000000e+00
	Manure (biogas treatment)	0.000000e+00

	Products of forestry, logging and related servi...	1.238077e+04
	Fish and other fishing products; services incid...	2.386407e+02
	Anthracite	2.667174e-02
	Coking Coal	4.366926e+01
	Other Bituminous Coal	2.264521e+02
	Sub-Bituminous Coal	4.841874e-01
	Patent Fuel	8.809831e-08
	Lignite/Brown Coal	4.872934e-03
	BKB/Peat Briquettes	3.160311e-07
	Peat	1.722028e+01
	Crude petroleum and services related to crude o...	0.000000e+00
	Natural gas and services related to natural gas...	4.791725e+02
	Natural Gas Liquids	0.000000e+00
...		...
RoW	Research and development services (73)	3.462010e+05
	Other business services (74)	2.201337e+05
	Public administration and defence services; com...	3.275685e+06
	Education services (80)	8.225607e+05
	Health and social work services (85)	3.295652e+06
	Food waste for treatment: incineration	1.342045e+04
	Paper waste for treatment: incineration	1.393753e+04
	Plastic waste for treatment: incineration	4.470866e+03
	Intert/metal waste for treatment: incineration	5.542830e+03
	Textiles waste for treatment: incineration	3.036051e+03
	Wood waste for treatment: incineration	3.808402e+03
	Oil/hazardous waste for treatment: incineration	5.447366e+03
	Food waste for treatment: biogasification and l...	4.332499e+02
	Paper waste for treatment: biogasification and ...	2.542250e+01
	Sewage sludge for treatment: biogasification an...	1.994383e+03
	Food waste for treatment: composting and land a...	7.971113e+03
	Paper and wood waste for treatment: composting ...	6.964815e+02
	Food waste for treatment: waste water treatment	5.248527e+04
	Other waste for treatment: waste water treatment	1.106008e+05
	Food waste for treatment: landfill	4.814291e+04
	Paper for treatment: landfill	2.414196e+04
	Plastic waste for treatment: landfill	1.451133e+04
	Inert/metal/hazardous waste for treatment: land...	2.446406e+04
	Textiles waste for treatment: landfill	6.552352e+03
	Wood waste for treatment: landfill	8.298375e+03
	Membership organisation services n.e.c. (91)	3.141803e+05
	Recreational, cultural and sporting services (92)	3.784661e+05
	Other services (93)	3.445721e+05
	Private households with employed persons (95)	7.675513e+04
	Extra-territorial organizations and bodies	0.000000e+00

		Water footprint [Mm3]
region	sector	
SE	Paddy rice	1.015887e-01

	Wheat	5.769528e+01
	Cereal grains nec	4.018649e-01
	Vegetables, fruit, nuts	2.586223e+02
	Oil seeds	1.434237e+01
	Sugar cane, sugar beet	1.076404e-02
	Plant-based fibers	5.000705e-06
	Crops nec	1.235689e+02
	Cattle	3.939932e+00
	Pigs	1.689726e+00
	Poultry	4.016469e+00
	Meat animals nec	4.297415e+00
	Animal products nec	1.583188e+00
	Raw milk	5.469038e-02
	Wool, silk-worm cocoons	4.997246e-06
	Manure (conventional treatment)	0.000000e+00
	Manure (biogas treatment)	0.000000e+00
	Products of forestry, logging and related servi...	3.170483e+00
	Fish and other fishing products; services incid...	1.295690e+00
	Anthracite	2.077617e-06
	Coking Coal	4.218110e-03
	Other Bituminous Coal	4.593687e-02
	Sub-Bituminous Coal	7.408850e-06
	Patent Fuel	2.188845e-10
	Lignite/Brown Coal	4.868900e-07
	BKB/Peat Briquettes	2.422454e-09
	Peat	3.911056e-03
	Crude petroleum and services related to crude o...	0.000000e+00
	Natural gas and services related to natural gas...	6.455364e-02
	Natural Gas Liquids	0.000000e+00
...		...
RoW	Research and development services (73)	3.700297e+03
	Other business services (74)	2.725800e+03
	Public administration and defence services; com...	3.028880e+04
	Education services (80)	9.854652e+03
	Health and social work services (85)	2.944414e+04
	Food waste for treatment: incineration	8.191814e+01
	Paper waste for treatment: incineration	1.075123e+02
	Plastic waste for treatment: incineration	2.831316e+01
	Intert/metal waste for treatment: incineration	4.016528e+01
	Textiles waste for treatment: incineration	1.744441e+01
	Wood waste for treatment: incineration	2.209017e+01
	Oil/hazardous waste for treatment: incineration	3.307149e+01
	Food waste for treatment: biogasification and l...	2.611769e+00
	Paper waste for treatment: biogasification and ...	2.325305e-01
	Sewage sludge for treatment: biogasification an...	1.685175e+01
	Food waste for treatment: composting and land a...	4.590584e+01
	Paper and wood waste for treatment: composting ...	6.068476e+00
	Food waste for treatment: waste water treatment	3.321603e+02

```
Other waste for treatment: waste water treatment          9.559626e+02
Food waste for treatment: landfill                        4.749070e+02
Paper for treatment: landfill                             2.187981e+02
Plastic waste for treatment: landfill                     1.255966e+02
Inert/metal/hazardous waste for treatment: land...        2.428656e+02
Textiles waste for treatment: landfill                    4.478365e+01
Wood waste for treatment: landfill                        6.021628e+01
Membership organisation services n.e.c. (91)              4.125552e+03
Recreational, cultural and sporting services (92)         6.237635e+03
Other services (93)                                       4.652915e+03
Private households with employed persons (95)             1.133608e+03
Extra-territorial organizations and bodies                0.000000e+00

[600 rows x 4 columns]
```

In [45]:
```
# Slicing only the environmental footprints for all products in Sweden
stressor_aggr_indicator.xs('SE')
```

Out[45]:
```
                                                  Carbon Footprint [t CO2-eq.]  \
sector
Paddy rice                                                        4.300361e+02
Wheat                                                             2.708281e+05
Cereal grains nec                                                 3.951627e+04
Vegetables, fruit, nuts                                           7.062354e+05
Oil seeds                                                         1.847766e+05
Sugar cane, sugar beet                                            8.788258e+01
Plant-based fibers                                                4.240767e-01
Crops nec                                                         3.812132e+05
Cattle                                                            1.352361e+05
Pigs                                                              3.745667e+04
Poultry                                                           1.681942e+05
Meat animals nec                                                  3.446644e+05
Animal products nec                                               3.198690e+05
Raw milk                                                          6.566744e+03
Wool, silk-worm cocoons                                           1.707951e+00
Manure (conventional treatment)                                   0.000000e+00
Manure (biogas treatment)                                         0.000000e+00
Products of forestry, logging and related servi...                1.768469e+05
Fish and other fishing products; services incid...                8.836144e+04
Anthracite                                                        6.103460e+00
Coking Coal                                                       7.129399e+03
Other Bituminous Coal                                             7.552722e+04
Sub-Bituminous Coal                                               1.891207e+01
Patent Fuel                                                       4.032836e-04
Lignite/Brown Coal                                                2.357234e-01
BKB/Peat Briquettes                                               9.876062e-04
Peat                                                              4.506679e+02
Crude petroleum and services related to crude o...                0.000000e+00
```

Natural gas and services related to natural gas...	1.221635e+05
Natural Gas Liquids	0.000000e+00
...	...
Research and development services (73)	1.728424e+05
Other business services (74)	5.584061e+05
Public administration and defence services; com...	2.322562e+06
Education services (80)	1.608930e+06
Health and social work services (85)	2.788107e+06
Food waste for treatment: incineration	0.000000e+00
Paper waste for treatment: incineration	0.000000e+00
Plastic waste for treatment: incineration	0.000000e+00
Intert/metal waste for treatment: incineration	0.000000e+00
Textiles waste for treatment: incineration	0.000000e+00
Wood waste for treatment: incineration	0.000000e+00
Oil/hazardous waste for treatment: incineration	0.000000e+00
Food waste for treatment: biogasification and l...	0.000000e+00
Paper waste for treatment: biogasification and ...	0.000000e+00
Sewage sludge for treatment: biogasification an...	0.000000e+00
Food waste for treatment: composting and land a...	0.000000e+00
Paper and wood waste for treatment: composting ...	0.000000e+00
Food waste for treatment: waste water treatment	0.000000e+00
Other waste for treatment: waste water treatment	0.000000e+00
Food waste for treatment: landfill	0.000000e+00
Paper for treatment: landfill	0.000000e+00
Plastic waste for treatment: landfill	0.000000e+00
Inert/metal/hazardous waste for treatment: land...	0.000000e+00
Textiles waste for treatment: landfill	0.000000e+00
Wood waste for treatment: landfill	0.000000e+00
Membership organisation services n.e.c. (91)	2.102769e+06
Recreational, cultural and sporting services (92)	1.079669e+06
Other services (93)	3.994310e+05
Private households with employed persons (95)	1.106690e+03
Extra-territorial organizations and bodies	0.000000e+00

	Land footprint [km2] \
sector	
Paddy rice	4.059979e-01
Wheat	4.528841e+03
Cereal grains nec	7.696287e+02
Vegetables, fruit, nuts	3.492989e+03
Oil seeds	3.182918e+03
Sugar cane, sugar beet	1.347821e+00
Plant-based fibers	2.246884e-03
Crops nec	1.616982e+03
Cattle	1.556224e+03
Pigs	8.311629e+01
Poultry	4.835901e+02
Meat animals nec	1.756265e+03

Animal products nec	1.682377e+02
Raw milk	3.126695e+01
Wool, silk-worm cocoons	3.055607e-04
Manure (conventional treatment)	0.000000e+00
Manure (biogas treatment)	0.000000e+00
Products of forestry, logging and related servi...	8.427092e+04
Fish and other fishing products; services incid...	3.863772e+02
Anthracite	1.078962e-04
Coking Coal	4.419943e-01
Other Bituminous Coal	4.130683e+00
Sub-Bituminous Coal	6.719669e-04
Patent Fuel	3.521535e-08
Lignite/Brown Coal	2.283058e-05
BKB/Peat Briquettes	2.207442e-07
Peat	3.809134e+00
Crude petroleum and services related to crude o...	0.000000e+00
Natural gas and services related to natural gas...	2.923347e+01
Natural Gas Liquids	0.000000e+00
...	...
Research and development services (73)	4.518658e+02
Other business services (74)	1.144656e+03
Public administration and defence services; com...	4.676813e+03
Education services (80)	4.951641e+03
Health and social work services (85)	8.285974e+03
Food waste for treatment: incineration	0.000000e+00
Paper waste for treatment: incineration	0.000000e+00
Plastic waste for treatment: incineration	0.000000e+00
Intert/metal waste for treatment: incineration	0.000000e+00
Textiles waste for treatment: incineration	0.000000e+00
Wood waste for treatment: incineration	0.000000e+00
Oil/hazardous waste for treatment: incineration	0.000000e+00
Food waste for treatment: biogasification and l...	0.000000e+00
Paper waste for treatment: biogasification and ...	0.000000e+00
Sewage sludge for treatment: biogasification an...	0.000000e+00
Food waste for treatment: composting and land a...	0.000000e+00
Paper and wood waste for treatment: composting ...	0.000000e+00
Food waste for treatment: waste water treatment	0.000000e+00
Other waste for treatment: waste water treatment	0.000000e+00
Food waste for treatment: landfill	0.000000e+00
Paper for treatment: landfill	0.000000e+00
Plastic waste for treatment: landfill	0.000000e+00
Inert/metal/hazardous waste for treatment: land...	0.000000e+00
Textiles waste for treatment: landfill	0.000000e+00
Wood waste for treatment: landfill	0.000000e+00
Membership organisation services n.e.c. (91)	5.950826e+03
Recreational, cultural and sporting services (92)	3.919332e+03
Other services (93)	1.357583e+03
Private households with employed persons (95)	1.269133e+00

```
Extra-territorial organizations and bodies                  0.000000e+00

                                                   Material footprint [kt]  \
sector
Paddy rice                                                      2.334721e-01
Wheat                                                           1.234295e+03
Cereal grains nec                                               1.719641e+02
Vegetables, fruit, nuts                                         4.115869e+03
Oil seeds                                                       5.764728e+02
Sugar cane, sugar beet                                          3.899792e+00
Plant-based fibers                                              6.882919e-04
Crops nec                                                       4.346631e+02
Cattle                                                          6.266705e+02
Pigs                                                            1.156021e+02
Poultry                                                         7.598228e+02
Meat animals nec                                                1.241676e+03
Animal products nec                                             5.992581e+01
Raw milk                                                        7.154615e+00
Wool, silk-worm cocoons                                         5.517894e-04
Manure (conventional treatment)                                 0.000000e+00
Manure (biogas treatment)                                       0.000000e+00
Products of forestry, logging and related servi...              1.238077e+04
Fish and other fishing products; services incid...              2.386407e+02
Anthracite                                                      2.667174e-02
Coking Coal                                                     4.366926e+01
Other Bituminous Coal                                           2.264521e+02
Sub-Bituminous Coal                                             4.841874e-01
Patent Fuel                                                     8.809831e-08
Lignite/Brown Coal                                              4.872934e-03
BKB/Peat Briquettes                                             3.160311e-07
Peat                                                            1.722028e+01
Crude petroleum and services related to crude o...              0.000000e+00
Natural gas and services related to natural gas...              4.791725e+02
Natural Gas Liquids                                             0.000000e+00
...                                                                      ...
Research and development services (73)                          2.652802e+02
Other business services (74)                                    1.006760e+03
Public administration and defence services; com...              5.090749e+03
Education services (80)                                         3.594597e+03
Health and social work services (85)                            6.158870e+03
Food waste for treatment: incineration                          0.000000e+00
Paper waste for treatment: incineration                         0.000000e+00
Plastic waste for treatment: incineration                       0.000000e+00
Intert/metal waste for treatment: incineration                  0.000000e+00
Textiles waste for treatment: incineration                      0.000000e+00
Wood waste for treatment: incineration                          0.000000e+00
Oil/hazardous waste for treatment: incineration                 0.000000e+00
Food waste for treatment: biogasification and l...              0.000000e+00
```

Paper waste for treatment: biogasification and ...	0.000000e+00
Sewage sludge for treatment: biogasification an...	0.000000e+00
Food waste for treatment: composting and land a...	0.000000e+00
Paper and wood waste for treatment: composting ...	0.000000e+00
Food waste for treatment: waste water treatment	0.000000e+00
Other waste for treatment: waste water treatment	0.000000e+00
Food waste for treatment: landfill	0.000000e+00
Paper for treatment: landfill	0.000000e+00
Plastic waste for treatment: landfill	0.000000e+00
Inert/metal/hazardous waste for treatment: land...	0.000000e+00
Textiles waste for treatment: landfill	0.000000e+00
Wood waste for treatment: landfill	0.000000e+00
Membership organisation services n.e.c. (91)	4.787108e+03
Recreational, cultural and sporting services (92)	2.580996e+03
Other services (93)	6.818344e+02
Private households with employed persons (95)	2.673500e+00
Extra-territorial organizations and bodies	0.000000e+00

sector	Water footprint [Mm3]
Paddy rice	1.015887e-01
Wheat	5.769528e+01
Cereal grains nec	4.018649e-01
Vegetables, fruit, nuts	2.586223e+02
Oil seeds	1.434237e+01
Sugar cane, sugar beet	1.076404e-02
Plant-based fibers	5.000705e-06
Crops nec	1.235689e+02
Cattle	3.939932e+00
Pigs	1.689726e+00
Poultry	4.016469e+00
Meat animals nec	4.297415e+00
Animal products nec	1.583188e+00
Raw milk	5.469038e-02
Wool, silk-worm cocoons	4.997246e-06
Manure (conventional treatment)	0.000000e+00
Manure (biogas treatment)	0.000000e+00
Products of forestry, logging and related servi...	3.170483e+00
Fish and other fishing products; services incid...	1.295690e+00
Anthracite	2.077617e-06
Coking Coal	4.218110e-03
Other Bituminous Coal	4.593687e-02
Sub-Bituminous Coal	7.408850e-06
Patent Fuel	2.188845e-10
Lignite/Brown Coal	4.868900e-07
BKB/Peat Briquettes	2.422454e-09
Peat	3.911056e-03
Crude petroleum and services related to crude o...	0.000000e+00

```
Natural gas and services related to natural gas...          6.455364e-02
Natural Gas Liquids                                          0.000000e+00
...                                                                   ...
Research and development services (73)                       9.053517e+00
Other business services (74)                                 1.470276e+01
Public administration and defence services; com...           4.299405e+01
Education services (80)                                      3.729566e+01
Health and social work services (85)                         7.640824e+01
Food waste for treatment: incineration                       0.000000e+00
Paper waste for treatment: incineration                      0.000000e+00
Plastic waste for treatment: incineration                    0.000000e+00
Intert/metal waste for treatment: incineration               0.000000e+00
Textiles waste for treatment: incineration                   0.000000e+00
Wood waste for treatment: incineration                       0.000000e+00
Oil/hazardous waste for treatment: incineration              0.000000e+00
Food waste for treatment: biogasification and l...           0.000000e+00
Paper waste for treatment: biogasification and ...           0.000000e+00
Sewage sludge for treatment: biogasification an...           0.000000e+00
Food waste for treatment: composting and land a...           0.000000e+00
Paper and wood waste for treatment: composting ...           0.000000e+00
Food waste for treatment: waste water treatment              0.000000e+00
Other waste for treatment: waste water treatment             0.000000e+00
Food waste for treatment: landfill                           0.000000e+00
Paper for treatment: landfill                                0.000000e+00
Plastic waste for treatment: landfill                        0.000000e+00
Inert/metal/hazardous waste for treatment: land...           0.000000e+00
Textiles waste for treatment: landfill                       0.000000e+00
Wood waste for treatment: landfill                           0.000000e+00
Membership organisation services n.e.c. (91)                 3.830950e+01
Recreational, cultural and sporting services (92)            3.244526e+01
Other services (93)                                          5.181718e+00
Private households with employed persons (95)                2.147139e-02
Extra-territorial organizations and bodies                   0.000000e+00

[200 rows x 4 columns]
```

In [46]:
```
# Grouping the environmental footprints as a total per region
World_Overview = stressor_aggr_indicator.groupby(level=0).sum()
World_Overview
```

Out[46]:
```
        Carbon Footprint [t CO2-eq.]  Land footprint [km2]  \
region
EU                      5.069025e+09          6.393946e+06
RoW                     3.238914e+10          5.797612e+07
SE                      9.426935e+07          3.330935e+05

        Material footprint [kt]  Water footprint [Mm3]
region
```

```
EU                  1.012672e+07          1.096098e+05
RoW                 6.806164e+07          1.048253e+06
SE                  2.791771e+05          1.998759e+03
```

4.3 RO 1.3 Visualisation of Swedish Consumption-based Emissions

The results of RO 1.0 shall be visualised in different ways. The first try is a nested treemap of all the regions, showing the total of all the four environmental footprints in four subplots.

```
In [47]: # Preperation of the data needed. All data saved in the according variable.

         index = [SE]

         # Extracting the data out of the satellite accounts from Exiobase
         D_imp = SE_aggr.satellite.D_imp.SE
         D_exp = SE_aggr.satellite.D_exp.SE
         D_cba = SE_aggr.satellite.D_cba.SE
         D_pba = SE_aggr.satellite.D_pba.SE

         # Multiply the data with characterisation file, transposing is needed. Result are the
         D_imp_chr = D_imp.T.dot(characterisation.T)
         D_exp_chr = D_exp.T.dot(characterisation.T)
         D_cba_chr = D_cba.T.dot(characterisation.T)
         D_pba_chr = D_pba.T.dot(characterisation.T)

         # Extracting the correct row from each of the variables to receive the value needed.

         # For the Carbon footprint
         Footprint_gwp = D_cba_chr.iloc[:,0].sum()
         Territorial_gwp = D_pba_chr.iloc[:,0].sum()
         Imports_gwp = D_imp_chr.iloc[:,0].sum()
         Exports_gwp = D_exp_chr.iloc[:,0].sum()
         df_carbon = pd.DataFrame({'Footprint': Footprint_gwp, 'Territorial': Territorial_gwp,

         # For the Land footprint
         Footprint_land = D_cba_chr.iloc[:,1].sum()
         Territorial_land = D_pba_chr.iloc[:,1].sum()
         Imports_land = D_imp_chr.iloc[:,1].sum()
         Exports_land = D_exp_chr.iloc[:,1].sum()
         df_land = pd.DataFrame({'Footprint': Footprint_land, 'Territorial': Territorial_land,

         # For the Material footprint
         Footprint_material = D_cba_chr.iloc[:,2].sum()
         Territorial_material = D_pba_chr.iloc[:,2].sum()
         Imports_material = D_imp_chr.iloc[:,2].sum()
         Exports_material = D_exp_chr.iloc[:,2].sum()
         df_material = pd.DataFrame({'Footprint': Footprint_material, 'Territorial': Territori
```

```
# For the Water footprint
Footprint_water = D_cba_chr.iloc[:,3].sum()
Territorial_water = D_pba_chr.iloc[:,3].sum()
Imports_water = D_imp_chr.iloc[:,3].sum()
Exports_water = D_exp_chr.iloc[:,3].sum()
df_water = pd.DataFrame({'Footprint': Footprint_water, 'Territorial': Territorial_wate

# Checking the results:

Footprint_gwp == Territorial_gwp+Imports_gwp-Exports_gwp
Footprint_gwp
```

Out[47]: 94269345.61441697

In [48]:
```
#Creating four subplots (Carbon footprint, Land footprint, Material footprint and Wat

accounts = ['Footprint','Territorial','Imports', 'Exports']
position = [0,1,2,3]

fig = plt.figure()

plt.subplot(2, 2, 1)
plt.bar(x=position, height = [Footprint_gwp, Territorial_gwp, Imports_gwp, Exports_gwp
plt.title('Carbon footprint',fontsize=14,fontname='STIXGeneral')
plt.ylabel('[t CO2-eq.]',fontname='STIXGeneral')
plt.xticks(position, accounts, fontname='STIXGeneral')
plt.ticklabel_format(axis='y',style='plain')

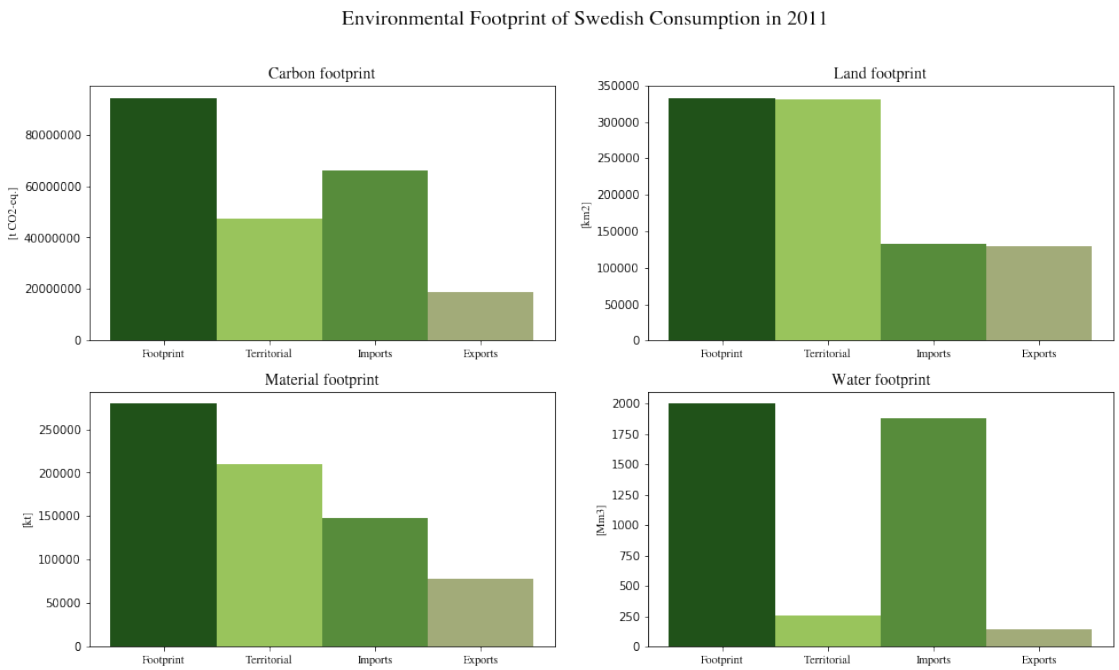

plt.subplot(2, 2, 2)
plt.bar(x=position, height = [Footprint_land, Territorial_land, Imports_land, Exports_
plt.title('Land footprint',fontsize=14,fontname='STIXGeneral')
plt.ylabel('[km2]',fontname='STIXGeneral')
plt.xticks(position, accounts,fontname='STIXGeneral')

plt.subplot(2, 2, 3)
plt.bar(x=position, height = [Footprint_material, Territorial_material, Imports_materi
plt.title('Material footprint',fontsize=14,fontname='STIXGeneral')
plt.ylabel('[kt]',fontname='STIXGeneral')
plt.xticks(position, accounts,fontname='STIXGeneral')

plt.subplot(2, 2, 4)
plt.bar(x=position, height = [Footprint_water, Territorial_water, Imports_water, Expo
plt.title('Water footprint',fontsize=14,fontname='STIXGeneral')
plt.ylabel('[Mm3]',fontname='STIXGeneral')
plt.xticks(position, accounts,fontname='STIXGeneral')

# # Create the legend
```

```
# fig.legend(loc="center right",   # Position of legend
#            borderaxespad=0.1,    # Small spacing around legend box
#            title="Legend"  # Title for the legend
#            )
fig.suptitle('Environmental Footprint of Swedish Consumption in 2011',fontsize=18, fo
# Adjust the scaling factor to fit your legend text completely outside the plot
# (smaller value results in more space being made for the legend)
plt.subplots_adjust(right=0.90)
plt.show()
```



This code constructes normalised stacked barcharts withthe four enironmental footprints for the whole exiobase, aggregate to the three regions SE,EU & RoW.

```
In [49]: d = {'Carbon footprint': [Footprint_gwp, Territorial_gwp,Imports_gwp,Exports_gwp], 'La
         df = pd.DataFrame(data=d,index=['Footprint','Territorial','Imports','Exports'])
         df
```

```
Out[49]:              Carbon footprint  Land footprint  Material Footprint  \
         Footprint        9.426935e+07   333093.492886       279177.080463
         Territorial      4.711316e+07   330214.788647       209545.575857
         Imports          6.593233e+07   132004.223123       147767.010839
         Exports          1.877614e+07   129125.518884        78135.506233

                       Water footprint
         Footprint         1998.758571
         Territorial        257.842298
         Imports           1883.966173
         Exports            143.049900
```

5 RO 2.0 Develop a methodological framework to link Swedish household expenditure surveys to environmentally extended input-output tables to analysis the environmental impacts of Swedish Households

5.1 RO 2.1 Adjusting Household Budget Survey

Read the .csv file with the Swedish average expenditure data and the tax multiplier

In [50]:
```
Swedish_HBS = pd.read_csv('C:/Users/tobia/Dropbox/KTH Sustainable Technology/4_AL250X_
# Command for display all columns and rows
#with pd.option_context('display.max_rows', None, 'display.max_columns', None):
 #   display(Swedish_HBS)
Swedish_HBS
```

Out[50]:
```
Division                                                    Group
Food                                                        Bread and cereals
                                                            Meat
                                                            Fish and seafood
                                                            Milk, cheese and eggs
...
Housing                                                     Rental for housing (incl. garage)
```

```
                                                        Maintenance
                                                        Household's insurance
                                                        Water supply and other services
                                                        Electricity, gas and other fuels
                                                        Interests rate (gross)
                                                        Real estate fee
                                                        Rent for a leasehold site
Furnishings, household equipment and maintenanc... Furniture and carpets
                                                        Textiles
                                                        Household appliance
Health and hospital services                            NaN
Transport                                               Purchase of motorcars
                                                        Purchase of other vehicles
                                                        Operation of motorcars
                                                        Interests & vehicle tax, motorcars
                                                        Operation of other vehicles
                                                        Interests & vehicle tax, other vehi
                                                        Local travel, transportation servic
Leisure and culture                                     Weekend cottage
                                                        Radio and tv
                                                        Sport and hobby
                                                        Watch, camera, photographic and cin
                                                        Vacation, hotels
                                                        Other leisure
                                                        Entertainment
                                                        Books, newspapers, television fees
                                                        Mobil phones
                                                        Telephone
Taxable benefits                                        NaN

[120 rows x 1 columns]
```

In [51]:
```
# Multiplying the HBS_final with the vector of 1.17 to address the problem of underre

Swedish_HBS_ur = Swedish_HBS*1.17
Swedish_HBS_ur
```

Out[51]:
```
Division                                                Group
Food                                                    Bread and cereals
```

	Meat
	Fish and seafood
	Milk, cheese and eggs
...	
Housing	Rental for housing (incl. garage)
	Maintenance
	Household's insurance
	Water supply and other services
	Electricity, gas and other fuels
	Interests rate (gross)
	Real estate fee
	Rent for a leasehold site
Furnishings, household equipment and maintenanc...	Furniture and carpets
	Textiles
	Household appliance
Health and hospital services	NaN
Transport	Purchase of motorcars
	Purchase of other vehicles
	Operation of motorcars
	Interests & vehicle tax, motorcars
	Operation of other vehicles
	Interests & vehicle tax, other vehi
	Local travel, transportation servic
Leisure and culture	Weekend cottage
	Radio and tv
	Sport and hobby
	Watch, camera, photographic and cin
	Vacation, hotels
	Other leisure
	Entertainment

```
                                                    Books, newspapers, television fees
                                                    Mobil phones
                                                    Telephone
Taxable benefits                                    NaN

[120 rows x 1 columns]
```

In [52]:
```
# Importing the vector out of the tax_multiplier.csv file
Tax_Multiplier = pd.read_csv('C:/Users/tobia/Dropbox/KTH Sustainable Technology/4_AL25
Tax_Multiplier
```

Out[52]:
```
Division                                            Group
Food                                                Bread and cereals
```

```
                                                    Meat
```

```
                                                    Fish and seafood
```

```
                                                    Milk, cheese and eggs
...
Housing                                             Rental for housing (incl. garage)
                                                    Maintenance
                                                    Household's insurance
                                                    Water supply and other services
```

```
                                                        Electricity, gas and other fuels
                                                        Interests rate (gross)
                                                        Real estate fee
                                                        Rent for a leasehold site
Furnishings, household equipment and maintenanc... Furniture and carpets
                                                        Textiles
                                                        Household appliance
Health and hospital services                            NaN
Transport                                               Purchase of motorcars
                                                        Purchase of other vehicles
                                                        Operation of motorcars
                                                        Interests & vehicle tax, motorcars
                                                        Operation of other vehicles
                                                        Interests & vehicle tax, other vehi
                                                        Local travel, transportation servic
Leisure and culture                                     Weekend cottage
                                                        Radio and tv
                                                        Sport and hobby
                                                        Watch, camera, photographic and cin
                                                        Vacation, hotels
                                                        Other leisure
                                                        Entertainment
                                                        Books, newspapers, television fees
                                                        Mobil phones
                                                        Telephone
Taxable benefits                                        NaN

[120 rows x 1 columns]
```

In [53]:
```
# Multiply the Swedish_HBS with the average exchange rate from SEK to EUR: 0.10218
Swedish_HBS_Euro = Swedish_HBS_ur.multiply(0.10218)
Swedish_HBS_Euro
```

Out[53]:
```
Division                                                Group
Food                                                    Bread and cereals
```

```
                                                        Meat
```

	Fish and seafood
	Milk, cheese and eggs
...	
Housing	Rental for housing (incl. garage)
	Maintenance
	Household's insurance
	Water supply and other services
	Electricity, gas and other fuels
	Interests rate (gross)
	Real estate fee
	Rent for a leasehold site
Furnishings, household equipment and maintenanc...	Furniture and carpets
	Textiles
	Household appliance
Health and hospital services	NaN
Transport	Purchase of motorcars
	Purchase of other vehicles
	Operation of motorcars
	Interests & vehicle tax, motorcars
	Operation of other vehicles
	Interests & vehicle tax, other vehi
	Local travel, transportation servic
Leisure and culture	Weekend cottage
	Radio and tv
	Sport and hobby
	Watch, camera, photographic and cin
	Vacation, hotels
	Other leisure
	Entertainment
	Books, newspapers, television fees
	Mobil phones
	Telephone
Taxable benefits	NaN

```
[120 rows x 1 columns]
```

In [54]:
```
# Adjusting the Swedish HBS in Euros to inflation, according to the CPI: 1.05
Swedish_HBS_Euro_adjusted = Swedish_HBS_Euro.multiply(1.05)
Swedish_HBS_Euro_adjusted
```

Out[54]:
```
Division                                                Group
Food                                                    Bread and cereals
                                                        Meat
                                                        Fish and seafood
                                                        Milk, cheese and eggs
...
Housing                                                 Rental for housing (incl. garage)
                                                        Maintenance
                                                        Household's insurance
                                                        Water supply and other services
                                                        Electricity, gas and other fuels
                                                        Interests rate (gross)
                                                        Real estate fee
                                                        Rent for a leasehold site
```

```
Furnishings, household equipment and maintenanc... Furniture and carpets
                                                   Textiles
                                                   Household appliance
Health and hospital services                       NaN
Transport                                          Purchase of motorcars
                                                   Purchase of other vehicles
                                                   Operation of motorcars
                                                   Interests & vehicle tax, motorcars
                                                   Operation of other vehicles
                                                   Interests & vehicle tax, other vehi
                                                   Local travel, transportation servic
Leisure and culture                                Weekend cottage
                                                   Radio and tv
                                                   Sport and hobby
                                                   Watch, camera, photographic and cin
                                                   Vacation, hotels
                                                   Other leisure
                                                   Entertainment
                                                   Books, newspapers, television fees
                                                   Mobil phones
                                                   Telephone
Taxable benefits                                   NaN

[120 rows x 1 columns]
```

In [55]:
```
# Renaming the header of the Swedish_HBS_Euro_adjusted to Expenditure in Euro []
Swedish_HBS_Euro_adjusted.rename(columns={'Expenditure [SEK]':'Expenditure [EUR]'}, in
Swedish_HBS_Euro_adjusted
```

Out[55]:
```
Division                                           Group
Food                                               Bread and cereals
```

```
                                                   Meat
```

```
                                                   Fish and seafood
                                                   Milk, cheese and eggs
...
Housing                                            Rental for housing (incl. garage)
                                                   Maintenance
                                                   Household's insurance
                                                   Water supply and other services
                                                   Electricity, gas and other fuels
                                                   Interests rate (gross)
                                                   Real estate fee
                                                   Rent for a leasehold site
Furnishings, household equipment and maintenanc... Furniture and carpets
                                                   Textiles
                                                   Household appliance
Health and hospital services                       NaN
Transport                                          Purchase of motorcars
                                                   Purchase of other vehicles
                                                   Operation of motorcars
                                                   Interests & vehicle tax, motorcars
                                                   Operation of other vehicles
                                                   Interests & vehicle tax, other vehi
                                                   Local travel, transportation servic
Leisure and culture                                Weekend cottage
                                                   Radio and tv
                                                   Sport and hobby
                                                   Watch, camera, photographic and cin
                                                   Vacation, hotels
                                                   Other leisure
                                                   Entertainment
                                                   Books, newspapers, television fees
                                                   Mobil phones
                                                   Telephone
Taxable benefits                                   NaN

[120 rows x 1 columns]
```

In [56]:
```
# Multipyling the Tax_Multiplier vector with the adjusted HBS in Euro
```

```
Swedish_HBS_final = pd.DataFrame(Swedish_HBS_Euro_adjusted.values*Tax_Multiplier.valu
Swedish_HBS_final.groupby(level=0, sort=False).sum()
```

Out[56]:

Division	Expenditure [EUR]
Food	3719.891639
Non-alcoholic beverages	289.163014
Meals out	1163.376776
Alcoholic beverages (with light beer)	390.643541
Tobacco	201.849233
Consumer goods	582.450523
Household services	1082.554593
Clothing and footwear	1547.806675
Housing	6988.402053
Furnishings, household equipment and maintenanc...	1650.945966
Health and hospital services	812.167001
Transport	4765.644665
Leisure and culture	5755.968395
Taxable benefits	189.798533

Multipyling the adjusted final demand file HBS with the concordance matrix to map all the EXIOBASE products to the Swedish HBS cateogries.

In [57]:
```
concordance = pd.read_csv('C:/Users/tobia/Dropbox/KTH Sustainable Technology/4_AL250X_
concordance.index
concordance.columns
concordance_t =concordance.T
concordance_t
```

Out[57]:

Division	Group
Food	Bread and cereals

	Fish and seafood
	Milk, cheese and eggs
...	
Housing	Rental for housing (incl. garage)
	Maintenance
	Household's insurance
	Water supply and other services
	Electricity, gas and other fuels
	Interests rate (gross)
	Real estate fee
	Rent for a leasehold site
Furnishings, household equipment and maintenanc...	Furniture and carpets
	Textiles
	Household appliance
Health and hospital services	Unnamed: 102_level_1
Transport	Purchase of motorcars
	Purchase of other vehicles
	Operation of motorcars
	Interests & vehicle tax, motorcars
	Operation of other vehicles
	Interests & vehicle tax, other vehi
	Local travel, transportation servic
Leisure and culture	Weekend cottage
	Radio and tv
	Sport and hobby
	Watch, camera, photographic and cin
	Vacation, hotels
	Other leisure
	Entertainment
	Books, newspapers, television fees
	Mobil phones
	Telephone
Taxable benefits	Unnamed: 120_level_1

Division	Group
Food	Bread and cereals

	Meat
	Fish and seafood
	Milk, cheese and eggs
...	
Housing	Rental for housing (incl. garage)
	Maintenance
	Household's insurance
	Water supply and other services
	Electricity, gas and other fuels
	Interests rate (gross)
	Real estate fee
	Rent for a leasehold site
Furnishings, household equipment and maintenanc...	Furniture and carpets
	Textiles
	Household appliance
Health and hospital services	Unnamed: 102_level_1
Transport	Purchase of motorcars
	Purchase of other vehicles
	Operation of motorcars
	Interests & vehicle tax, motorcars
	Operation of other vehicles
	Interests & vehicle tax, other vehi
	Local travel, transportation servic

Leisure and culture	Weekend cottage
	Radio and tv
	Sport and hobby
	Watch, camera, photographic and ci
	Vacation, hotels
	Other leisure
	Entertainment
	Books, newspapers, television fees
	Mobil phones
	Telephone
Taxable benefits	Unnamed: 120_level_1

Division	Group
Food	Bread and cereals
	Meat
	Fish and seafood
	Milk, cheese and eggs
...	
Housing	Rental for housing (incl. garage)
	Maintenance
	Household's insurance

	Water supply and other services
	Electricity, gas and other fuels
	Interests rate (gross)
	Real estate fee
	Rent for a leasehold site
Furnishings, household equipment and maintenanc...	Furniture and carpets
	Textiles
	Household appliance
Health and hospital services	Unnamed: 102_level_1
Transport	Purchase of motorcars
	Purchase of other vehicles
	Operation of motorcars
	Interests & vehicle tax, motorcars
	Operation of other vehicles
	Interests & vehicle tax, other vehi
	Local travel, transportation servic
Leisure and culture	Weekend cottage
	Radio and tv
	Sport and hobby
	Watch, camera, photographic and cin
	Vacation, hotels
	Other leisure
	Entertainment
	Books, newspapers, television fees
	Mobil phones
	Telephone
Taxable benefits	Unnamed: 120_level_1

Division	Group
Food	Bread and cereals
	Meat

	Fish and seafood
	Milk, cheese and eggs
...	
Housing	Rental for housing (incl. garage)
	Maintenance
	Household's insurance
	Water supply and other services
	Electricity, gas and other fuels
	Interests rate (gross)
	Real estate fee
	Rent for a leasehold site
Furnishings, household equipment and maintenanc...	Furniture and carpets
	Textiles
	Household appliance
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Transport	Purchase of motorcars
	Purchase of other vehicles
	Operation of motorcars
	Interests & vehicle tax, motorcars
	Operation of other vehicles
	Interests & vehicle tax, other vehi
	Local travel, transportation servic
Leisure and culture	Weekend cottage
	Radio and tv
	Sport and hobby
	Watch, camera, photographic and cin
	Vacation, hotels
	Other leisure
	Entertainment
	Books, newspapers, television fees
	Mobil phones
	Telephone
Taxable benefits	Unnamed: 120_level_1

Division	Group
Food	Bread and cereals

	Meat
	Fish and seafood
	Milk, cheese and eggs
...	
Housing	Rental for housing (incl. garage)
	Maintenance
	Household's insurance
	Water supply and other services
	Electricity, gas and other fuels
	Interests rate (gross)
	Real estate fee
	Rent for a leasehold site
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	Operation of motorcars
	Interests & vehicle tax, motorcars
	Operation of other vehicles
	Interests & vehicle tax, other vehi
	Local travel, transportation servic

Leisure and culture	Weekend cottage
	Radio and tv
	Sport and hobby
	Watch, camera, photographic and ci
	Vacation, hotels
	Other leisure
	Entertainment
	Books, newspapers, television fees
	Mobil phones
	Telephone
Taxable benefits	Unnamed: 120_level_1

Division	Group
Food	Bread and cereals
	Meat
	Fish and seafood
	Milk, cheese and eggs
...	
Housing	Rental for housing (incl. garage)
	Maintenance
	Household's insurance

	Water supply and other services
	Electricity, gas and other fuels
	Interests rate (gross)
	Real estate fee
	Rent for a leasehold site
Furnishings, household equipment and maintenanc...	Furniture and carpets
	Textiles
	Household appliance
Health and hospital services	Unnamed: 102_level_1
Transport	Purchase of motorcars
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	Operation of motorcars
	Interests & vehicle tax, motorcars
	Operation of other vehicles
	Interests & vehicle tax, other vehi
	Local travel, transportation servic
Leisure and culture	Weekend cottage
	Radio and tv
	Sport and hobby
	Watch, camera, photographic and cin
	Vacation, hotels
	Other leisure
	Entertainment
	Books, newspapers, television fees
	Mobil phones
	Telephone
Taxable benefits	Unnamed: 120_level_1

Division	Group
Food	Bread and cereals
	Meat

	Fish and seafood
	Milk, cheese and eggs
...	
Housing	Rental for housing (incl. garage)
	Maintenance
	Household's insurance
	Water supply and other services
	Electricity, gas and other fuels
	Interests rate (gross)
	Real estate fee
	Rent for a leasehold site
Furnishings, household equipment and maintenanc...	Furniture and carpets
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Health and hospital services	Unnamed: 102_level_1
Transport	Purchase of motorcars
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	Interests & vehicle tax, motorcars
	Operation of other vehicles
	Interests & vehicle tax, other vehi
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Leisure and culture	Weekend cottage
	Radio and tv
	Sport and hobby
	Watch, camera, photographic and cin
	Vacation, hotels
	Other leisure
	Entertainment
	Books, newspapers, television fees
	Mobil phones
	Telephone
Taxable benefits	Unnamed: 120_level_1

Division	Group
Food	Bread and cereals

	Meat
	Fish and seafood
	Milk, cheese and eggs
...	
Housing	Rental for housing (incl. garage)
	Maintenance
	Household's insurance
	Water supply and other services
	Electricity, gas and other fuels
	Interests rate (gross)
	Real estate fee
	Rent for a leasehold site
Furnishings, household equipment and maintenanc...	Furniture and carpets
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	Household appliance
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	Operation of motorcars
	Interests & vehicle tax, motorcars
	Operation of other vehicles
	Interests & vehicle tax, other vehi
	Local travel, transportation servic

Leisure and culture	Weekend cottage
	Radio and tv
	Sport and hobby
	Watch, camera, photographic and ci
	Vacation, hotels
	Other leisure
	Entertainment
	Books, newspapers, television fees
	Mobil phones
	Telephone
Taxable benefits	Unnamed: 120_level_1

Division	Group
Food	Bread and cereals
	Meat
	Fish and seafood
	Milk, cheese and eggs
...	
Housing	Rental for housing (incl. garage)
	Maintenance
	Household's insurance

	Water supply and other services
	Electricity, gas and other fuels
	Interests rate (gross)
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	Rent for a leasehold site
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	Entertainment
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	Telephone
Taxable benefits	Unnamed: 120_level_1

Division	Group
Food	Bread and cereals
	Meat

	Fish and seafood
	Milk, cheese and eggs
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Housing	Rental for housing (incl. garage)
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	Entertainment
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	Mobil phones
	Telephone
Taxable benefits	Unnamed: 120_level_1

Division	Group
Food	Bread and cereals

	Meat
	Fish and seafood
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Housing	Rental for housing (incl. garage)
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Division	Group
Food	Bread and cereals
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	Fish and seafood
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Housing	Rental for housing (incl. garage)
	Maintenance
	Household's insurance

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	Books, newspapers, television fees
	Mobil phones
	Telephone
Taxable benefits	Unnamed: 120_level_1

Division	Group
Food	Bread and cereals
	Meat

	Fish and seafood
	Milk, cheese and eggs
...	
Housing	Rental for housing (incl. garage)
	Maintenance
	Household's insurance
	Water supply and other services
	Electricity, gas and other fuels
	Interests rate (gross)
	Real estate fee
	Rent for a leasehold site
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	Radio and tv
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	Other leisure
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	Books, newspapers, television fees
	Mobil phones
	Telephone
Taxable benefits	Unnamed: 120_level_1

Division	Group
Food	Bread and cereals

	Meat
	Fish and seafood
	Milk, cheese and eggs
...	
Housing	Rental for housing (incl. garage)
	Maintenance
	Household's insurance
	Water supply and other services
	Electricity, gas and other fuels
	Interests rate (gross)
	Real estate fee
	Rent for a leasehold site
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	Textiles
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	Vacation, hotels
	Other leisure
	Entertainment
	Books, newspapers, television fees
	Mobil phones
	Telephone
Taxable benefits	Unnamed: 120_level_1

Division	Group
Food	Bread and cereals
	Meat
	Fish and seafood
	Milk, cheese and eggs
...	
Housing	Rental for housing (incl. garage)
	Maintenance
	Household's insurance

	Water supply and other services
	Electricity, gas and other fuels
	Interests rate (gross)
	Real estate fee
	Rent for a leasehold site
Furnishings, household equipment and maintenanc...	Furniture and carpets
	Textiles
	Household appliance
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Transport	Purchase of motorcars
	Purchase of other vehicles
	Operation of motorcars
	Interests & vehicle tax, motorcars
	Operation of other vehicles
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	Local travel, transportation servic
Leisure and culture	Weekend cottage
	Radio and tv
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	Vacation, hotels
	Other leisure
	Entertainment
	Books, newspapers, television fees
	Mobil phones
	Telephone
Taxable benefits	Unnamed: 120_level_1

Division	Group
Food	Bread and cereals
	Meat

	Fish and seafood
	Milk, cheese and eggs
...	
Housing	Rental for housing (incl. garage)
	Maintenance
	Household's insurance
	Water supply and other services
	Electricity, gas and other fuels
	Interests rate (gross)
	Real estate fee
	Rent for a leasehold site
Furnishings, household equipment and maintenanc...	Furniture and carpets
	Textiles
	Household appliance
Health and hospital services	Unnamed: 102_level_1
Transport	Purchase of motorcars
	Purchase of other vehicles
	Operation of motorcars
	Interests & vehicle tax, motorcars
	Operation of other vehicles
	Interests & vehicle tax, other vehi
	Local travel, transportation servic
Leisure and culture	Weekend cottage
	Radio and tv
	Sport and hobby
	Watch, camera, photographic and cin
	Vacation, hotels
	Other leisure
	Entertainment
	Books, newspapers, television fees
	Mobil phones
	Telephone
Taxable benefits	Unnamed: 120_level_1

Division	Group
Food	Bread and cereals

	Meat
	Fish and seafood
	Milk, cheese and eggs
...	
Housing	Rental for housing (incl. garage)
	Maintenance
	Household's insurance
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	Electricity, gas and other fuels
	Interests rate (gross)
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	Rent for a leasehold site
Furnishings, household equipment and maintenanc...	Furniture and carpets
	Textiles
	Household appliance
Health and hospital services	Unnamed: 102_level_1
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	Operation of motorcars
	Interests & vehicle tax, motorcars
	Operation of other vehicles
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	Local travel, transportation servic

Leisure and culture	Weekend cottage
	Radio and tv
	Sport and hobby
	Watch, camera, photographic and ci
	Vacation, hotels
	Other leisure
	Entertainment
	Books, newspapers, television fees
	Mobil phones
	Telephone
Taxable benefits	Unnamed: 120_level_1

Division	Group
Food	Bread and cereals
	Meat
	Fish and seafood
	Milk, cheese and eggs
...	
Housing	Rental for housing (incl. garage)
	Maintenance
	Household's insurance

	Water supply and other services
	Electricity, gas and other fuels
	Interests rate (gross)
	Real estate fee
	Rent for a leasehold site
Furnishings, household equipment and maintenanc...	Furniture and carpets
	Textiles
	Household appliance
Health and hospital services	Unnamed: 102_level_1
Transport	Purchase of motorcars
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	Operation of motorcars
	Interests & vehicle tax, motorcars
	Operation of other vehicles
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	Local travel, transportation servic
Leisure and culture	Weekend cottage
	Radio and tv
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	Other leisure
	Entertainment
	Books, newspapers, television fees
	Mobil phones
	Telephone
Taxable benefits	Unnamed: 120_level_1

Division	Group
Food	Bread and cereals
	Meat

	Fish and seafood
	Milk, cheese and eggs
...	
Housing	Rental for housing (incl. garage)
	Maintenance
	Household's insurance
	Water supply and other services
	Electricity, gas and other fuels
	Interests rate (gross)
	Real estate fee
	Rent for a leasehold site
Furnishings, household equipment and maintenanc...	Furniture and carpets
	Textiles
	Household appliance
Health and hospital services	Unnamed: 102_level_1
Transport	Purchase of motorcars
	Purchase of other vehicles
	Operation of motorcars
	Interests & vehicle tax, motorcars
	Operation of other vehicles
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	Local travel, transportation servic
Leisure and culture	Weekend cottage
	Radio and tv
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	Vacation, hotels
	Other leisure
	Entertainment
	Books, newspapers, television fees
	Mobil phones
	Telephone
Taxable benefits	Unnamed: 120_level_1

Division	Group
Food	Bread and cereals

	Meat
	Fish and seafood
	Milk, cheese and eggs
...	
Housing	Rental for housing (incl. garage)
	Maintenance
	Household's insurance
	Water supply and other services
	Electricity, gas and other fuels
	Interests rate (gross)
	Real estate fee
	Rent for a leasehold site
Furnishings, household equipment and maintenanc...	Furniture and carpets
	Textiles
	Household appliance
Health and hospital services	Unnamed: 102_level_1
Transport	Purchase of motorcars
	Purchase of other vehicles
	Operation of motorcars
	Interests & vehicle tax, motorcars
	Operation of other vehicles
	Interests & vehicle tax, other vehi
	Local travel, transportation servic

Leisure and culture	Weekend cottage
	Radio and tv
	Sport and hobby
	Watch, camera, photographic and ci
	Vacation, hotels
	Other leisure
	Entertainment
	Books, newspapers, television fees
	Mobil phones
	Telephone
Taxable benefits	Unnamed: 120_level_1

Division	Group
Food	Bread and cereals
	Meat
	Fish and seafood
	Milk, cheese and eggs
...	
Housing	Rental for housing (incl. garage)
	Maintenance
	Household's insurance

```
                                                    Water supply and other services
                                                    Electricity, gas and other fuels
                                                    Interests rate (gross)
                                                    Real estate fee
                                                    Rent for a leasehold site
Furnishings, household equipment and maintenanc... Furniture and carpets
                                                    Textiles
                                                    Household appliance
Health and hospital services                        Unnamed: 102_level_1
Transport                                           Purchase of motorcars
                                                    Purchase of other vehicles
                                                    Operation of motorcars
                                                    Interests & vehicle tax, motorcars
                                                    Operation of other vehicles
                                                    Interests & vehicle tax, other vehi
                                                    Local travel, transportation servic
Leisure and culture                                 Weekend cottage
                                                    Radio and tv
                                                    Sport and hobby
                                                    Watch, camera, photographic and cin
                                                    Vacation, hotels
                                                    Other leisure
                                                    Entertainment
                                                    Books, newspapers, television fees
                                                    Mobil phones
                                                    Telephone
Taxable benefits                                    Unnamed: 120_level_1

[120 rows x 200 columns]
```

5.2 2.2 RO2: Adjusting the Multiplier File

Adjusting the multiplier file M.txt according to the share of the products and its contribution to the final demand Y.txt

```
In [58]: # Isolating the Swedish Final Demand for Households
         FD_SE=SE_aggr.Y.iloc[:,0]
         FD_SE

         # Group the Final Demand on the Index level 1 (Products) and sum them up.

         FD_SE_Tot=FD_SE.groupby(level=1, sort=False).sum()
         FD_SE_Tot
         #Check if adding the wheat by grouping and ccompare with the FD_SE_Tot:
         SE_aggr.Y.iloc[[1,201,401],0]

         print('The sum of the final demand of wheat is the same:', SE_aggr.Y.iloc[[1,201,401]
```

```
# Another way of checking if all the results are correct

Check = 0
for product in range (0,200):
    if FD_SE_Tot.iloc[product] == SE_aggr.Y.iloc[[product, (200+product), (400+product
        Check+=1
        continue
Check
if Check == 200:print('The aggregation worked!')
```

```
The sum of the final demand of wheat is the same: True
The aggregation worked!
```

After summing up the Final demand file, a new file `Y_ratio` was created with the fraction of the final demand from SE, EU, or Row.

```
In [59]: # Multiply the FD_SE_Tot_600 times 3 and create a dataframe with 600 products and the

         FD_SE_Tot_600 = pd.concat(3*[FD_SE_Tot])
         # Test if the products
             # FD_SE_Tot_600
             # FD_SE_Tot_600[0]==FD_SE_Tot_600[200] == FD_SE_Tot_600[400]
         FD_SE_Tot_600.index = FD_SE.index

         Y_Ratios = FD_SE.div(FD_SE_Tot_600)
         Y_Ratios[0]+Y_Ratios[200]+Y_Ratios[400]

         Y_Ratios
```

```
Out[59]: region  sector
         SE      Paddy rice
                 Wheat
                 Cereal grains nec
                 Vegetables, fruit, nuts
                 Oil seeds
                 Sugar cane, sugar beet
                 Plant-based fibers
                 Crops nec
                 Cattle
                 Pigs
                 Poultry
                 Meat animals nec
                 Animal products nec
                 Raw milk
                 Wool, silk-worm cocoons
                 Manure (conventional treatment)
                 Manure (biogas treatment)
                 Products of forestry, logging and related services (02)
```

```
            Fish and other fishing products; services incidental of fishing (05)
            Anthracite
            Coking Coal
            Other Bituminous Coal
            Sub-Bituminous Coal
            Patent Fuel
            Lignite/Brown Coal
            BKB/Peat Briquettes
            Peat
            Crude petroleum and services related to crude oil extraction, excluding surve
            Natural gas and services related to natural gas extraction, excluding surveyi
            Natural Gas Liquids

RoW         Research and development services (73)
            Other business services (74)
            Public administration and defence services; compulsory social security servic
            Education services (80)
            Health and social work services (85)
            Food waste for treatment: incineration
            Paper waste for treatment: incineration
            Plastic waste for treatment: incineration
            Intert/metal waste for treatment: incineration
            Textiles waste for treatment: incineration
            Wood waste for treatment: incineration
            Oil/hazardous waste for treatment: incineration
            Food waste for treatment: biogasification and land application
            Paper waste for treatment: biogasification and land application
            Sewage sludge for treatment: biogasification and land application
            Food waste for treatment: composting and land application
            Paper and wood waste for treatment: composting and land application
            Food waste for treatment: waste water treatment
            Other waste for treatment: waste water treatment
            Food waste for treatment: landfill
            Paper for treatment: landfill
            Plastic waste for treatment: landfill
            Inert/metal/hazardous waste for treatment: landfill
            Textiles waste for treatment: landfill
            Wood waste for treatment: landfill
            Membership organisation services n.e.c. (91)
            Recreational, cultural and sporting services (92)
            Other services (93)
            Private households with employed persons (95)
            Extra-territorial organizations and bodies
Name: (SE, Final consumption expenditure by households), Length: 600, dtype: float64
```

In [60]:
```
# Filling missing values with 0

Y_Ratios = Y_Ratios.fillna(0)
```

```
        #Test if it worked
            #Check if it adds up to 1
            #Y_Ratios[0]+Y_Ratios[200]+Y_Ratios[400]

    # Compare the different products and check if the sum up correctily

    Ones = 0
    Zeroes = 0
    Almost = 0

    for product in range(0,200):
        if Y_Ratios[[product,(200+product),(400+product)]].sum() == 1:
            Ones+=1
        elif Y_Ratios[[product,(200+product),(400+product)]].sum() == 0:
            if concordance.T.sum()[product] !=0:
                print(str(Y_Ratios[[product,(200+product),(400+product)]])+ "\n")
                print( "There's a problem with the " + str(concordance.index[product])+" I
               #print ("It's used in " + str(concordance.T.index[product]))
                print("It accounts for " + str(concordance.T.sum()[product])+ " of one HBS
            Zeroes+=1
        elif Y_Ratios[[product,(200+product),(400+product)]].sum() != 1:
            #print(Y_Ratios[[product,(200+product),(400+product)]].sum())
            #print(Y_Ratios[[product,(200+product),(400+product)]])
            Almost+=1

    print("\n" + str(Ones) + " Products that add up to 1\n")
    print(str(Zeroes) + " Products are unconsumed by households (add up to 0)\n")
    print(str(Almost) + " Products are close to 1, but have a rounding error.\n")
```

```
91 Products that add up to 1

84 Products are unconsumed by households (add up to 0)

25 Products are close to 1, but have a rounding error.
```

The errors occuring above are used EXIOBASE products with no Final Demand for Households in Sweden and therefore no Multiplier.

```
region  sector
SE      Plant-based fibers    0.0
EU      Plant-based fibers    0.0
RoW     Plant-based fibers    0.0
Name: (SE, Final consumption expenditure by households), dtype: float64

There's a problem with the Plant-based fibers Exiobase product:
It accounts for 0.30000000000000004 of one HBS product or more
```

Plant-based fibres replaced with `paper and paper products`

```
region  sector
SE      Plastics, basic    0.0
EU      Plastics, basic    0.0
RoW     Plastics, basic    0.0
Name: (SE, Final consumption expenditure by households), dtype: float64

There's a problem with the Plastics, basic Exiobase product:
It accounts for 0.5 of one HBS product or more
```

`Plastics, basic` is replaced with `Rubber and plastic products (25)`

```
region  sector
SE      Collected and purified water, distribution services of water (41)    0.0
EU      Collected and purified water, distribution services of water (41)    0.0
RoW     Collected and purified water, distribution services of water (41)    0.0
Name: (SE, Final consumption expenditure by households), dtype: float64

There's a problem with the Collected and purified water, distribution services of water (41) E
It accounts for 1.0 of one HBS product or more
```

`Collected and purified water, distribution services of water (41)` is replaced with `Steam and hot water supply services`

In this step it needs to be checked if the 84 products unconsumed by households, are used in the concordance file. Otherwise it will be multiplied with 0 and end the impacts will result as zero.

After creating the ratios, we can multiply the ratio file with the `M.txt` file to create the adjusted multiplier file. Multiplier file is in 1MEUR so the matrix need to be divided by 0ne million to adjust it to multiplier per Eur.

```
In [61]: M_SE_EUR = SE_aggr.satellite.M.divide(1000000)
         M_SE_adj = M_SE_EUR.multiply(Y_Ratios)
         M_SE_adj.T

         M_SE_adj_grouped = M_SE_adj.T.groupby(level=1,sort=False).sum()
         M_SE_adj_grouped
```

```
Out[61]:                                                    Taxes less subsidies on products p
         sector
         Paddy rice
         Wheat
         Cereal grains nec
         Vegetables, fruit, nuts
         Oil seeds
         Sugar cane, sugar beet
         Plant-based fibers
         Crops nec
         Cattle
         Pigs
```

```
Poultry
Meat animals nec
Animal products nec
Raw milk
Wool, silk-worm cocoons
Manure (conventional treatment)
Manure (biogas treatment)
Products of forestry, logging and related servi...
Fish and other fishing products; services incid...
Anthracite
Coking Coal
Other Bituminous Coal
Sub-Bituminous Coal
Patent Fuel
Lignite/Brown Coal
BKB/Peat Briquettes
Peat
Crude petroleum and services related to crude o...
Natural gas and services related to natural gas...
Natural Gas Liquids
...
Research and development services (73)
Other business services (74)
Public administration and defence services; com...
Education services (80)
Health and social work services (85)
Food waste for treatment: incineration
Paper waste for treatment: incineration
Plastic waste for treatment: incineration
Intert/metal waste for treatment: incineration
Textiles waste for treatment: incineration
Wood waste for treatment: incineration
Oil/hazardous waste for treatment: incineration
Food waste for treatment: biogasification and l...
Paper waste for treatment: biogasification and ...
Sewage sludge for treatment: biogasification an...
Food waste for treatment: composting and land a...
Paper and wood waste for treatment: composting ...
Food waste for treatment: waste water treatment
Other waste for treatment: waste water treatment
Food waste for treatment: landfill
Paper for treatment: landfill
Plastic waste for treatment: landfill
Inert/metal/hazardous waste for treatment: land...
Textiles waste for treatment: landfill
Wood waste for treatment: landfill
Membership organisation services n.e.c. (91)
Recreational, cultural and sporting services (92)
```

```
Other services (93)
Private households with employed persons (95)
Extra-territorial organizations and bodies

                                                        Other net taxes on production  \
sector
Paddy rice                                                                 2.641363e-08
Wheat                                                                     -4.398411e-08
Cereal grains nec                                                         -6.348476e-08
Vegetables, fruit, nuts                                                    3.599368e-09
Oil seeds                                                                 -2.360716e-08
Sugar cane, sugar beet                                                     0.000000e+00
Plant-based fibers                                                         0.000000e+00
Crops nec                                                                  3.915386e-10
Cattle                                                                     0.000000e+00
Pigs                                                                       0.000000e+00
Poultry                                                                   -3.691324e-09
Meat animals nec                                                          -2.315331e-08
Animal products nec                                                       -2.555921e-08
Raw milk                                                                   2.939331e-08
Wool, silk-worm cocoons                                                    0.000000e+00
Manure (conventional treatment)                                            0.000000e+00
Manure (biogas treatment)                                                  0.000000e+00
Products of forestry, logging and related servi...                         2.280529e-08
Fish and other fishing products; services incid...                        2.494743e-08
Anthracite                                                                 2.176570e-08
Coking Coal                                                               -2.541927e-08
Other Bituminous Coal                                                      7.200670e-08
Sub-Bituminous Coal                                                        5.583321e-08
Patent Fuel                                                               -5.633273e-09
Lignite/Brown Coal                                                         3.477513e-08
BKB/Peat Briquettes                                                        1.927770e-08
Peat                                                                       3.585287e-08
Crude petroleum and services related to crude o...                         0.000000e+00
Natural gas and services related to natural gas...                         0.000000e+00
Natural Gas Liquids                                                        0.000000e+00
...                                                                                   ...
Research and development services (73)                                     0.000000e+00
Other business services (74)                                               2.579489e-08
Public administration and defence services; com...                         3.496059e-08
Education services (80)                                                    4.787449e-08
Health and social work services (85)                                       5.228150e-08
Food waste for treatment: incineration                                     0.000000e+00
Paper waste for treatment: incineration                                    0.000000e+00
Plastic waste for treatment: incineration                                  0.000000e+00
Intert/metal waste for treatment: incineration                             0.000000e+00
Textiles waste for treatment: incineration                                 0.000000e+00
Wood waste for treatment: incineration                                     0.000000e+00
```

Oil/hazardous waste for treatment: incineration	0.000000e+00
Food waste for treatment: biogasification and l...	0.000000e+00
Paper waste for treatment: biogasification and ...	0.000000e+00
Sewage sludge for treatment: biogasification an...	0.000000e+00
Food waste for treatment: composting and land a...	0.000000e+00
Paper and wood waste for treatment: composting ...	0.000000e+00
Food waste for treatment: waste water treatment	0.000000e+00
Other waste for treatment: waste water treatment	0.000000e+00
Food waste for treatment: landfill	0.000000e+00
Paper for treatment: landfill	0.000000e+00
Plastic waste for treatment: landfill	0.000000e+00
Inert/metal/hazardous waste for treatment: land...	0.000000e+00
Textiles waste for treatment: landfill	0.000000e+00
Wood waste for treatment: landfill	0.000000e+00
Membership organisation services n.e.c. (91)	2.240339e-08
Recreational, cultural and sporting services (92)	1.713359e-08
Other services (93)	2.979429e-08
Private households with employed persons (95)	4.458611e-08
Extra-territorial organizations and bodies	0.000000e+00

	Compensation of employees; wages,
sector	
Paddy rice	
Wheat	
Cereal grains nec	
Vegetables, fruit, nuts	
Oil seeds	
Sugar cane, sugar beet	
Plant-based fibers	
Crops nec	
Cattle	
Pigs	
Poultry	
Meat animals nec	
Animal products nec	
Raw milk	
Wool, silk-worm cocoons	
Manure (conventional treatment)	
Manure (biogas treatment)	
Products of forestry, logging and related servi...	
Fish and other fishing products; services incid...	
Anthracite	
Coking Coal	
Other Bituminous Coal	
Sub-Bituminous Coal	
Patent Fuel	
Lignite/Brown Coal	
BKB/Peat Briquettes	

Peat
Crude petroleum and services related to crude o...
Natural gas and services related to natural gas...
Natural Gas Liquids
...
Research and development services (73)
Other business services (74)
Public administration and defence services; com...
Education services (80)
Health and social work services (85)
Food waste for treatment: incineration
Paper waste for treatment: incineration
Plastic waste for treatment: incineration
Intert/metal waste for treatment: incineration
Textiles waste for treatment: incineration
Wood waste for treatment: incineration
Oil/hazardous waste for treatment: incineration
Food waste for treatment: biogasification and l...
Paper waste for treatment: biogasification and ...
Sewage sludge for treatment: biogasification an...
Food waste for treatment: composting and land a...
Paper and wood waste for treatment: composting ...
Food waste for treatment: waste water treatment
Other waste for treatment: waste water treatment
Food waste for treatment: landfill
Paper for treatment: landfill
Plastic waste for treatment: landfill
Inert/metal/hazardous waste for treatment: land...
Textiles waste for treatment: landfill
Wood waste for treatment: landfill
Membership organisation services n.e.c. (91)
Recreational, cultural and sporting services (92)
Other services (93)
Private households with employed persons (95)
Extra-territorial organizations and bodies

Compensation of employees; wages,
sector
Paddy rice
Wheat
Cereal grains nec
Vegetables, fruit, nuts
Oil seeds
Sugar cane, sugar beet
Plant-based fibers
Crops nec
Cattle
Pigs

```
Poultry
Meat animals nec
Animal products nec
Raw milk
Wool, silk-worm cocoons
Manure (conventional treatment)
Manure (biogas treatment)
Products of forestry, logging and related servi...
Fish and other fishing products; services incid...
Anthracite
Coking Coal
Other Bituminous Coal
Sub-Bituminous Coal
Patent Fuel
Lignite/Brown Coal
BKB/Peat Briquettes
Peat
Crude petroleum and services related to crude o...
Natural gas and services related to natural gas...
Natural Gas Liquids
...
Research and development services (73)
Other business services (74)
Public administration and defence services; com...
Education services (80)
Health and social work services (85)
Food waste for treatment: incineration
Paper waste for treatment: incineration
Plastic waste for treatment: incineration
Intert/metal waste for treatment: incineration
Textiles waste for treatment: incineration
Wood waste for treatment: incineration
Oil/hazardous waste for treatment: incineration
Food waste for treatment: biogasification and l...
Paper waste for treatment: biogasification and ...
Sewage sludge for treatment: biogasification an...
Food waste for treatment: composting and land a...
Paper and wood waste for treatment: composting ...
Food waste for treatment: waste water treatment
Other waste for treatment: waste water treatment
Food waste for treatment: landfill
Paper for treatment: landfill
Plastic waste for treatment: landfill
Inert/metal/hazardous waste for treatment: land...
Textiles waste for treatment: landfill
Wood waste for treatment: landfill
Membership organisation services n.e.c. (91)
Recreational, cultural and sporting services (92)
```

```
Other services (93)
Private households with employed persons (95)
Extra-territorial organizations and bodies

                                                        Compensation of employees; wages,
sector
Paddy rice
Wheat
Cereal grains nec
Vegetables, fruit, nuts
Oil seeds
Sugar cane, sugar beet
Plant-based fibers
Crops nec
Cattle
Pigs
Poultry
Meat animals nec
Animal products nec
Raw milk
Wool, silk-worm cocoons
Manure (conventional treatment)
Manure (biogas treatment)
Products of forestry, logging and related servi...
Fish and other fishing products; services incid...
Anthracite
Coking Coal
Other Bituminous Coal
Sub-Bituminous Coal
Patent Fuel
Lignite/Brown Coal
BKB/Peat Briquettes
Peat
Crude petroleum and services related to crude o...
Natural gas and services related to natural gas...
Natural Gas Liquids
...
Research and development services (73)
Other business services (74)
Public administration and defence services; com...
Education services (80)
Health and social work services (85)
Food waste for treatment: incineration
Paper waste for treatment: incineration
Plastic waste for treatment: incineration
Intert/metal waste for treatment: incineration
Textiles waste for treatment: incineration
Wood waste for treatment: incineration
```

Oil/hazardous waste for treatment: incineration
Food waste for treatment: biogasification and l...
Paper waste for treatment: biogasification and ...
Sewage sludge for treatment: biogasification an...
Food waste for treatment: composting and land a...
Paper and wood waste for treatment: composting ...
Food waste for treatment: waste water treatment
Other waste for treatment: waste water treatment
Food waste for treatment: landfill
Paper for treatment: landfill
Plastic waste for treatment: landfill
Inert/metal/hazardous waste for treatment: land...
Textiles waste for treatment: landfill
Wood waste for treatment: landfill
Membership organisation services n.e.c. (91)
Recreational, cultural and sporting services (92)
Other services (93)
Private households with employed persons (95)
Extra-territorial organizations and bodies

Operating surplus: Consumption of

sector
Paddy rice
Wheat
Cereal grains nec
Vegetables, fruit, nuts
Oil seeds
Sugar cane, sugar beet
Plant-based fibers
Crops nec
Cattle
Pigs
Poultry
Meat animals nec
Animal products nec
Raw milk
Wool, silk-worm cocoons
Manure (conventional treatment)
Manure (biogas treatment)
Products of forestry, logging and related servi...
Fish and other fishing products; services incid...
Anthracite
Coking Coal
Other Bituminous Coal
Sub-Bituminous Coal
Patent Fuel
Lignite/Brown Coal
BKB/Peat Briquettes

Peat
Crude petroleum and services related to crude o...
Natural gas and services related to natural gas...
Natural Gas Liquids
...
Research and development services (73)
Other business services (74)
Public administration and defence services; com...
Education services (80)
Health and social work services (85)
Food waste for treatment: incineration
Paper waste for treatment: incineration
Plastic waste for treatment: incineration
Intert/metal waste for treatment: incineration
Textiles waste for treatment: incineration
Wood waste for treatment: incineration
Oil/hazardous waste for treatment: incineration
Food waste for treatment: biogasification and l...
Paper waste for treatment: biogasification and ...
Sewage sludge for treatment: biogasification an...
Food waste for treatment: composting and land a...
Paper and wood waste for treatment: composting ...
Food waste for treatment: waste water treatment
Other waste for treatment: waste water treatment
Food waste for treatment: landfill
Paper for treatment: landfill
Plastic waste for treatment: landfill
Inert/metal/hazardous waste for treatment: land...
Textiles waste for treatment: landfill
Wood waste for treatment: landfill
Membership organisation services n.e.c. (91)
Recreational, cultural and sporting services (92)
Other services (93)
Private households with employed persons (95)
Extra-territorial organizations and bodies

	Operating surplus: Rents on land
sector	
Paddy rice	0.0
Wheat	0.0
Cereal grains nec	0.0
Vegetables, fruit, nuts	0.0
Oil seeds	0.0
Sugar cane, sugar beet	0.0
Plant-based fibers	0.0
Crops nec	0.0
Cattle	0.0
Pigs	0.0

Poultry	0.0
Meat animals nec	0.0
Animal products nec	0.0
Raw milk	0.0
Wool, silk-worm cocoons	0.0
Manure (conventional treatment)	0.0
Manure (biogas treatment)	0.0
Products of forestry, logging and related servi...	0.0
Fish and other fishing products; services incid...	0.0
Anthracite	0.0
Coking Coal	0.0
Other Bituminous Coal	0.0
Sub-Bituminous Coal	0.0
Patent Fuel	0.0
Lignite/Brown Coal	0.0
BKB/Peat Briquettes	0.0
Peat	0.0
Crude petroleum and services related to crude o...	0.0
Natural gas and services related to natural gas...	0.0
Natural Gas Liquids	0.0
...	...
Research and development services (73)	0.0
Other business services (74)	0.0
Public administration and defence services; com...	0.0
Education services (80)	0.0
Health and social work services (85)	0.0
Food waste for treatment: incineration	0.0
Paper waste for treatment: incineration	0.0
Plastic waste for treatment: incineration	0.0
Intert/metal waste for treatment: incineration	0.0
Textiles waste for treatment: incineration	0.0
Wood waste for treatment: incineration	0.0
Oil/hazardous waste for treatment: incineration	0.0
Food waste for treatment: biogasification and l...	0.0
Paper waste for treatment: biogasification and ...	0.0
Sewage sludge for treatment: biogasification an...	0.0
Food waste for treatment: composting and land a...	0.0
Paper and wood waste for treatment: composting ...	0.0
Food waste for treatment: waste water treatment	0.0
Other waste for treatment: waste water treatment	0.0
Food waste for treatment: landfill	0.0
Paper for treatment: landfill	0.0
Plastic waste for treatment: landfill	0.0
Inert/metal/hazardous waste for treatment: land...	0.0
Textiles waste for treatment: landfill	0.0
Wood waste for treatment: landfill	0.0
Membership organisation services n.e.c. (91)	0.0
Recreational, cultural and sporting services (92)	0.0

```
Other services (93)                                                    0.0
Private households with employed persons (95)                          0.0
Extra-territorial organizations and bodies                             0.0

                                                   Operating surplus: Royalties on re
sector
Paddy rice
Wheat
Cereal grains nec
Vegetables, fruit, nuts
Oil seeds
Sugar cane, sugar beet
Plant-based fibers
Crops nec
Cattle
Pigs
Poultry
Meat animals nec
Animal products nec
Raw milk
Wool, silk-worm cocoons
Manure (conventional treatment)
Manure (biogas treatment)
Products of forestry, logging and related servi...
Fish and other fishing products; services incid...
Anthracite
Coking Coal
Other Bituminous Coal
Sub-Bituminous Coal
Patent Fuel
Lignite/Brown Coal
BKB/Peat Briquettes
Peat
Crude petroleum and services related to crude o...
Natural gas and services related to natural gas...
Natural Gas Liquids
...
Research and development services (73)
Other business services (74)
Public administration and defence services; com...
Education services (80)
Health and social work services (85)
Food waste for treatment: incineration
Paper waste for treatment: incineration
Plastic waste for treatment: incineration
Intert/metal waste for treatment: incineration
Textiles waste for treatment: incineration
Wood waste for treatment: incineration
```

Oil/hazardous waste for treatment: incineration
Food waste for treatment: biogasification and l...
Paper waste for treatment: biogasification and ...
Sewage sludge for treatment: biogasification an...
Food waste for treatment: composting and land a...
Paper and wood waste for treatment: composting ...
Food waste for treatment: waste water treatment
Other waste for treatment: waste water treatment
Food waste for treatment: landfill
Paper for treatment: landfill
Plastic waste for treatment: landfill
Inert/metal/hazardous waste for treatment: land...
Textiles waste for treatment: landfill
Wood waste for treatment: landfill
Membership organisation services n.e.c. (91)
Recreational, cultural and sporting services (92)
Other services (93)
Private households with employed persons (95)
Extra-territorial organizations and bodies

Operating surplus: Remaining net c

sector
Paddy rice
Wheat
Cereal grains nec
Vegetables, fruit, nuts
Oil seeds
Sugar cane, sugar beet
Plant-based fibers
Crops nec
Cattle
Pigs
Poultry
Meat animals nec
Animal products nec
Raw milk
Wool, silk-worm cocoons
Manure (conventional treatment)
Manure (biogas treatment)
Products of forestry, logging and related servi...
Fish and other fishing products; services incid...
Anthracite
Coking Coal
Other Bituminous Coal
Sub-Bituminous Coal
Patent Fuel
Lignite/Brown Coal
BKB/Peat Briquettes

```
Peat
Crude petroleum and services related to crude o...
Natural gas and services related to natural gas...
Natural Gas Liquids
...
Research and development services (73)
Other business services (74)
Public administration and defence services; com...
Education services (80)
Health and social work services (85)
Food waste for treatment: incineration
Paper waste for treatment: incineration
Plastic waste for treatment: incineration
Intert/metal waste for treatment: incineration
Textiles waste for treatment: incineration
Wood waste for treatment: incineration
Oil/hazardous waste for treatment: incineration
Food waste for treatment: biogasification and l...
Paper waste for treatment: biogasification and ...
Sewage sludge for treatment: biogasification an...
Food waste for treatment: composting and land a...
Paper and wood waste for treatment: composting ...
Food waste for treatment: waste water treatment
Other waste for treatment: waste water treatment
Food waste for treatment: landfill
Paper for treatment: landfill
Plastic waste for treatment: landfill
Inert/metal/hazardous waste for treatment: land...
Textiles waste for treatment: landfill
Wood waste for treatment: landfill
Membership organisation services n.e.c. (91)
Recreational, cultural and sporting services (92)
Other services (93)
Private households with employed persons (95)
Extra-territorial organizations and bodies

                                                    Employment: Low-skilled male  \
sector
Paddy rice                                                           3.687352e-09
Wheat                                                                3.017314e-09
Cereal grains nec                                                    1.625998e-09
Vegetables, fruit, nuts                                              2.715066e-08
Oil seeds                                                            7.205884e-09
Sugar cane, sugar beet                                               0.000000e+00
Plant-based fibers                                                   0.000000e+00
Crops nec                                                            5.593639e-08
Cattle                                                               0.000000e+00
Pigs                                                                 0.000000e+00
```

Poultry	1.521051e-09
Meat animals nec	8.727094e-09
Animal products nec	2.534018e-09
Raw milk	1.982679e-09
Wool, silk-worm cocoons	0.000000e+00
Manure (conventional treatment)	0.000000e+00
Manure (biogas treatment)	0.000000e+00
Products of forestry, logging and related servi...	1.264852e-09
Fish and other fishing products; services incid...	3.424611e-08
Anthracite	3.734005e-08
Coking Coal	1.932032e-09
Other Bituminous Coal	6.236234e-09
Sub-Bituminous Coal	2.497848e-08
Patent Fuel	1.707444e-09
Lignite/Brown Coal	2.070166e-08
BKB/Peat Briquettes	1.546848e-08
Peat	1.902149e-09
Crude petroleum and services related to crude o...	0.000000e+00
Natural gas and services related to natural gas...	0.000000e+00
Natural Gas Liquids	0.000000e+00
...	...
Research and development services (73)	0.000000e+00
Other business services (74)	1.165100e-09
Public administration and defence services; com...	8.962452e-10
Education services (80)	4.972228e-10
Health and social work services (85)	4.657604e-10
Food waste for treatment: incineration	0.000000e+00
Paper waste for treatment: incineration	0.000000e+00
Plastic waste for treatment: incineration	0.000000e+00
Intert/metal waste for treatment: incineration	0.000000e+00
Textiles waste for treatment: incineration	0.000000e+00
Wood waste for treatment: incineration	0.000000e+00
Oil/hazardous waste for treatment: incineration	0.000000e+00
Food waste for treatment: biogasification and l...	0.000000e+00
Paper waste for treatment: biogasification and ...	0.000000e+00
Sewage sludge for treatment: biogasification an...	0.000000e+00
Food waste for treatment: composting and land a...	0.000000e+00
Paper and wood waste for treatment: composting ...	0.000000e+00
Food waste for treatment: waste water treatment	0.000000e+00
Other waste for treatment: waste water treatment	0.000000e+00
Food waste for treatment: landfill	0.000000e+00
Paper for treatment: landfill	0.000000e+00
Plastic waste for treatment: landfill	0.000000e+00
Inert/metal/hazardous waste for treatment: land...	0.000000e+00
Textiles waste for treatment: landfill	0.000000e+00
Wood waste for treatment: landfill	0.000000e+00
Membership organisation services n.e.c. (91)	8.863009e-10
Recreational, cultural and sporting services (92)	9.988702e-10

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Other services (93)                                                6.038437e-10
Private households with employed persons (95)                      1.621984e-10
Extra-territorial organizations and bodies                         0.000000e+00

                                                          ...  \
sector                                                    ...
Paddy rice                                                ...
Wheat                                                     ...
Cereal grains nec                                         ...
Vegetables, fruit, nuts                                   ...
Oil seeds                                                 ...
Sugar cane, sugar beet                                    ...
Plant-based fibers                                        ...
Crops nec                                                 ...
Cattle                                                    ...
Pigs                                                      ...
Poultry                                                   ...
Meat animals nec                                          ...
Animal products nec                                       ...
Raw milk                                                  ...
Wool, silk-worm cocoons                                   ...
Manure (conventional treatment)                           ...
Manure (biogas treatment)                                 ...
Products of forestry, logging and related servi...        ...
Fish and other fishing products; services incid...        ...
Anthracite                                                ...
Coking Coal                                               ...
Other Bituminous Coal                                     ...
Sub-Bituminous Coal                                       ...
Patent Fuel                                               ...
Lignite/Brown Coal                                        ...
BKB/Peat Briquettes                                       ...
Peat                                                      ...
Crude petroleum and services related to crude o...        ...
Natural gas and services related to natural gas...        ...
Natural Gas Liquids                                       ...
...                                                       ...
Research and development services (73)                    ...
Other business services (74)                              ...
Public administration and defence services; com...        ...
Education services (80)                                   ...
Health and social work services (85)                      ...
Food waste for treatment: incineration                    ...
Paper waste for treatment: incineration                   ...
Plastic waste for treatment: incineration                 ...
Intert/metal waste for treatment: incineration            ...
Textiles waste for treatment: incineration                ...
Wood waste for treatment: incineration                    ...
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Oil/hazardous waste for treatment: incineration     ...
Food waste for treatment: biogasification and l...  ...
Paper waste for treatment: biogasification and ...  ...
Sewage sludge for treatment: biogasification an...  ...
Food waste for treatment: composting and land a...  ...
Paper and wood waste for treatment: composting ...  ...
Food waste for treatment: waste water treatment     ...
Other waste for treatment: waste water treatment    ...
Food waste for treatment: landfill                  ...
Paper for treatment: landfill                       ...
Plastic waste for treatment: landfill               ...
Inert/metal/hazardous waste for treatment: land...  ...
Textiles waste for treatment: landfill              ...
Wood waste for treatment: landfill                  ...
Membership organisation services n.e.c. (91)        ...
Recreational, cultural and sporting services (92)   ...
Other services (93)                                 ...
Private households with employed persons (95)       ...
Extra-territorial organizations and bodies          ...

                                                    Water Withdrawal Blue - Electrici
sector
Paddy rice
Wheat
Cereal grains nec
Vegetables, fruit, nuts
Oil seeds
Sugar cane, sugar beet
Plant-based fibers
Crops nec
Cattle
Pigs
Poultry
Meat animals nec
Animal products nec
Raw milk
Wool, silk-worm cocoons
Manure (conventional treatment)
Manure (biogas treatment)
Products of forestry, logging and related servi...
Fish and other fishing products; services incid...
Anthracite
Coking Coal
Other Bituminous Coal
Sub-Bituminous Coal
Patent Fuel
Lignite/Brown Coal
BKB/Peat Briquettes
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Peat
Crude petroleum and services related to crude o...
Natural gas and services related to natural gas...
Natural Gas Liquids
...
Research and development services (73)
Other business services (74)
Public administration and defence services; com...
Education services (80)
Health and social work services (85)
Food waste for treatment: incineration
Paper waste for treatment: incineration
Plastic waste for treatment: incineration
Intert/metal waste for treatment: incineration
Textiles waste for treatment: incineration
Wood waste for treatment: incineration
Oil/hazardous waste for treatment: incineration
Food waste for treatment: biogasification and l...
Paper waste for treatment: biogasification and ...
Sewage sludge for treatment: biogasification an...
Food waste for treatment: composting and land a...
Paper and wood waste for treatment: composting ...
Food waste for treatment: waste water treatment
Other waste for treatment: waste water treatment
Food waste for treatment: landfill
Paper for treatment: landfill
Plastic waste for treatment: landfill
Inert/metal/hazardous waste for treatment: land...
Textiles waste for treatment: landfill
Wood waste for treatment: landfill
Membership organisation services n.e.c. (91)
Recreational, cultural and sporting services (92)
Other services (93)
Private households with employed persons (95)
Extra-territorial organizations and bodies

Water Withdrawal Blue - Electrici

sector
Paddy rice
Wheat
Cereal grains nec
Vegetables, fruit, nuts
Oil seeds
Sugar cane, sugar beet
Plant-based fibers
Crops nec
Cattle
Pigs

```
Poultry
Meat animals nec
Animal products nec
Raw milk
Wool, silk-worm cocoons
Manure (conventional treatment)
Manure (biogas treatment)
Products of forestry, logging and related servi...
Fish and other fishing products; services incid...
Anthracite
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Crude petroleum and services related to crude o...
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...
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Other business services (74)
Public administration and defence services; com...
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Plastic waste for treatment: incineration
Intert/metal waste for treatment: incineration
Textiles waste for treatment: incineration
Wood waste for treatment: incineration
Oil/hazardous waste for treatment: incineration
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Paper waste for treatment: biogasification and ...
Sewage sludge for treatment: biogasification an...
Food waste for treatment: composting and land a...
Paper and wood waste for treatment: composting ...
Food waste for treatment: waste water treatment
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Paper for treatment: landfill
Plastic waste for treatment: landfill
Inert/metal/hazardous waste for treatment: land...
Textiles waste for treatment: landfill
Wood waste for treatment: landfill
Membership organisation services n.e.c. (91)
Recreational, cultural and sporting services (92)
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Other services (93)
Private households with employed persons (95)
Extra-territorial organizations and bodies

                                                        Water Withdrawal Blue - Electricit
sector
Paddy rice
Wheat
Cereal grains nec
Vegetables, fruit, nuts
Oil seeds
Sugar cane, sugar beet
Plant-based fibers
Crops nec
Cattle
Pigs
Poultry
Meat animals nec
Animal products nec
Raw milk
Wool, silk-worm cocoons
Manure (conventional treatment)
Manure (biogas treatment)
Products of forestry, logging and related servi...
Fish and other fishing products; services incid...
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Coking Coal
Other Bituminous Coal
Sub-Bituminous Coal
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Lignite/Brown Coal
BKB/Peat Briquettes
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Crude petroleum and services related to crude o...
Natural gas and services related to natural gas...
Natural Gas Liquids
...
Research and development services (73)
Other business services (74)
Public administration and defence services; com...
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Health and social work services (85)
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Paper waste for treatment: incineration
Plastic waste for treatment: incineration
Intert/metal waste for treatment: incineration
Textiles waste for treatment: incineration
Wood waste for treatment: incineration
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Oil/hazardous waste for treatment: incineration
Food waste for treatment: biogasification and l...
Paper waste for treatment: biogasification and ...
Sewage sludge for treatment: biogasification an...
Food waste for treatment: composting and land a...
Paper and wood waste for treatment: composting ...
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Textiles waste for treatment: landfill
Wood waste for treatment: landfill
Membership organisation services n.e.c. (91)
Recreational, cultural and sporting services (92)
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Private households with employed persons (95)
Extra-territorial organizations and bodies

                                                    Water Withdrawal Blue - Electrici
sector
Paddy rice
Wheat
Cereal grains nec
Vegetables, fruit, nuts
Oil seeds
Sugar cane, sugar beet
Plant-based fibers
Crops nec
Cattle
Pigs
Poultry
Meat animals nec
Animal products nec
Raw milk
Wool, silk-worm cocoons
Manure (conventional treatment)
Manure (biogas treatment)
Products of forestry, logging and related servi...
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Coking Coal
Other Bituminous Coal
Sub-Bituminous Coal
Patent Fuel
Lignite/Brown Coal
BKB/Peat Briquettes
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Peat
Crude petroleum and services related to crude o...
Natural gas and services related to natural gas...
Natural Gas Liquids
...
Research and development services (73)
Other business services (74)
Public administration and defence services; com...
Education services (80)
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Wood waste for treatment: incineration
Oil/hazardous waste for treatment: incineration
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Sewage sludge for treatment: biogasification an...
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Other waste for treatment: waste water treatment
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Plastic waste for treatment: landfill
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Textiles waste for treatment: landfill
Wood waste for treatment: landfill
Membership organisation services n.e.c. (91)
Recreational, cultural and sporting services (92)
Other services (93)
Private households with employed persons (95)
Extra-territorial organizations and bodies

                                                        Water Withdrawal Blue - Electrici
sector
Paddy rice
Wheat
Cereal grains nec
Vegetables, fruit, nuts
Oil seeds
Sugar cane, sugar beet
Plant-based fibers
Crops nec
Cattle
Pigs
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Poultry
Meat animals nec
Animal products nec
Raw milk
Wool, silk-worm cocoons
Manure (conventional treatment)
Manure (biogas treatment)
Products of forestry, logging and related servi...
Fish and other fishing products; services incid...
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Coking Coal
Other Bituminous Coal
Sub-Bituminous Coal
Patent Fuel
Lignite/Brown Coal
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Crude petroleum and services related to crude o...
Natural gas and services related to natural gas...
Natural Gas Liquids
...
Research and development services (73)
Other business services (74)
Public administration and defence services; com...
Education services (80)
Health and social work services (85)
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Textiles waste for treatment: incineration
Wood waste for treatment: incineration
Oil/hazardous waste for treatment: incineration
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Paper waste for treatment: biogasification and ...
Sewage sludge for treatment: biogasification an...
Food waste for treatment: composting and land a...
Paper and wood waste for treatment: composting ...
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Paper for treatment: landfill
Plastic waste for treatment: landfill
Inert/metal/hazardous waste for treatment: land...
Textiles waste for treatment: landfill
Wood waste for treatment: landfill
Membership organisation services n.e.c. (91)
Recreational, cultural and sporting services (92)
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Other services (93)
Private households with employed persons (95)
Extra-territorial organizations and bodies

                                                        Water Withdrawal Blue - Electricit
sector
Paddy rice
Wheat
Cereal grains nec
Vegetables, fruit, nuts
Oil seeds
Sugar cane, sugar beet
Plant-based fibers
Crops nec
Cattle
Pigs
Poultry
Meat animals nec
Animal products nec
Raw milk
Wool, silk-worm cocoons
Manure (conventional treatment)
Manure (biogas treatment)
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Sub-Bituminous Coal
Patent Fuel
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Crude petroleum and services related to crude o...
Natural gas and services related to natural gas...
Natural Gas Liquids
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Plastic waste for treatment: incineration
Intert/metal waste for treatment: incineration
Textiles waste for treatment: incineration
Wood waste for treatment: incineration
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Oil/hazardous waste for treatment: incineration
Food waste for treatment: biogasification and l...
Paper waste for treatment: biogasification and ...
Sewage sludge for treatment: biogasification an...
Food waste for treatment: composting and land a...
Paper and wood waste for treatment: composting ...
Food waste for treatment: waste water treatment
Other waste for treatment: waste water treatment
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Paper for treatment: landfill
Plastic waste for treatment: landfill
Inert/metal/hazardous waste for treatment: land...
Textiles waste for treatment: landfill
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Membership organisation services n.e.c. (91)
Recreational, cultural and sporting services (92)
Other services (93)
Private households with employed persons (95)
Extra-territorial organizations and bodies

Water Withdrawal Blue - Electrici

sector
Paddy rice
Wheat
Cereal grains nec
Vegetables, fruit, nuts
Oil seeds
Sugar cane, sugar beet
Plant-based fibers
Crops nec
Cattle
Pigs
Poultry
Meat animals nec
Animal products nec
Raw milk
Wool, silk-worm cocoons
Manure (conventional treatment)
Manure (biogas treatment)
Products of forestry, logging and related servi...
Fish and other fishing products; services incid...
Anthracite
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Lignite/Brown Coal
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Peat
Crude petroleum and services related to crude o...
Natural gas and services related to natural gas...
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...
Research and development services (73)
Other business services (74)
Public administration and defence services; com...
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Plastic waste for treatment: incineration
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Textiles waste for treatment: incineration
Wood waste for treatment: incineration
Oil/hazardous waste for treatment: incineration
Food waste for treatment: biogasification and l...
Paper waste for treatment: biogasification and ...
Sewage sludge for treatment: biogasification an...
Food waste for treatment: composting and land a...
Paper and wood waste for treatment: composting ...
Food waste for treatment: waste water treatment
Other waste for treatment: waste water treatment
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Paper for treatment: landfill
Plastic waste for treatment: landfill
Inert/metal/hazardous waste for treatment: land...
Textiles waste for treatment: landfill
Wood waste for treatment: landfill
Membership organisation services n.e.c. (91)
Recreational, cultural and sporting services (92)
Other services (93)
Private households with employed persons (95)
Extra-territorial organizations and bodies

Water Withdrawal Blue - Electrici

sector
Paddy rice
Wheat
Cereal grains nec
Vegetables, fruit, nuts
Oil seeds
Sugar cane, sugar beet
Plant-based fibers
Crops nec
Cattle
Pigs

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Poultry
Meat animals nec
Animal products nec
Raw milk
Wool, silk-worm cocoons
Manure (conventional treatment)
Manure (biogas treatment)
Products of forestry, logging and related servi...
Fish and other fishing products; services incid...
Anthracite
Coking Coal
Other Bituminous Coal
Sub-Bituminous Coal
Patent Fuel
Lignite/Brown Coal
BKB/Peat Briquettes
Peat
Crude petroleum and services related to crude o...
Natural gas and services related to natural gas...
Natural Gas Liquids
...
Research and development services (73)
Other business services (74)
Public administration and defence services; com...
Education services (80)
Health and social work services (85)
Food waste for treatment: incineration
Paper waste for treatment: incineration
Plastic waste for treatment: incineration
Intert/metal waste for treatment: incineration
Textiles waste for treatment: incineration
Wood waste for treatment: incineration
Oil/hazardous waste for treatment: incineration
Food waste for treatment: biogasification and l...
Paper waste for treatment: biogasification and ...
Sewage sludge for treatment: biogasification an...
Food waste for treatment: composting and land a...
Paper and wood waste for treatment: composting ...
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Paper for treatment: landfill
Plastic waste for treatment: landfill
Inert/metal/hazardous waste for treatment: land...
Textiles waste for treatment: landfill
Wood waste for treatment: landfill
Membership organisation services n.e.c. (91)
Recreational, cultural and sporting services (92)
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Other services (93)
Private households with employed persons (95)
Extra-territorial organizations and bodies

                                                        Water Withdrawal Blue - Electricit
sector
Paddy rice
Wheat
Cereal grains nec
Vegetables, fruit, nuts
Oil seeds
Sugar cane, sugar beet
Plant-based fibers
Crops nec
Cattle
Pigs
Poultry
Meat animals nec
Animal products nec
Raw milk
Wool, silk-worm cocoons
Manure (conventional treatment)
Manure (biogas treatment)
Products of forestry, logging and related servi...
Fish and other fishing products; services incid...
Anthracite
Coking Coal
Other Bituminous Coal
Sub-Bituminous Coal
Patent Fuel
Lignite/Brown Coal
BKB/Peat Briquettes
Peat
Crude petroleum and services related to crude o...
Natural gas and services related to natural gas...
Natural Gas Liquids
...
Research and development services (73)
Other business services (74)
Public administration and defence services; com...
Education services (80)
Health and social work services (85)
Food waste for treatment: incineration
Paper waste for treatment: incineration
Plastic waste for treatment: incineration
Intert/metal waste for treatment: incineration
Textiles waste for treatment: incineration
Wood waste for treatment: incineration
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Oil/hazardous waste for treatment: incineration
Food waste for treatment: biogasification and l...
Paper waste for treatment: biogasification and ...
Sewage sludge for treatment: biogasification an...
Food waste for treatment: composting and land a...
Paper and wood waste for treatment: composting ...
Food waste for treatment: waste water treatment
Other waste for treatment: waste water treatment
Food waste for treatment: landfill
Paper for treatment: landfill
Plastic waste for treatment: landfill
Inert/metal/hazardous waste for treatment: land...
Textiles waste for treatment: landfill
Wood waste for treatment: landfill
Membership organisation services n.e.c. (91)
Recreational, cultural and sporting services (92)
Other services (93)
Private households with employed persons (95)
Extra-territorial organizations and bodies

Water Withdrawal Blue - Domestic -
sector
Paddy rice
Wheat
Cereal grains nec
Vegetables, fruit, nuts
Oil seeds
Sugar cane, sugar beet
Plant-based fibers
Crops nec
Cattle
Pigs
Poultry
Meat animals nec
Animal products nec
Raw milk
Wool, silk-worm cocoons
Manure (conventional treatment)
Manure (biogas treatment)
Products of forestry, logging and related servi...
Fish and other fishing products; services incid...
Anthracite
Coking Coal
Other Bituminous Coal
Sub-Bituminous Coal
Patent Fuel
Lignite/Brown Coal
BKB/Peat Briquettes

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Peat
Crude petroleum and services related to crude o...
Natural gas and services related to natural gas...
Natural Gas Liquids
...
Research and development services (73)
Other business services (74)
Public administration and defence services; com...
Education services (80)
Health and social work services (85)
Food waste for treatment: incineration
Paper waste for treatment: incineration
Plastic waste for treatment: incineration
Intert/metal waste for treatment: incineration
Textiles waste for treatment: incineration
Wood waste for treatment: incineration
Oil/hazardous waste for treatment: incineration
Food waste for treatment: biogasification and l...
Paper waste for treatment: biogasification and ...
Sewage sludge for treatment: biogasification an...
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Food waste for treatment: landfill
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Plastic waste for treatment: landfill
Inert/metal/hazardous waste for treatment: land...
Textiles waste for treatment: landfill
Wood waste for treatment: landfill
Membership organisation services n.e.c. (91)
Recreational, cultural and sporting services (92)
Other services (93)
Private households with employed persons (95)
Extra-territorial organizations and bodies

[200 rows x 1104 columns]
```

In [62]:
```
# The name of 'sector' can be removed
M_SE = M_SE_adj_grouped.rename_axis(index=None, columns=None)
M_SE_t = M_SE.T
M_SE_t.columns.equals(concordance.index)
M_SE_t
```

Out[62]:
```
                                                   Paddy rice  \
Taxes less subsidies on products purchased: Total   1.286576e-08
Other net taxes on production                       2.641363e-08
Compensation of employees; wages, salaries, & e...  2.696499e-08
```

```
Compensation of employees; wages, salaries, & e...  1.058415e-07
Compensation of employees; wages, salaries, & e...  6.599903e-08
Operating surplus: Consumption of fixed capital     2.177788e-07
Operating surplus: Rents on land                    0.000000e+00
Operating surplus: Royalties on resources           0.000000e+00
Operating surplus: Remaining net operating surplus  5.441363e-07
Employment: Low-skilled male                        3.687352e-09
Employment: Low-skilled female                      4.332195e-09
Employment: Medium-skilled male                     8.257774e-09
Employment: Medium-skilled female                   3.872956e-09
Employment: High-skilled male                       1.908515e-09
Employment: High-skilled female                     9.342513e-10
Employment hours: Low-skilled male                  8.059526e-09
Employment hours: Low-skilled female                8.724138e-09
Employment hours: Medium-skilled male               1.748952e-08
Employment hours: Medium-skilled female             7.659921e-09
Employment hours: High-skilled male                 3.967161e-09
Employment hours: High-skilled female               1.865429e-09
Employment: Vulnerable employment                   1.057630e-08
Employment hours: Vulnerable employment             2.323560e-08
CO2 - combustion - air                              4.468342e-01
CH4 - combustion - air                              7.326865e-05
N2O - combustion - air                              1.015171e-05
SOx - combustion - air                              2.778339e-04
NOx - combustion - air                              2.878789e-03
NH3 - combustion - air                              2.151591e-05
CO - combustion - air                               1.704346e-02
...                                                          ...
Water Withdrawal Blue - Manufacturing - Radio, ...  1.115415e-12
Water Withdrawal Blue - Manufacturing - Medical...  4.578449e-13
Water Withdrawal Blue - Manufacturing - Motor v...  2.948804e-13
Water Withdrawal Blue - Manufacturing - Other t...  5.860375e-13
Water Withdrawal Blue - Manufacturing - Furnitu...  2.957819e-12
Water Withdrawal Blue - Electricity - tower - E...  1.531758e-11
Water Withdrawal Blue - Electricity - tower - E...  2.658087e-12
Water Withdrawal Blue - Electricity - tower - E...  3.441234e-12
Water Withdrawal Blue - Electricity - tower - E...  0.000000e+00
Water Withdrawal Blue - Electricity - tower - E...  0.000000e+00
Water Withdrawal Blue - Electricity - tower - E...  9.278462e-14
Water Withdrawal Blue - Electricity - tower - E...  1.091748e-12
Water Withdrawal Blue - Electricity - tower - E...  0.000000e+00
Water Withdrawal Blue - Electricity - tower - E...  1.369385e-14
Water Withdrawal Blue - Electricity - tower - E...  0.000000e+00
Water Withdrawal Blue - Electricity - tower - E...  1.814648e-13
Water Withdrawal Blue - Electricity - tower - E...  0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...  2.919032e-10
Water Withdrawal Blue - Electricity - once-thro...  1.854608e-10
Water Withdrawal Blue - Electricity - once-thro...  1.467518e-10
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Water Withdrawal Blue - Electricity - once-thro...	0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...	0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...	4.374888e-11
Water Withdrawal Blue - Electricity - once-thro...	6.376641e-12
Water Withdrawal Blue - Electricity - once-thro...	0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...	0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...	0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...	4.087753e-13
Water Withdrawal Blue - Electricity - once-thro...	0.000000e+00
Water Withdrawal Blue - Domestic - domestic Wat...	0.000000e+00

	Wheat \
Taxes less subsidies on products purchased: Total	1.081643e-07
Other net taxes on production	-4.398411e-08
Compensation of employees; wages, salaries, & e...	2.063733e-08
Compensation of employees; wages, salaries, & e...	2.734194e-07
Compensation of employees; wages, salaries, & e...	1.387227e-07
Operating surplus: Consumption of fixed capital	3.227823e-07
Operating surplus: Rents on land	0.000000e+00
Operating surplus: Royalties on resources	0.000000e+00
Operating surplus: Remaining net operating surplus	1.802581e-07
Employment: Low-skilled male	3.017314e-09
Employment: Low-skilled female	1.611603e-09
Employment: Medium-skilled male	1.674273e-08
Employment: Medium-skilled female	7.688687e-09
Employment: High-skilled male	2.391295e-09
Employment: High-skilled female	1.238258e-09
Employment hours: Low-skilled male	6.660698e-09
Employment hours: Low-skilled female	3.463992e-09
Employment hours: Medium-skilled male	3.565420e-08
Employment hours: Medium-skilled female	1.601031e-08
Employment hours: High-skilled male	4.823410e-09
Employment hours: High-skilled female	2.418129e-09
Employment: Vulnerable employment	1.906687e-08
Employment hours: Vulnerable employment	4.158251e-08
CO2 - combustion - air	4.815248e-01
CH4 - combustion - air	8.348358e-05
N2O - combustion - air	9.189937e-06
SOx - combustion - air	7.482133e-04
NOx - combustion - air	3.162938e-03
NH3 - combustion - air	4.618447e-05
CO - combustion - air	3.540455e-03
...	...
Water Withdrawal Blue - Manufacturing - Radio, ...	7.742666e-12
Water Withdrawal Blue - Manufacturing - Medical...	2.273975e-12
Water Withdrawal Blue - Manufacturing - Motor v...	3.726540e-13
Water Withdrawal Blue - Manufacturing - Other t...	2.120196e-13
Water Withdrawal Blue - Manufacturing - Furnitu...	9.494766e-12

Water Withdrawal Blue - Electricity - tower - E...	6.741299e-11
Water Withdrawal Blue - Electricity - tower - E...	1.184906e-11
Water Withdrawal Blue - Electricity - tower - E...	1.272679e-11
Water Withdrawal Blue - Electricity - tower - E...	0.000000e+00
Water Withdrawal Blue - Electricity - tower - E...	0.000000e+00
Water Withdrawal Blue - Electricity - tower - E...	4.919660e-13
Water Withdrawal Blue - Electricity - tower - E...	6.939248e-12
Water Withdrawal Blue - Electricity - tower - E...	0.000000e+00
Water Withdrawal Blue - Electricity - tower - E...	1.413234e-14
Water Withdrawal Blue - Electricity - tower - E...	0.000000e+00
Water Withdrawal Blue - Electricity - tower - E...	7.272904e-13
Water Withdrawal Blue - Electricity - tower - E...	0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...	1.326356e-09
Water Withdrawal Blue - Electricity - once-thro...	8.671046e-10
Water Withdrawal Blue - Electricity - once-thro...	5.506916e-10
Water Withdrawal Blue - Electricity - once-thro...	0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...	0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...	2.073640e-10
Water Withdrawal Blue - Electricity - once-thro...	3.063934e-11
Water Withdrawal Blue - Electricity - once-thro...	0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...	0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...	0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...	2.598625e-12
Water Withdrawal Blue - Electricity - once-thro...	0.000000e+00
Water Withdrawal Blue - Domestic - domestic Wat...	0.000000e+00

	Cereal grains nec \
Taxes less subsidies on products purchased: Total	1.011903e-07
Other net taxes on production	-6.348476e-08
Compensation of employees; wages, salaries, & e...	2.098466e-08
Compensation of employees; wages, salaries, & e...	3.239947e-07
Compensation of employees; wages, salaries, & e...	1.421072e-07
Operating surplus: Consumption of fixed capital	3.015066e-07
Operating surplus: Rents on land	0.000000e+00
Operating surplus: Royalties on resources	0.000000e+00
Operating surplus: Remaining net operating surplus	1.737013e-07
Employment: Low-skilled male	1.625998e-09
Employment: Low-skilled female	9.056524e-10
Employment: Medium-skilled male	1.260775e-08
Employment: Medium-skilled female	4.997056e-09
Employment: High-skilled male	2.335048e-09
Employment: High-skilled female	1.206597e-09
Employment hours: Low-skilled male	3.585431e-09
Employment hours: Low-skilled female	1.915504e-09
Employment hours: Medium-skilled male	2.680960e-08
Employment hours: Medium-skilled female	1.033266e-08
Employment hours: High-skilled male	4.722665e-09
Employment hours: High-skilled female	2.368303e-09

Employment: Vulnerable employment	1.120957e-08
Employment hours: Vulnerable employment	2.487018e-08
CO2 - combustion - air	4.182635e-01
CH4 - combustion - air	7.546753e-05
N2O - combustion - air	8.087898e-06
SOx - combustion - air	6.671111e-04
NOx - combustion - air	2.715637e-03
NH3 - combustion - air	4.288698e-05
CO - combustion - air	2.855217e-03
...	...
Water Withdrawal Blue - Manufacturing - Radio, ...	8.298610e-12
Water Withdrawal Blue - Manufacturing - Medical...	1.982488e-12
Water Withdrawal Blue - Manufacturing - Motor v...	2.691734e-13
Water Withdrawal Blue - Manufacturing - Other t...	1.904228e-13
Water Withdrawal Blue - Manufacturing - Furnitu...	7.979034e-12
Water Withdrawal Blue - Electricity - tower - E...	5.791633e-11
Water Withdrawal Blue - Electricity - tower - E...	1.004869e-11
Water Withdrawal Blue - Electricity - tower - E...	1.090416e-11
Water Withdrawal Blue - Electricity - tower - E...	0.000000e+00
Water Withdrawal Blue - Electricity - tower - E...	0.000000e+00
Water Withdrawal Blue - Electricity - tower - E...	3.924083e-13
Water Withdrawal Blue - Electricity - tower - E...	9.570718e-12
Water Withdrawal Blue - Electricity - tower - E...	0.000000e+00
Water Withdrawal Blue - Electricity - tower - E...	1.257727e-14
Water Withdrawal Blue - Electricity - tower - E...	0.000000e+00
Water Withdrawal Blue - Electricity - tower - E...	6.211168e-13
Water Withdrawal Blue - Electricity - tower - E...	0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...	1.091693e-09
Water Withdrawal Blue - Electricity - once-thro...	7.253600e-10
Water Withdrawal Blue - Electricity - once-thro...	4.707302e-10
Water Withdrawal Blue - Electricity - once-thro...	0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...	0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...	1.692215e-10
Water Withdrawal Blue - Electricity - once-thro...	3.700662e-11
Water Withdrawal Blue - Electricity - once-thro...	0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...	0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...	0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...	2.041620e-12
Water Withdrawal Blue - Electricity - once-thro...	0.000000e+00
Water Withdrawal Blue - Domestic - domestic Wat...	0.000000e+00

	Vegetables, fruit, nuts \
Taxes less subsidies on products purchased: Total	3.782212e-08
Other net taxes on production	3.599368e-09
Compensation of employees; wages, salaries, & e...	4.040890e-08
Compensation of employees; wages, salaries, & e...	2.768941e-07
Compensation of employees; wages, salaries, & e...	9.237096e-08
Operating surplus: Consumption of fixed capital	1.895675e-07

Operating surplus: Rents on land	0.000000e+00
Operating surplus: Royalties on resources	0.000000e+00
Operating surplus: Remaining net operating surplus	3.593371e-07
Employment: Low-skilled male	2.715066e-08
Employment: Low-skilled female	1.496520e-08
Employment: Medium-skilled male	8.345271e-08
Employment: Medium-skilled female	4.522592e-08
Employment: High-skilled male	3.865576e-09
Employment: High-skilled female	1.542822e-09
Employment hours: Low-skilled male	5.942480e-08
Employment hours: Low-skilled female	3.261568e-08
Employment hours: Medium-skilled male	1.671237e-07
Employment hours: Medium-skilled female	8.876009e-08
Employment hours: High-skilled male	8.161338e-09
Employment hours: High-skilled female	3.171591e-09
Employment: Vulnerable employment	1.280202e-07
Employment hours: Vulnerable employment	2.532836e-07
CO2 - combustion - air	2.502203e-01
CH4 - combustion - air	5.498881e-05
N2O - combustion - air	5.292014e-06
SOx - combustion - air	4.358253e-04
NOx - combustion - air	1.155260e-03
NH3 - combustion - air	6.369092e-05
CO - combustion - air	1.984918e-03
...	...
Water Withdrawal Blue - Manufacturing - Radio, ...	4.943325e-12
Water Withdrawal Blue - Manufacturing - Medical...	2.043710e-12
Water Withdrawal Blue - Manufacturing - Motor v...	8.913864e-13
Water Withdrawal Blue - Manufacturing - Other t...	2.205009e-13
Water Withdrawal Blue - Manufacturing - Furnitu...	1.512700e-11
Water Withdrawal Blue - Electricity - tower - E...	1.094966e-10
Water Withdrawal Blue - Electricity - tower - E...	3.015275e-11
Water Withdrawal Blue - Electricity - tower - E...	2.392091e-11
Water Withdrawal Blue - Electricity - tower - E...	0.000000e+00
Water Withdrawal Blue - Electricity - tower - E...	0.000000e+00
Water Withdrawal Blue - Electricity - tower - E...	9.412391e-13
Water Withdrawal Blue - Electricity - tower - E...	1.579322e-11
Water Withdrawal Blue - Electricity - tower - E...	0.000000e+00
Water Withdrawal Blue - Electricity - tower - E...	5.192292e-14
Water Withdrawal Blue - Electricity - tower - E...	0.000000e+00
Water Withdrawal Blue - Electricity - tower - E...	1.633502e-12
Water Withdrawal Blue - Electricity - tower - E...	0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...	1.735252e-09
Water Withdrawal Blue - Electricity - once-thro...	2.015129e-09
Water Withdrawal Blue - Electricity - once-thro...	1.008091e-09
Water Withdrawal Blue - Electricity - once-thro...	0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...	0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...	4.511342e-10

Water Withdrawal Blue - Electricity - once-thro...	7.919461e-11
Water Withdrawal Blue - Electricity - once-thro...	0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...	0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...	0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...	2.150868e-12
Water Withdrawal Blue - Electricity - once-thro...	0.000000e+00
Water Withdrawal Blue - Domestic - domestic Wat...	0.000000e+00

	Oil seeds \
Taxes less subsidies on products purchased: Total	3.301464e-08
Other net taxes on production	-2.360716e-08
Compensation of employees; wages, salaries, & e...	3.010021e-08
Compensation of employees; wages, salaries, & e...	4.886395e-07
Compensation of employees; wages, salaries, & e...	1.134341e-07
Operating surplus: Consumption of fixed capital	1.933613e-07
Operating surplus: Rents on land	0.000000e+00
Operating surplus: Royalties on resources	0.000000e+00
Operating surplus: Remaining net operating surplus	1.650574e-07
Employment: Low-skilled male	7.205884e-09
Employment: Low-skilled female	3.979874e-09
Employment: Medium-skilled male	3.361933e-08
Employment: Medium-skilled female	1.387598e-08
Employment: High-skilled male	3.044640e-09
Employment: High-skilled female	1.317860e-09
Employment hours: Low-skilled male	1.601513e-08
Employment hours: Low-skilled female	8.709298e-09
Employment hours: Medium-skilled male	7.039065e-08
Employment hours: Medium-skilled female	2.819866e-08
Employment hours: High-skilled male	6.416969e-09
Employment hours: High-skilled female	2.691760e-09
Employment: Vulnerable employment	3.792255e-08
Employment hours: Vulnerable employment	8.094371e-08
CO2 - combustion - air	2.645135e-01
CH4 - combustion - air	4.228911e-05
N2O - combustion - air	5.558264e-06
SOx - combustion - air	3.864175e-04
NOx - combustion - air	1.985576e-03
NH3 - combustion - air	2.888873e-05
CO - combustion - air	2.350409e-03
...	...
Water Withdrawal Blue - Manufacturing - Radio, ...	3.345123e-12
Water Withdrawal Blue - Manufacturing - Medical...	1.154613e-12
Water Withdrawal Blue - Manufacturing - Motor v...	2.193689e-13
Water Withdrawal Blue - Manufacturing - Other t...	1.183540e-13
Water Withdrawal Blue - Manufacturing - Furnitu...	5.341479e-12
Water Withdrawal Blue - Electricity - tower - E...	3.747733e-11
Water Withdrawal Blue - Electricity - tower - E...	6.262974e-12
Water Withdrawal Blue - Electricity - tower - E...	8.353278e-12

Water Withdrawal Blue - Electricity - tower - E...	0.000000e+00
Water Withdrawal Blue - Electricity - tower - E...	0.000000e+00
Water Withdrawal Blue - Electricity - tower - E...	2.335139e-13
Water Withdrawal Blue - Electricity - tower - E...	4.523116e-12
Water Withdrawal Blue - Electricity - tower - E...	0.000000e+00
Water Withdrawal Blue - Electricity - tower - E...	3.357104e-14
Water Withdrawal Blue - Electricity - tower - E...	0.000000e+00
Water Withdrawal Blue - Electricity - tower - E...	3.968041e-13
Water Withdrawal Blue - Electricity - tower - E...	0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...	6.799024e-10
Water Withdrawal Blue - Electricity - once-thro...	4.420576e-10
Water Withdrawal Blue - Electricity - once-thro...	3.576416e-10
Water Withdrawal Blue - Electricity - once-thro...	0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...	0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...	1.090632e-10
Water Withdrawal Blue - Electricity - once-thro...	2.003973e-11
Water Withdrawal Blue - Electricity - once-thro...	0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...	0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...	0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...	1.447809e-12
Water Withdrawal Blue - Electricity - once-thro...	0.000000e+00
Water Withdrawal Blue - Domestic - domestic Wat...	0.000000e+00

	Sugar cane, sugar beet \
Taxes less subsidies on products purchased: Total	0.0
Other net taxes on production	0.0
Compensation of employees; wages, salaries, & e...	0.0
Compensation of employees; wages, salaries, & e...	0.0
Compensation of employees; wages, salaries, & e...	0.0
Operating surplus: Consumption of fixed capital	0.0
Operating surplus: Rents on land	0.0
Operating surplus: Royalties on resources	0.0
Operating surplus: Remaining net operating surplus	0.0
Employment: Low-skilled male	0.0
Employment: Low-skilled female	0.0
Employment: Medium-skilled male	0.0
Employment: Medium-skilled female	0.0
Employment: High-skilled male	0.0
Employment: High-skilled female	0.0
Employment hours: Low-skilled male	0.0
Employment hours: Low-skilled female	0.0
Employment hours: Medium-skilled male	0.0
Employment hours: Medium-skilled female	0.0
Employment hours: High-skilled male	0.0
Employment hours: High-skilled female	0.0
Employment: Vulnerable employment	0.0
Employment hours: Vulnerable employment	0.0
CO2 - combustion - air	0.0

CH4 - combustion - air	0.0
N2O - combustion - air	0.0
SOx - combustion - air	0.0
NOx - combustion - air	0.0
NH3 - combustion - air	0.0
CO - combustion - air	0.0
...	...
Water Withdrawal Blue - Manufacturing - Radio, ...	0.0
Water Withdrawal Blue - Manufacturing - Medical...	0.0
Water Withdrawal Blue - Manufacturing - Motor v...	0.0
Water Withdrawal Blue - Manufacturing - Other t...	0.0
Water Withdrawal Blue - Manufacturing - Furnitu...	0.0
Water Withdrawal Blue - Electricity - tower - E...	0.0
Water Withdrawal Blue - Electricity - tower - E...	0.0
Water Withdrawal Blue - Electricity - tower - E...	0.0
Water Withdrawal Blue - Electricity - tower - E...	0.0
Water Withdrawal Blue - Electricity - tower - E...	0.0
Water Withdrawal Blue - Electricity - tower - E...	0.0
Water Withdrawal Blue - Electricity - tower - E...	0.0
Water Withdrawal Blue - Electricity - tower - E...	0.0
Water Withdrawal Blue - Electricity - tower - E...	0.0
Water Withdrawal Blue - Electricity - tower - E...	0.0
Water Withdrawal Blue - Electricity - tower - E...	0.0
Water Withdrawal Blue - Electricity - tower - E...	0.0
Water Withdrawal Blue - Electricity - once-thro...	0.0
Water Withdrawal Blue - Electricity - once-thro...	0.0
Water Withdrawal Blue - Electricity - once-thro...	0.0
Water Withdrawal Blue - Electricity - once-thro...	0.0
Water Withdrawal Blue - Electricity - once-thro...	0.0
Water Withdrawal Blue - Electricity - once-thro...	0.0
Water Withdrawal Blue - Electricity - once-thro...	0.0
Water Withdrawal Blue - Electricity - once-thro...	0.0
Water Withdrawal Blue - Electricity - once-thro...	0.0
Water Withdrawal Blue - Electricity - once-thro...	0.0
Water Withdrawal Blue - Electricity - once-thro...	0.0
Water Withdrawal Blue - Electricity - once-thro...	0.0
Water Withdrawal Blue - Domestic - domestic Wat...	0.0

	Plant-based fibers \
Taxes less subsidies on products purchased: Total	0.0
Other net taxes on production	0.0
Compensation of employees; wages, salaries, & e...	0.0
Compensation of employees; wages, salaries, & e...	0.0
Compensation of employees; wages, salaries, & e...	0.0
Operating surplus: Consumption of fixed capital	0.0
Operating surplus: Rents on land	0.0
Operating surplus: Royalties on resources	0.0
Operating surplus: Remaining net operating surplus	0.0

Employment: Low-skilled male	0.0
Employment: Low-skilled female	0.0
Employment: Medium-skilled male	0.0
Employment: Medium-skilled female	0.0
Employment: High-skilled male	0.0
Employment: High-skilled female	0.0
Employment hours: Low-skilled male	0.0
Employment hours: Low-skilled female	0.0
Employment hours: Medium-skilled male	0.0
Employment hours: Medium-skilled female	0.0
Employment hours: High-skilled male	0.0
Employment hours: High-skilled female	0.0
Employment: Vulnerable employment	0.0
Employment hours: Vulnerable employment	0.0
CO2 - combustion - air	0.0
CH4 - combustion - air	0.0
N2O - combustion - air	0.0
SOx - combustion - air	0.0
NOx - combustion - air	0.0
NH3 - combustion - air	0.0
CO - combustion - air	0.0
...	...
Water Withdrawal Blue - Manufacturing - Radio, ...	0.0
Water Withdrawal Blue - Manufacturing - Medical...	0.0
Water Withdrawal Blue - Manufacturing - Motor v...	0.0
Water Withdrawal Blue - Manufacturing - Other t...	0.0
Water Withdrawal Blue - Manufacturing - Furnitu...	0.0
Water Withdrawal Blue - Electricity - tower - E...	0.0
Water Withdrawal Blue - Electricity - tower - E...	0.0
Water Withdrawal Blue - Electricity - tower - E...	0.0
Water Withdrawal Blue - Electricity - tower - E...	0.0
Water Withdrawal Blue - Electricity - tower - E...	0.0
Water Withdrawal Blue - Electricity - tower - E...	0.0
Water Withdrawal Blue - Electricity - tower - E...	0.0
Water Withdrawal Blue - Electricity - tower - E...	0.0
Water Withdrawal Blue - Electricity - tower - E...	0.0
Water Withdrawal Blue - Electricity - tower - E...	0.0
Water Withdrawal Blue - Electricity - tower - E...	0.0
Water Withdrawal Blue - Electricity - once-thro...	0.0
Water Withdrawal Blue - Electricity - once-thro...	0.0
Water Withdrawal Blue - Electricity - once-thro...	0.0
Water Withdrawal Blue - Electricity - once-thro...	0.0
Water Withdrawal Blue - Electricity - once-thro...	0.0
Water Withdrawal Blue - Electricity - once-thro...	0.0
Water Withdrawal Blue - Electricity - once-thro...	0.0
Water Withdrawal Blue - Electricity - once-thro...	0.0
Water Withdrawal Blue - Electricity - once-thro...	0.0

Water Withdrawal Blue - Electricity - once-thro...	0.0
Water Withdrawal Blue - Electricity - once-thro...	0.0
Water Withdrawal Blue - Electricity - once-thro...	0.0
Water Withdrawal Blue - Domestic - domestic Wat...	0.0

	Crops nec	Cattle	\
Taxes less subsidies on products purchased: Total	6.548772e-08	0.0	
Other net taxes on production	3.915386e-10	0.0	
Compensation of employees; wages, salaries, & e...	5.217912e-08	0.0	
Compensation of employees; wages, salaries, & e...	3.484955e-07	0.0	
Compensation of employees; wages, salaries, & e...	8.923283e-08	0.0	
Operating surplus: Consumption of fixed capital	1.582004e-07	0.0	
Operating surplus: Rents on land	0.000000e+00	0.0	
Operating surplus: Royalties on resources	0.000000e+00	0.0	
Operating surplus: Remaining net operating surplus	2.860129e-07	0.0	
Employment: Low-skilled male	5.593639e-08	0.0	
Employment: Low-skilled female	3.030499e-08	0.0	
Employment: Medium-skilled male	1.718349e-07	0.0	
Employment: Medium-skilled female	8.997168e-08	0.0	
Employment: High-skilled male	5.863357e-09	0.0	
Employment: High-skilled female	2.171369e-09	0.0	
Employment hours: Low-skilled male	1.225970e-07	0.0	
Employment hours: Low-skilled female	6.630151e-08	0.0	
Employment hours: Medium-skilled male	3.419946e-07	0.0	
Employment hours: Medium-skilled female	1.746908e-07	0.0	
Employment hours: High-skilled male	1.290971e-08	0.0	
Employment hours: High-skilled female	4.697923e-09	0.0	
Employment: Vulnerable employment	2.492707e-07	0.0	
Employment hours: Vulnerable employment	4.919016e-07	0.0	
CO2 - combustion - air	5.114512e-01	0.0	
CH4 - combustion - air	1.062837e-04	0.0	
N2O - combustion - air	1.104305e-05	0.0	
SOx - combustion - air	1.029907e-03	0.0	
NOx - combustion - air	3.486865e-03	0.0	
NH3 - combustion - air	3.027457e-04	0.0	
CO - combustion - air	1.202845e-02	0.0	
...	...	...	
Water Withdrawal Blue - Manufacturing - Radio, ...	6.412172e-12	0.0	
Water Withdrawal Blue - Manufacturing - Medical...	3.024987e-12	0.0	
Water Withdrawal Blue - Manufacturing - Motor v...	9.009731e-13	0.0	
Water Withdrawal Blue - Manufacturing - Other t...	4.571737e-13	0.0	
Water Withdrawal Blue - Manufacturing - Furnitu...	1.762047e-11	0.0	
Water Withdrawal Blue - Electricity - tower - E...	1.062469e-10	0.0	
Water Withdrawal Blue - Electricity - tower - E...	1.930164e-11	0.0	
Water Withdrawal Blue - Electricity - tower - E...	1.564602e-11	0.0	
Water Withdrawal Blue - Electricity - tower - E...	0.000000e+00	0.0	
Water Withdrawal Blue - Electricity - tower - E...	0.000000e+00	0.0	
Water Withdrawal Blue - Electricity - tower - E...	1.400772e-12	0.0	

Water Withdrawal Blue - Electricity - tower - E...	7.239134e-12	0.0
Water Withdrawal Blue - Electricity - tower - E...	0.000000e+00	0.0
Water Withdrawal Blue - Electricity - tower - E...	2.756504e-14	0.0
Water Withdrawal Blue - Electricity - tower - E...	0.000000e+00	0.0
Water Withdrawal Blue - Electricity - tower - E...	1.612489e-12	0.0
Water Withdrawal Blue - Electricity - tower - E...	0.000000e+00	0.0
Water Withdrawal Blue - Electricity - once-thro...	2.541162e-09	0.0
Water Withdrawal Blue - Electricity - once-thro...	1.538774e-09	0.0
Water Withdrawal Blue - Electricity - once-thro...	6.948489e-10	0.0
Water Withdrawal Blue - Electricity - once-thro...	0.000000e+00	0.0
Water Withdrawal Blue - Electricity - once-thro...	0.000000e+00	0.0
Water Withdrawal Blue - Electricity - once-thro...	5.280312e-10	0.0
Water Withdrawal Blue - Electricity - once-thro...	3.046918e-11	0.0
Water Withdrawal Blue - Electricity - once-thro...	0.000000e+00	0.0
Water Withdrawal Blue - Electricity - once-thro...	0.000000e+00	0.0
Water Withdrawal Blue - Electricity - once-thro...	0.000000e+00	0.0
Water Withdrawal Blue - Electricity - once-thro...	9.707924e-12	0.0
Water Withdrawal Blue - Electricity - once-thro...	0.000000e+00	0.0
Water Withdrawal Blue - Domestic - domestic Wat...	0.000000e+00	0.0

	Pigs	... \
Taxes less subsidies on products purchased: Total	0.0	...
Other net taxes on production	0.0	...
Compensation of employees; wages, salaries, & e...	0.0	...
Compensation of employees; wages, salaries, & e...	0.0	...
Compensation of employees; wages, salaries, & e...	0.0	...
Operating surplus: Consumption of fixed capital	0.0	...
Operating surplus: Rents on land	0.0	...
Operating surplus: Royalties on resources	0.0	...
Operating surplus: Remaining net operating surplus	0.0	...
Employment: Low-skilled male	0.0	...
Employment: Low-skilled female	0.0	...
Employment: Medium-skilled male	0.0	...
Employment: Medium-skilled female	0.0	...
Employment: High-skilled male	0.0	...
Employment: High-skilled female	0.0	...
Employment hours: Low-skilled male	0.0	...
Employment hours: Low-skilled female	0.0	...
Employment hours: Medium-skilled male	0.0	...
Employment hours: Medium-skilled female	0.0	...
Employment hours: High-skilled male	0.0	...
Employment hours: High-skilled female	0.0	...
Employment: Vulnerable employment	0.0	...
Employment hours: Vulnerable employment	0.0	...
CO2 - combustion - air	0.0	...
CH4 - combustion - air	0.0	...
N2O - combustion - air	0.0	...
SOx - combustion - air	0.0	...

NOx - combustion - air	0.0	...
NH3 - combustion - air	0.0	...
CO - combustion - air	0.0	...
...	...	...
Water Withdrawal Blue - Manufacturing - Radio, ...	0.0	...
Water Withdrawal Blue - Manufacturing - Medical...	0.0	...
Water Withdrawal Blue - Manufacturing - Motor v...	0.0	...
Water Withdrawal Blue - Manufacturing - Other t...	0.0	...
Water Withdrawal Blue - Manufacturing - Furnitu...	0.0	...
Water Withdrawal Blue - Electricity - tower - E...	0.0	...
Water Withdrawal Blue - Electricity - tower - E...	0.0	...
Water Withdrawal Blue - Electricity - tower - E...	0.0	...
Water Withdrawal Blue - Electricity - tower - E...	0.0	...
Water Withdrawal Blue - Electricity - tower - E...	0.0	...
Water Withdrawal Blue - Electricity - tower - E...	0.0	...
Water Withdrawal Blue - Electricity - tower - E...	0.0	...
Water Withdrawal Blue - Electricity - tower - E...	0.0	...
Water Withdrawal Blue - Electricity - tower - E...	0.0	...
Water Withdrawal Blue - Electricity - tower - E...	0.0	...
Water Withdrawal Blue - Electricity - tower - E...	0.0	...
Water Withdrawal Blue - Electricity - once-thro...	0.0	...
Water Withdrawal Blue - Electricity - once-thro...	0.0	...
Water Withdrawal Blue - Electricity - once-thro...	0.0	...
Water Withdrawal Blue - Electricity - once-thro...	0.0	...
Water Withdrawal Blue - Electricity - once-thro...	0.0	...
Water Withdrawal Blue - Electricity - once-thro...	0.0	...
Water Withdrawal Blue - Electricity - once-thro...	0.0	...
Water Withdrawal Blue - Electricity - once-thro...	0.0	...
Water Withdrawal Blue - Electricity - once-thro...	0.0	...
Water Withdrawal Blue - Electricity - once-thro...	0.0	...
Water Withdrawal Blue - Electricity - once-thro...	0.0	...
Water Withdrawal Blue - Domestic - domestic Wat...	0.0	...

	Paper for treatment: landfill	\
Taxes less subsidies on products purchased: Total	0.0	
Other net taxes on production	0.0	
Compensation of employees; wages, salaries, & e...	0.0	
Compensation of employees; wages, salaries, & e...	0.0	
Compensation of employees; wages, salaries, & e...	0.0	
Operating surplus: Consumption of fixed capital	0.0	
Operating surplus: Rents on land	0.0	
Operating surplus: Royalties on resources	0.0	
Operating surplus: Remaining net operating surplus	0.0	
Employment: Low-skilled male	0.0	
Employment: Low-skilled female	0.0	
Employment: Medium-skilled male	0.0	

Employment: Medium-skilled female	0.0
Employment: High-skilled male	0.0
Employment: High-skilled female	0.0
Employment hours: Low-skilled male	0.0
Employment hours: Low-skilled female	0.0
Employment hours: Medium-skilled male	0.0
Employment hours: Medium-skilled female	0.0
Employment hours: High-skilled male	0.0
Employment hours: High-skilled female	0.0
Employment: Vulnerable employment	0.0
Employment hours: Vulnerable employment	0.0
CO2 - combustion - air	0.0
CH4 - combustion - air	0.0
N2O - combustion - air	0.0
SOx - combustion - air	0.0
NOx - combustion - air	0.0
NH3 - combustion - air	0.0
CO - combustion - air	0.0
...	...
Water Withdrawal Blue - Manufacturing - Radio, ...	0.0
Water Withdrawal Blue - Manufacturing - Medical...	0.0
Water Withdrawal Blue - Manufacturing - Motor v...	0.0
Water Withdrawal Blue - Manufacturing - Other t...	0.0
Water Withdrawal Blue - Manufacturing - Furnitu...	0.0
Water Withdrawal Blue - Electricity - tower - E...	0.0
Water Withdrawal Blue - Electricity - tower - E...	0.0
Water Withdrawal Blue - Electricity - tower - E...	0.0
Water Withdrawal Blue - Electricity - tower - E...	0.0
Water Withdrawal Blue - Electricity - tower - E...	0.0
Water Withdrawal Blue - Electricity - tower - E...	0.0
Water Withdrawal Blue - Electricity - tower - E...	0.0
Water Withdrawal Blue - Electricity - tower - E...	0.0
Water Withdrawal Blue - Electricity - tower - E...	0.0
Water Withdrawal Blue - Electricity - tower - E...	0.0
Water Withdrawal Blue - Electricity - tower - E...	0.0
Water Withdrawal Blue - Electricity - once-thro...	0.0
Water Withdrawal Blue - Electricity - once-thro...	0.0
Water Withdrawal Blue - Electricity - once-thro...	0.0
Water Withdrawal Blue - Electricity - once-thro...	0.0
Water Withdrawal Blue - Electricity - once-thro...	0.0
Water Withdrawal Blue - Electricity - once-thro...	0.0
Water Withdrawal Blue - Electricity - once-thro...	0.0
Water Withdrawal Blue - Electricity - once-thro...	0.0
Water Withdrawal Blue - Electricity - once-thro...	0.0
Water Withdrawal Blue - Electricity - once-thro...	0.0
Water Withdrawal Blue - Electricity - once-thro...	0.0
Water Withdrawal Blue - Electricity - once-thro...	0.0

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Water Withdrawal Blue - Domestic - domestic Wat...                          0.0

                                                        Plastic waste for treatment: landf
Taxes less subsidies on products purchased: Total
Other net taxes on production
Compensation of employees; wages, salaries, & e...
Compensation of employees; wages, salaries, & e...
Compensation of employees; wages, salaries, & e...
Operating surplus: Consumption of fixed capital
Operating surplus: Rents on land
Operating surplus: Royalties on resources
Operating surplus: Remaining net operating surplus
Employment: Low-skilled male
Employment: Low-skilled female
Employment: Medium-skilled male
Employment: Medium-skilled female
Employment: High-skilled male
Employment: High-skilled female
Employment hours: Low-skilled male
Employment hours: Low-skilled female
Employment hours: Medium-skilled male
Employment hours: Medium-skilled female
Employment hours: High-skilled male
Employment hours: High-skilled female
Employment: Vulnerable employment
Employment hours: Vulnerable employment
CO2 - combustion - air
CH4 - combustion - air
N2O - combustion - air
SOx - combustion - air
NOx - combustion - air
NH3 - combustion - air
CO - combustion - air
...
Water Withdrawal Blue - Manufacturing - Radio, ...
Water Withdrawal Blue - Manufacturing - Medical...
Water Withdrawal Blue - Manufacturing - Motor v...
Water Withdrawal Blue - Manufacturing - Other t...
Water Withdrawal Blue - Manufacturing - Furnitu...
Water Withdrawal Blue - Electricity - tower - E...
Water Withdrawal Blue - Electricity - tower - E...
Water Withdrawal Blue - Electricity - tower - E...
Water Withdrawal Blue - Electricity - tower - E...
Water Withdrawal Blue - Electricity - tower - E...
Water Withdrawal Blue - Electricity - tower - E...
Water Withdrawal Blue - Electricity - tower - E...
Water Withdrawal Blue - Electricity - tower - E...
Water Withdrawal Blue - Electricity - tower - E...
```

Water Withdrawal Blue - Electricity - tower - E...
Water Withdrawal Blue - Electricity - tower - E...
Water Withdrawal Blue - Electricity - tower - E...
Water Withdrawal Blue - Electricity - once-thro...
Water Withdrawal Blue - Electricity - once-thro...
Water Withdrawal Blue - Electricity - once-thro...
Water Withdrawal Blue - Electricity - once-thro...
Water Withdrawal Blue - Electricity - once-thro...
Water Withdrawal Blue - Electricity - once-thro...
Water Withdrawal Blue - Electricity - once-thro...
Water Withdrawal Blue - Electricity - once-thro...
Water Withdrawal Blue - Electricity - once-thro...
Water Withdrawal Blue - Electricity - once-thro...
Water Withdrawal Blue - Electricity - once-thro...
Water Withdrawal Blue - Electricity - once-thro...
Water Withdrawal Blue - Electricity - once-thro...
Water Withdrawal Blue - Domestic - domestic Wat...

Inert/metal/hazardous waste for tr
Taxes less subsidies on products purchased: Total
Other net taxes on production
Compensation of employees; wages, salaries, & e...
Compensation of employees; wages, salaries, & e...
Compensation of employees; wages, salaries, & e...
Operating surplus: Consumption of fixed capital
Operating surplus: Rents on land
Operating surplus: Royalties on resources
Operating surplus: Remaining net operating surplus
Employment: Low-skilled male
Employment: Low-skilled female
Employment: Medium-skilled male
Employment: Medium-skilled female
Employment: High-skilled male
Employment: High-skilled female
Employment hours: Low-skilled male
Employment hours: Low-skilled female
Employment hours: Medium-skilled male
Employment hours: Medium-skilled female
Employment hours: High-skilled male
Employment hours: High-skilled female
Employment: Vulnerable employment
Employment hours: Vulnerable employment
CO2 - combustion - air
CH4 - combustion - air
N2O - combustion - air
SOx - combustion - air
NOx - combustion - air
NH3 - combustion - air
CO - combustion - air

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...
Water Withdrawal Blue - Manufacturing - Radio, ...
Water Withdrawal Blue - Manufacturing - Medical...
Water Withdrawal Blue - Manufacturing - Motor v...
Water Withdrawal Blue - Manufacturing - Other t...
Water Withdrawal Blue - Manufacturing - Furnitu...
Water Withdrawal Blue - Electricity - tower - E...
Water Withdrawal Blue - Electricity - tower - E...
Water Withdrawal Blue - Electricity - tower - E...
Water Withdrawal Blue - Electricity - tower - E...
Water Withdrawal Blue - Electricity - tower - E...
Water Withdrawal Blue - Electricity - tower - E...
Water Withdrawal Blue - Electricity - tower - E...
Water Withdrawal Blue - Electricity - tower - E...
Water Withdrawal Blue - Electricity - tower - E...
Water Withdrawal Blue - Electricity - tower - E...
Water Withdrawal Blue - Electricity - tower - E...
Water Withdrawal Blue - Electricity - tower - E...
Water Withdrawal Blue - Electricity - once-thro...
Water Withdrawal Blue - Electricity - once-thro...
Water Withdrawal Blue - Electricity - once-thro...
Water Withdrawal Blue - Electricity - once-thro...
Water Withdrawal Blue - Electricity - once-thro...
Water Withdrawal Blue - Electricity - once-thro...
Water Withdrawal Blue - Electricity - once-thro...
Water Withdrawal Blue - Electricity - once-thro...
Water Withdrawal Blue - Electricity - once-thro...
Water Withdrawal Blue - Electricity - once-thro...
Water Withdrawal Blue - Electricity - once-thro...
Water Withdrawal Blue - Electricity - once-thro...
Water Withdrawal Blue - Domestic - domestic Wat...

                                                    Textiles waste for treatment: lan
Taxes less subsidies on products purchased: Total
Other net taxes on production
Compensation of employees; wages, salaries, & e...
Compensation of employees; wages, salaries, & e...
Compensation of employees; wages, salaries, & e...
Operating surplus: Consumption of fixed capital
Operating surplus: Rents on land
Operating surplus: Royalties on resources
Operating surplus: Remaining net operating surplus
Employment: Low-skilled male
Employment: Low-skilled female
Employment: Medium-skilled male
Employment: Medium-skilled female
Employment: High-skilled male
Employment: High-skilled female
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Employment hours: Low-skilled male
Employment hours: Low-skilled female
Employment hours: Medium-skilled male
Employment hours: Medium-skilled female
Employment hours: High-skilled male
Employment hours: High-skilled female
Employment: Vulnerable employment
Employment hours: Vulnerable employment
CO2 - combustion - air
CH4 - combustion - air
N2O - combustion - air
SOx - combustion - air
NOx - combustion - air
NH3 - combustion - air
CO - combustion - air
...
Water Withdrawal Blue - Manufacturing - Radio, ...
Water Withdrawal Blue - Manufacturing - Medical...
Water Withdrawal Blue - Manufacturing - Motor v...
Water Withdrawal Blue - Manufacturing - Other t...
Water Withdrawal Blue - Manufacturing - Furnitu...
Water Withdrawal Blue - Electricity - tower - E...
Water Withdrawal Blue - Electricity - tower - E...
Water Withdrawal Blue - Electricity - tower - E...
Water Withdrawal Blue - Electricity - tower - E...
Water Withdrawal Blue - Electricity - tower - E...
Water Withdrawal Blue - Electricity - tower - E...
Water Withdrawal Blue - Electricity - tower - E...
Water Withdrawal Blue - Electricity - tower - E...
Water Withdrawal Blue - Electricity - tower - E...
Water Withdrawal Blue - Electricity - tower - E...
Water Withdrawal Blue - Electricity - tower - E...
Water Withdrawal Blue - Electricity - tower - E...
Water Withdrawal Blue - Electricity - once-thro...
Water Withdrawal Blue - Electricity - once-thro...
Water Withdrawal Blue - Electricity - once-thro...
Water Withdrawal Blue - Electricity - once-thro...
Water Withdrawal Blue - Electricity - once-thro...
Water Withdrawal Blue - Electricity - once-thro...
Water Withdrawal Blue - Electricity - once-thro...
Water Withdrawal Blue - Electricity - once-thro...
Water Withdrawal Blue - Electricity - once-thro...
Water Withdrawal Blue - Electricity - once-thro...
Water Withdrawal Blue - Electricity - once-thro...
Water Withdrawal Blue - Electricity - once-thro...
Water Withdrawal Blue - Domestic - domestic Wat...
```

Wood waste for treatment: landfil

Taxes less subsidies on products purchased: Total	0.0
Other net taxes on production	0.0
Compensation of employees; wages, salaries, & e...	0.0
Compensation of employees; wages, salaries, & e...	0.0
Compensation of employees; wages, salaries, & e...	0.0
Operating surplus: Consumption of fixed capital	0.0
Operating surplus: Rents on land	0.0
Operating surplus: Royalties on resources	0.0
Operating surplus: Remaining net operating surplus	0.0
Employment: Low-skilled male	0.0
Employment: Low-skilled female	0.0
Employment: Medium-skilled male	0.0
Employment: Medium-skilled female	0.0
Employment: High-skilled male	0.0
Employment: High-skilled female	0.0
Employment hours: Low-skilled male	0.0
Employment hours: Low-skilled female	0.0
Employment hours: Medium-skilled male	0.0
Employment hours: Medium-skilled female	0.0
Employment hours: High-skilled male	0.0
Employment hours: High-skilled female	0.0
Employment: Vulnerable employment	0.0
Employment hours: Vulnerable employment	0.0
CO2 - combustion - air	0.0
CH4 - combustion - air	0.0
N2O - combustion - air	0.0
SOx - combustion - air	0.0
NOx - combustion - air	0.0
NH3 - combustion - air	0.0
CO - combustion - air	0.0
...	..
Water Withdrawal Blue - Manufacturing - Radio, ...	0.0
Water Withdrawal Blue - Manufacturing - Medical...	0.0
Water Withdrawal Blue - Manufacturing - Motor v...	0.0
Water Withdrawal Blue - Manufacturing - Other t...	0.0
Water Withdrawal Blue - Manufacturing - Furnitu...	0.0
Water Withdrawal Blue - Electricity - tower - E...	0.0
Water Withdrawal Blue - Electricity - tower - E...	0.0
Water Withdrawal Blue - Electricity - tower - E...	0.0
Water Withdrawal Blue - Electricity - tower - E...	0.0
Water Withdrawal Blue - Electricity - tower - E...	0.0
Water Withdrawal Blue - Electricity - tower - E...	0.0
Water Withdrawal Blue - Electricity - tower - E...	0.0
Water Withdrawal Blue - Electricity - tower - E...	0.0
Water Withdrawal Blue - Electricity - tower - E...	0.0
Water Withdrawal Blue - Electricity - tower - E...	0.0
Water Withdrawal Blue - Electricity - tower - E...	0.0
Water Withdrawal Blue - Electricity - tower - E...	0.0

Water Withdrawal Blue - Electricity - once-thro...	0.0
Water Withdrawal Blue - Electricity - once-thro...	0.0
Water Withdrawal Blue - Electricity - once-thro...	0.0
Water Withdrawal Blue - Electricity - once-thro...	0.0
Water Withdrawal Blue - Electricity - once-thro...	0.0
Water Withdrawal Blue - Electricity - once-thro...	0.0
Water Withdrawal Blue - Electricity - once-thro...	0.0
Water Withdrawal Blue - Electricity - once-thro...	0.0
Water Withdrawal Blue - Electricity - once-thro...	0.0
Water Withdrawal Blue - Electricity - once-thro...	0.0
Water Withdrawal Blue - Electricity - once-thro...	0.0
Water Withdrawal Blue - Electricity - once-thro...	0.0
Water Withdrawal Blue - Domestic - domestic Wat...	0.0

	Membership organisation services r
Taxes less subsidies on products purchased: Total	4
Other net taxes on production	2
Compensation of employees; wages, salaries, & e...	3
Compensation of employees; wages, salaries, & e...	2
Compensation of employees; wages, salaries, & e...	3
Operating surplus: Consumption of fixed capital	1
Operating surplus: Rents on land	0
Operating surplus: Royalties on resources	0
Operating surplus: Remaining net operating surplus	1
Employment: Low-skilled male	8
Employment: Low-skilled female	7
Employment: Medium-skilled male	4
Employment: Medium-skilled female	4
Employment: High-skilled male	3
Employment: High-skilled female	3
Employment hours: Low-skilled male	1
Employment hours: Low-skilled female	1
Employment hours: Medium-skilled male	9
Employment hours: Medium-skilled female	7
Employment hours: High-skilled male	5
Employment hours: High-skilled female	5
Employment: Vulnerable employment	4
Employment hours: Vulnerable employment	9
CO2 - combustion - air	1
CH4 - combustion - air	3
N2O - combustion - air	4
SOx - combustion - air	2
NOx - combustion - air	4
NH3 - combustion - air	6
CO - combustion - air	3
...	
Water Withdrawal Blue - Manufacturing - Radio, ...	5
Water Withdrawal Blue - Manufacturing - Medical...	1

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Water Withdrawal Blue - Manufacturing - Motor v...                                  1
Water Withdrawal Blue - Manufacturing - Other t...                                  2
Water Withdrawal Blue - Manufacturing - Furnitu...                                  5
Water Withdrawal Blue - Electricity - tower - E...                                  1
Water Withdrawal Blue - Electricity - tower - E...                                  3
Water Withdrawal Blue - Electricity - tower - E...                                  3
Water Withdrawal Blue - Electricity - tower - E...                                  0
Water Withdrawal Blue - Electricity - tower - E...                                  0
Water Withdrawal Blue - Electricity - tower - E...                                  1
Water Withdrawal Blue - Electricity - tower - E...                                  4
Water Withdrawal Blue - Electricity - tower - E...                                  0
Water Withdrawal Blue - Electricity - tower - E...                                  5
Water Withdrawal Blue - Electricity - tower - E...                                  0
Water Withdrawal Blue - Electricity - tower - E...                                  1
Water Withdrawal Blue - Electricity - tower - E...                                  0
Water Withdrawal Blue - Electricity - once-thro...                                  3
Water Withdrawal Blue - Electricity - once-thro...                                  2
Water Withdrawal Blue - Electricity - once-thro...                                  1
Water Withdrawal Blue - Electricity - once-thro...                                  0
Water Withdrawal Blue - Electricity - once-thro...                                  0
Water Withdrawal Blue - Electricity - once-thro...                                  6
Water Withdrawal Blue - Electricity - once-thro...                                  1
Water Withdrawal Blue - Electricity - once-thro...                                  0
Water Withdrawal Blue - Electricity - once-thro...                                  0
Water Withdrawal Blue - Electricity - once-thro...                                  0
Water Withdrawal Blue - Electricity - once-thro...                                  7
Water Withdrawal Blue - Electricity - once-thro...                                  0
Water Withdrawal Blue - Domestic - domestic Wat...                                  0

                                                     Recreational, cultural and sportin
Taxes less subsidies on products purchased: Total
Other net taxes on production
Compensation of employees; wages, salaries, & e...
Compensation of employees; wages, salaries, & e...
Compensation of employees; wages, salaries, & e...
Operating surplus: Consumption of fixed capital
Operating surplus: Rents on land
Operating surplus: Royalties on resources
Operating surplus: Remaining net operating surplus
Employment: Low-skilled male
Employment: Low-skilled female
Employment: Medium-skilled male
Employment: Medium-skilled female
Employment: High-skilled male
Employment: High-skilled female
Employment hours: Low-skilled male
Employment hours: Low-skilled female
Employment hours: Medium-skilled male
```

```
Employment hours: Medium-skilled female
Employment hours: High-skilled male
Employment hours: High-skilled female
Employment: Vulnerable employment
Employment hours: Vulnerable employment
CO2 - combustion - air
CH4 - combustion - air
N2O - combustion - air
SOx - combustion - air
NOx - combustion - air
NH3 - combustion - air
CO - combustion - air
...
Water Withdrawal Blue - Manufacturing - Radio, ...
Water Withdrawal Blue - Manufacturing - Medical...
Water Withdrawal Blue - Manufacturing - Motor v...
Water Withdrawal Blue - Manufacturing - Other t...
Water Withdrawal Blue - Manufacturing - Furnitu...
Water Withdrawal Blue - Electricity - tower - E...
Water Withdrawal Blue - Electricity - tower - E...
Water Withdrawal Blue - Electricity - tower - E...
Water Withdrawal Blue - Electricity - tower - E...
Water Withdrawal Blue - Electricity - tower - E...
Water Withdrawal Blue - Electricity - tower - E...
Water Withdrawal Blue - Electricity - tower - E...
Water Withdrawal Blue - Electricity - tower - E...
Water Withdrawal Blue - Electricity - tower - E...
Water Withdrawal Blue - Electricity - tower - E...
Water Withdrawal Blue - Electricity - tower - E...
Water Withdrawal Blue - Electricity - once-thro...
Water Withdrawal Blue - Electricity - once-thro...
Water Withdrawal Blue - Electricity - once-thro...
Water Withdrawal Blue - Electricity - once-thro...
Water Withdrawal Blue - Electricity - once-thro...
Water Withdrawal Blue - Electricity - once-thro...
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Water Withdrawal Blue - Electricity - once-thro...
Water Withdrawal Blue - Electricity - once-thro...
Water Withdrawal Blue - Electricity - once-thro...
Water Withdrawal Blue - Electricity - once-thro...
Water Withdrawal Blue - Electricity - once-thro...
Water Withdrawal Blue - Domestic - domestic Wat...

                                                 Other services (93)  \
Taxes less subsidies on products purchased: Total         2.752545e-08
Other net taxes on production                             2.979429e-08
Compensation of employees; wages, salaries, & e...        1.954562e-08
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Compensation of employees; wages, salaries, & e...	1.849134e-07
Compensation of employees; wages, salaries, & e...	2.265964e-07
Operating surplus: Consumption of fixed capital	9.198943e-08
Operating surplus: Rents on land	0.000000e+00
Operating surplus: Royalties on resources	0.000000e+00
Operating surplus: Remaining net operating surplus	4.196354e-07
Employment: Low-skilled male	6.038437e-10
Employment: Low-skilled female	4.869671e-10
Employment: Medium-skilled male	3.456101e-09
Employment: Medium-skilled female	2.753553e-09
Employment: High-skilled male	1.973825e-09
Employment: High-skilled female	1.862655e-09
Employment hours: Low-skilled male	1.314449e-09
Employment hours: Low-skilled female	9.710378e-10
Employment hours: Medium-skilled male	7.086594e-09
Employment hours: Medium-skilled female	5.300809e-09
Employment hours: High-skilled male	3.689943e-09
Employment hours: High-skilled female	3.380505e-09
Employment: Vulnerable employment	3.473413e-09
Employment hours: Vulnerable employment	7.245417e-09
CO2 - combustion - air	7.696662e-02
CH4 - combustion - air	1.676496e-05
N2O - combustion - air	2.261890e-06
SOx - combustion - air	1.498653e-04
NOx - combustion - air	3.502875e-04
NH3 - combustion - air	8.546348e-05
CO - combustion - air	2.905015e-04
...	...
Water Withdrawal Blue - Manufacturing - Radio, ...	1.435694e-11
Water Withdrawal Blue - Manufacturing - Medical...	1.309104e-11
Water Withdrawal Blue - Manufacturing - Motor v...	1.076598e-13
Water Withdrawal Blue - Manufacturing - Other t...	1.852517e-13
Water Withdrawal Blue - Manufacturing - Furnitu...	1.086165e-11
Water Withdrawal Blue - Electricity - tower - E...	1.452128e-11
Water Withdrawal Blue - Electricity - tower - E...	2.631234e-12
Water Withdrawal Blue - Electricity - tower - E...	2.877296e-12
Water Withdrawal Blue - Electricity - tower - E...	0.000000e+00
Water Withdrawal Blue - Electricity - tower - E...	0.000000e+00
Water Withdrawal Blue - Electricity - tower - E...	1.176050e-13
Water Withdrawal Blue - Electricity - tower - E...	5.589310e-12
Water Withdrawal Blue - Electricity - tower - E...	0.000000e+00
Water Withdrawal Blue - Electricity - tower - E...	5.158235e-15
Water Withdrawal Blue - Electricity - tower - E...	0.000000e+00
Water Withdrawal Blue - Electricity - tower - E...	1.540954e-13
Water Withdrawal Blue - Electricity - tower - E...	0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...	2.962990e-10
Water Withdrawal Blue - Electricity - once-thro...	1.917182e-10
Water Withdrawal Blue - Electricity - once-thro...	1.245686e-10

Water Withdrawal Blue - Electricity - once-thro...	0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...	0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...	4.983584e-11
Water Withdrawal Blue - Electricity - once-thro...	1.832865e-11
Water Withdrawal Blue - Electricity - once-thro...	0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...	0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...	0.000000e+00
Water Withdrawal Blue - Electricity - once-thro...	5.891735e-13
Water Withdrawal Blue - Electricity - once-thro...	0.000000e+00
Water Withdrawal Blue - Domestic - domestic Wat...	0.000000e+00

	Private households with employed p
Taxes less subsidies on products purchased: Total	[illegible]
Other net taxes on production	[illegible]
Compensation of employees; wages, salaries, & e...	[illegible]
Compensation of employees; wages, salaries, & e...	[illegible]
Compensation of employees; wages, salaries, & e...	[illegible]
Operating surplus: Consumption of fixed capital	[illegible]
Operating surplus: Rents on land	[illegible]
Operating surplus: Royalties on resources	[illegible]
Operating surplus: Remaining net operating surplus	[illegible]
Employment: Low-skilled male	[illegible]
Employment: Low-skilled female	[illegible]
Employment: Medium-skilled male	[illegible]
Employment: Medium-skilled female	[illegible]
Employment: High-skilled male	[illegible]
Employment: High-skilled female	[illegible]
Employment hours: Low-skilled male	[illegible]
Employment hours: Low-skilled female	[illegible]
Employment hours: Medium-skilled male	[illegible]
Employment hours: Medium-skilled female	[illegible]
Employment hours: High-skilled male	[illegible]
Employment hours: High-skilled female	[illegible]
Employment: Vulnerable employment	[illegible]
Employment hours: Vulnerable employment	[illegible]
CO2 - combustion - air	[illegible]
CH4 - combustion - air	[illegible]
N2O - combustion - air	[illegible]
SOx - combustion - air	[illegible]
NOx - combustion - air	[illegible]
NH3 - combustion - air	[illegible]
CO - combustion - air	[illegible]
...	
Water Withdrawal Blue - Manufacturing - Radio, ...	[illegible]
Water Withdrawal Blue - Manufacturing - Medical...	[illegible]
Water Withdrawal Blue - Manufacturing - Motor v...	[illegible]
Water Withdrawal Blue - Manufacturing - Other t...	[illegible]
Water Withdrawal Blue - Manufacturing - Furnitu...	[illegible]

Water Withdrawal Blue - Electricity - tower - E...	3
Water Withdrawal Blue - Electricity - tower - E...	5
Water Withdrawal Blue - Electricity - tower - E...	6
Water Withdrawal Blue - Electricity - tower - E...	0
Water Withdrawal Blue - Electricity - tower - E...	0
Water Withdrawal Blue - Electricity - tower - E...	3
Water Withdrawal Blue - Electricity - tower - E...	6
Water Withdrawal Blue - Electricity - tower - E...	0
Water Withdrawal Blue - Electricity - tower - E...	6
Water Withdrawal Blue - Electricity - tower - E...	0
Water Withdrawal Blue - Electricity - tower - E...	2
Water Withdrawal Blue - Electricity - tower - E...	0
Water Withdrawal Blue - Electricity - once-thro...	7
Water Withdrawal Blue - Electricity - once-thro...	4
Water Withdrawal Blue - Electricity - once-thro...	2
Water Withdrawal Blue - Electricity - once-thro...	0
Water Withdrawal Blue - Electricity - once-thro...	0
Water Withdrawal Blue - Electricity - once-thro...	1
Water Withdrawal Blue - Electricity - once-thro...	2
Water Withdrawal Blue - Electricity - once-thro...	0
Water Withdrawal Blue - Electricity - once-thro...	0
Water Withdrawal Blue - Electricity - once-thro...	0
Water Withdrawal Blue - Electricity - once-thro...	1
Water Withdrawal Blue - Electricity - once-thro...	0
Water Withdrawal Blue - Domestic - domestic Wat...	0

	Extra-territorial organizations ar
Taxes less subsidies on products purchased: Total	
Other net taxes on production	
Compensation of employees; wages, salaries, & e...	
Compensation of employees; wages, salaries, & e...	
Compensation of employees; wages, salaries, & e...	
Operating surplus: Consumption of fixed capital	
Operating surplus: Rents on land	
Operating surplus: Royalties on resources	
Operating surplus: Remaining net operating surplus	
Employment: Low-skilled male	
Employment: Low-skilled female	
Employment: Medium-skilled male	
Employment: Medium-skilled female	
Employment: High-skilled male	
Employment: High-skilled female	
Employment hours: Low-skilled male	
Employment hours: Low-skilled female	
Employment hours: Medium-skilled male	
Employment hours: Medium-skilled female	
Employment hours: High-skilled male	
Employment hours: High-skilled female	

```
Employment: Vulnerable employment
Employment hours: Vulnerable employment
CO2 - combustion - air
CH4 - combustion - air
N2O - combustion - air
SOx - combustion - air
NOx - combustion - air
NH3 - combustion - air
CO - combustion - air
...
Water Withdrawal Blue - Manufacturing - Radio, ...
Water Withdrawal Blue - Manufacturing - Medical...
Water Withdrawal Blue - Manufacturing - Motor v...
Water Withdrawal Blue - Manufacturing - Other t...
Water Withdrawal Blue - Manufacturing - Furnitu...
Water Withdrawal Blue - Electricity - tower - E...
Water Withdrawal Blue - Electricity - tower - E...
Water Withdrawal Blue - Electricity - tower - E...
Water Withdrawal Blue - Electricity - tower - E...
Water Withdrawal Blue - Electricity - tower - E...
Water Withdrawal Blue - Electricity - tower - E...
Water Withdrawal Blue - Electricity - tower - E...
Water Withdrawal Blue - Electricity - tower - E...
Water Withdrawal Blue - Electricity - tower - E...
Water Withdrawal Blue - Electricity - tower - E...
Water Withdrawal Blue - Electricity - tower - E...
Water Withdrawal Blue - Electricity - once-thro...
Water Withdrawal Blue - Electricity - once-thro...
Water Withdrawal Blue - Electricity - once-thro...
Water Withdrawal Blue - Electricity - once-thro...
Water Withdrawal Blue - Electricity - once-thro...
Water Withdrawal Blue - Electricity - once-thro...
Water Withdrawal Blue - Electricity - once-thro...
Water Withdrawal Blue - Electricity - once-thro...
Water Withdrawal Blue - Electricity - once-thro...
Water Withdrawal Blue - Electricity - once-thro...
Water Withdrawal Blue - Electricity - once-thro...
Water Withdrawal Blue - Domestic - domestic Wat...

[1104 rows x 200 columns]
```

In [85]:
```
M_file=M_SE.dot(characterisation_HBS.T)
M_file.to_csv('C:/Users/tobia/Dropbox/KTH Sustainable Technology/4_AL250X_Degree Proje
```

In [64]:
```
# Searching for differences in the two indexes
len(list(set(concordance.index) - set(M_SE.columns)))
```

```
for var in range(0,200):
    if concordance.index[var] != M_SE_t.columns[var]:
        print(concordance.index[var] + " \n " + M_SE_t.columns[var] + '\n \n')
```

In [65]:
```
# Multiplying the two matrices to get stressors per HBS category
HBS_stressor = M_SE_t.dot(concordance)
HBS_stressor_t = HBS_stressor.T
HBS_stressor_t.index = Swedish_HBS_final.index.copy()
HBS_stressor_t
```

Out[65]:

Division	Group
Food	Bread and cereals
	Meat
	Fish and seafood
	Milk, cheese and eggs
...	
Housing	Rental for housing (incl. garage)
	Maintenance
	Household's insurance
	Water supply and other services

	Electricity, gas and other fuels
	Interests rate (gross)
	Real estate fee
	Rent for a leasehold site
Furnishings, household equipment and maintenanc...	Furniture and carpets
	Textiles
	Household appliance
Health and hospital services	NaN
Transport	Purchase of motorcars
	Purchase of other vehicles
	Operation of motorcars
	Interests & vehicle tax, motorcars
	Operation of other vehicles
	Interests & vehicle tax, other vehi
	Local travel, transportation servic
Leisure and culture	Weekend cottage
	Radio and tv
	Sport and hobby
	Watch, camera, photographic and cin
	Vacation, hotels
	Other leisure
	Entertainment
	Books, newspapers, television fees
	Mobil phones
	Telephone
Taxable benefits	NaN

Division	Group
Food	Bread and cereals
	Meat

	Fish and seafood
	Milk, cheese and eggs
...	
Housing	Rental for housing (incl. garage)
	Maintenance
	Household's insurance
	Water supply and other services
	Electricity, gas and other fuels
	Interests rate (gross)
	Real estate fee
	Rent for a leasehold site
Furnishings, household equipment and maintenanc...	Furniture and carpets
	Textiles
	Household appliance
Health and hospital services	NaN
Transport	Purchase of motorcars
	Purchase of other vehicles
	Operation of motorcars
	Interests & vehicle tax, motorcars
	Operation of other vehicles
	Interests & vehicle tax, other vehi
	Local travel, transportation servic
Leisure and culture	Weekend cottage
	Radio and tv
	Sport and hobby
	Watch, camera, photographic and ci
	Vacation, hotels
	Other leisure
	Entertainment
	Books, newspapers, television fees
	Mobil phones
	Telephone
Taxable benefits	NaN

Division	Group
Food	Bread and cereals

	Meat
	Fish and seafood
	Milk, cheese and eggs
...	
Housing	Rental for housing (incl. garage)
	Maintenance
	Household's insurance
	Water supply and other services
	Electricity, gas and other fuels
	Interests rate (gross)
	Real estate fee
	Rent for a leasehold site
Furnishings, household equipment and maintenanc...	Furniture and carpets
	Textiles
	Household appliance
Health and hospital services	NaN
Transport	Purchase of motorcars
	Purchase of other vehicles
	Operation of motorcars
	Interests & vehicle tax, motorcars
	Operation of other vehicles
	Interests & vehicle tax, other vehi
	Local travel, transportation servic
Leisure and culture	Weekend cottage

	Radio and tv
	Sport and hobby
	Watch, camera, photographic and ci
	Vacation, hotels
	Other leisure
	Entertainment
	Books, newspapers, television fees
	Mobil phones
	Telephone
Taxable benefits	NaN

Division	Group
Food	Bread and cereals
	Meat
	Fish and seafood
	Milk, cheese and eggs
...	
Housing	Rental for housing (incl. garage)
	Maintenance
	Household's insurance
	Water supply and other services

	Electricity, gas and other fuels
	Interests rate (gross)
	Real estate fee
	Rent for a leasehold site
Furnishings, household equipment and maintenanc...	Furniture and carpets
	Textiles
	Household appliance
Health and hospital services	NaN
Transport	Purchase of motorcars
	Purchase of other vehicles
	Operation of motorcars
	Interests & vehicle tax, motorcars
	Operation of other vehicles
	Interests & vehicle tax, other vehi
	Local travel, transportation servic
Leisure and culture	Weekend cottage
	Radio and tv
	Sport and hobby
	Watch, camera, photographic and cin
	Vacation, hotels
	Other leisure
	Entertainment
	Books, newspapers, television fees
	Mobil phones
	Telephone
Taxable benefits	NaN

Division	Group
Food	Bread and cereals
	Meat

	Fish and seafood
	Milk, cheese and eggs
...	
Housing	Rental for housing (incl. garage)
	Maintenance
	Household's insurance
	Water supply and other services
	Electricity, gas and other fuels
	Interests rate (gross)
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	Rent for a leasehold site
Furnishings, household equipment and maintenanc...	Furniture and carpets
	Textiles
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Health and hospital services	NaN
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	Operation of motorcars
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	Operation of other vehicles
	Interests & vehicle tax, other vehi
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	Radio and tv
	Sport and hobby
	Watch, camera, photographic and ci
	Vacation, hotels
	Other leisure
	Entertainment
	Books, newspapers, television fees
	Mobil phones
	Telephone
Taxable benefits	NaN

Division	Group
Food	Bread and cereals

	Meat
	Fish and seafood
	Milk, cheese and eggs
...	
Housing	Rental for housing (incl. garage)
	Maintenance
	Household's insurance
	Water supply and other services
	Electricity, gas and other fuels
	Interests rate (gross)
	Real estate fee
	Rent for a leasehold site
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	Textiles
	Household appliance
Health and hospital services	NaN
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	Operation of motorcars
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	Operation of other vehicles
	Interests & vehicle tax, other vehi
	Local travel, transportation servic
Leisure and culture	Weekend cottage

	Radio and tv
	Sport and hobby
	Watch, camera, photographic and ci
	Vacation, hotels
	Other leisure
	Entertainment
	Books, newspapers, television fees
	Mobil phones
	Telephone
Taxable benefits	NaN

Division	Group
Food	Bread and cereals
	Meat
	Fish and seafood
	Milk, cheese and eggs
...	
Housing	Rental for housing (incl. garage)
	Maintenance
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	Water supply and other services

	Electricity, gas and other fuels
	Interests rate (gross)
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	Vacation, hotels
	Other leisure
	Entertainment
	Books, newspapers, television fees
	Mobil phones
	Telephone
Taxable benefits	NaN

Division	Group
Food	Bread and cereals
	Meat

	Fish and seafood
	Milk, cheese and eggs
...	
Housing	Rental for housing (incl. garage)
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	Water supply and other services
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	Textiles
	Household appliance
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	Operation of other vehicles
	Interests & vehicle tax, other vehi
	Local travel, transportation servic
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	Sport and hobby
	Watch, camera, photographic and ci
	Vacation, hotels
	Other leisure
	Entertainment
	Books, newspapers, television fees
	Mobil phones
	Telephone
Taxable benefits	NaN

Division	Group
Food	Bread and cereals

	Meat
	Fish and seafood
	Milk, cheese and eggs
...	
Housing	Rental for housing (incl. garage)
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	Household's insurance
	Water supply and other services
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	Interests rate (gross)
	Real estate fee
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	Textiles
	Household appliance
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	Purchase of other vehicles
	Operation of motorcars
	Interests & vehicle tax, motorcars
	Operation of other vehicles
	Interests & vehicle tax, other vehi
	Local travel, transportation servic
Leisure and culture	Weekend cottage

	Radio and tv
	Sport and hobby
	Watch, camera, photographic and ci
	Vacation, hotels
	Other leisure
	Entertainment
	Books, newspapers, television fees
	Mobil phones
	Telephone
Taxable benefits	NaN

Division	Group
Food	Bread and cereals
	Meat
	Fish and seafood
	Milk, cheese and eggs
...	
Housing	Rental for housing (incl. garage)
	Maintenance
	Household's insurance
	Water supply and other services

	Electricity, gas and other fuels
	Interests rate (gross)
	Real estate fee
	Rent for a leasehold site
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	Textiles
	Household appliance
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	Interests & vehicle tax, other vehi
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Leisure and culture	Weekend cottage
	Radio and tv
	Sport and hobby
	Watch, camera, photographic and cin
	Vacation, hotels
	Other leisure
	Entertainment
	Books, newspapers, television fees
	Mobil phones
	Telephone
Taxable benefits	NaN

Division	Group
Food	Bread and cereals
	Meat

	Fish and seafood
	Milk, cheese and eggs
...	
Housing	Rental for housing (incl. garage)
	Maintenance
	Household's insurance
	Water supply and other services
	Electricity, gas and other fuels
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	Local travel, transportation servic
Leisure and culture	Weekend cottage
	Radio and tv
	Sport and hobby
	Watch, camera, photographic and ci
	Vacation, hotels
	Other leisure
	Entertainment
	Books, newspapers, television fees
	Mobil phones
	Telephone
Taxable benefits	NaN

Division	Group
Food	Bread and cereals

	Meat
	Fish and seafood
	Milk, cheese and eggs
...	
Housing	Rental for housing (incl. garage)
	Maintenance
	Household's insurance
	Water supply and other services
	Electricity, gas and other fuels
	Interests rate (gross)
	Real estate fee
	Rent for a leasehold site
Furnishings, household equipment and maintenanc...	Furniture and carpets
	Textiles
	Household appliance
Health and hospital services	NaN
Transport	Purchase of motorcars
	Purchase of other vehicles
	Operation of motorcars
	Interests & vehicle tax, motorcars
	Operation of other vehicles
	Interests & vehicle tax, other vehi
	Local travel, transportation servic
Leisure and culture	Weekend cottage

	Radio and tv
	Sport and hobby
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	Vacation, hotels
	Other leisure
	Entertainment
	Books, newspapers, television fees
	Mobil phones
	Telephone
Taxable benefits	NaN

Division	Group
Food	Bread and cereals
	Meat
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Housing	Rental for housing (incl. garage)
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	Household's insurance
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	Electricity, gas and other fuels
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	Vacation, hotels
	Other leisure
	Entertainment
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	Mobil phones
	Telephone
Taxable benefits	NaN

Division	Group
Food	Bread and cereals
	Meat

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	Milk, cheese and eggs
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Housing	Rental for housing (incl. garage)
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	Household's insurance
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	Local travel, transportation servic
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	Vacation, hotels
	Other leisure
	Entertainment
	Books, newspapers, television fees
	Mobil phones
	Telephone
Taxable benefits	NaN

Division	Group
Food	Bread and cereals

	Meat
	Fish and seafood
	Milk, cheese and eggs
...	
Housing	Rental for housing (incl. garage)
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	Entertainment
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	Mobil phones
	Telephone
Taxable benefits	NaN

Division	Group
Food	Bread and cereals

	Meat
	Fish and seafood
	Milk, cheese and eggs
...	
Housing	Rental for housing (incl. garage)
	Maintenance
	Household's insurance
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	Mobil phones
	Telephone
Taxable benefits	NaN

Division	Group
Food	Bread and cereals
	Meat
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	Entertainment
	Books, newspapers, television fees
	Mobil phones
	Telephone
Taxable benefits	NaN

Division	Group
Food	Bread and cereals
	Meat

	Fish and seafood
	Milk, cheese and eggs
...	
Housing	Rental for housing (incl. garage)
	Maintenance
	Household's insurance
	Water supply and other services
	Electricity, gas and other fuels
	Interests rate (gross)
	Real estate fee
	Rent for a leasehold site
Furnishings, household equipment and maintenanc...	Furniture and carpets
	Textiles
	Household appliance
Health and hospital services	NaN
Transport	Purchase of motorcars
	Purchase of other vehicles
	Operation of motorcars
	Interests & vehicle tax, motorcars
	Operation of other vehicles
	Interests & vehicle tax, other vehi
	Local travel, transportation servic
Leisure and culture	Weekend cottage
	Radio and tv
	Sport and hobby
	Watch, camera, photographic and ci
	Vacation, hotels
	Other leisure
	Entertainment
	Books, newspapers, television fees
	Mobil phones
	Telephone
Taxable benefits	NaN

Division	Group
Food	Bread and cereals

	Meat
	Fish and seafood
	Milk, cheese and eggs
...	
Housing	Rental for housing (incl. garage)
	Maintenance
	Household's insurance
	Water supply and other services
	Electricity, gas and other fuels
	Interests rate (gross)
	Real estate fee
	Rent for a leasehold site
Furnishings, household equipment and maintenanc...	Furniture and carpets
	Textiles
	Household appliance
Health and hospital services	NaN
Transport	Purchase of motorcars
	Purchase of other vehicles
	Operation of motorcars
	Interests & vehicle tax, motorcars
	Operation of other vehicles
	Interests & vehicle tax, other vehi
	Local travel, transportation servic
Leisure and culture	Weekend cottage

```
                                                        Radio and tv
                                                        Sport and hobby
                                                        Watch, camera, photographic and ci
                                                        Vacation, hotels
                                                        Other leisure
                                                        Entertainment
                                                        Books, newspapers, television fees
                                                        Mobil phones
                                                        Telephone
Taxable benefits                                        NaN

[120 rows x 1104 columns]
```

5.3 RO 2.3 Muliplying Stressor/HBS and Final Demand/HBS

Multiplying the `HBS_stressor` (containing stressors in HBS) and the `HBS_final` to get the multipliers according to the HBS categories `FD_HBS`.

Check for same indices in the two matrices `HBS_final` and `HBS_stressor`.

```
In [66]: # This code block checks if both matrices contain the same names and amount of indice
         #If yes, it will print: 'Correct amount of strings in Tuples)' if not it will print t
         Success = 0
         Floats = 0
         for tup in range(0,120):
             for string in range(0, 4):
                 if isinstance(Swedish_HBS_final.index[tup][string], float) == True:
                     Floats += 1
                     continue
                 elif Swedish_HBS_final.index[tup][string] != HBS_stressor_t.index[tup][string
                     print("HBS_final: " + Swedish_HBS_final.index[tup][string] + " \n\n HBS_s
                 elif Swedish_HBS_final.index[tup][string] == HBS_stressor_t.index[tup][string
                     Success += 1
                     #print('String pair ' + str(Success) + ' are the same!')
         if Success + Floats == 480: print('Correct amount of strings in Tuples (480)')
```

```
Correct amount of strings in Tuples (480)
```

The matrices `Swedish_HBS_final` can be multiplied with `HBS_stressors` to get the total environmental stressors per HBS category in the variable `FD_HBS`.

```
In [67]: FD_HBS = HBS_stressor_t * Swedish_HBS_final.values
         FD_HBS
```

```
Out[67]:
         Division                                       Group
         Food                                           Bread and cereals
```

	Meat
	Fish and seafood
	Milk, cheese and eggs
...	
Housing	Rental for housing (incl. garage)
	Maintenance
	Household's insurance
	Water supply and other services
	Electricity, gas and other fuels
	Interests rate (gross)
	Real estate fee
	Rent for a leasehold site
Furnishings, household equipment and maintenanc...	Furniture and carpets
	Textiles
	Household appliance
Health and hospital services	NaN
Transport	Purchase of motorcars
	Purchase of other vehicles
	Operation of motorcars
	Interests & vehicle tax, motorcars
	Operation of other vehicles
	Interests & vehicle tax, other vehi
	Local travel, transportation servic
Leisure and culture	Weekend cottage

	Radio and tv
	Sport and hobby
	Watch, camera, photographic and ci
	Vacation, hotels
	Other leisure
	Entertainment
	Books, newspapers, television fees
	Mobil phones
	Telephone
Taxable benefits	NaN

Division	Group
Food	Bread and cereals
	Meat
	Fish and seafood
	Milk, cheese and eggs
...	
Housing	Rental for housing (incl. garage)
	Maintenance
	Household's insurance
	Water supply and other services

	Electricity, gas and other fuels
	Interests rate (gross)
	Real estate fee
	Rent for a leasehold site
Furnishings, household equipment and maintenanc...	Furniture and carpets
	Textiles
	Household appliance
Health and hospital services	NaN
Transport	Purchase of motorcars
	Purchase of other vehicles
	Operation of motorcars
	Interests & vehicle tax, motorcars
	Operation of other vehicles
	Interests & vehicle tax, other vehi
	Local travel, transportation servic
Leisure and culture	Weekend cottage
	Radio and tv
	Sport and hobby
	Watch, camera, photographic and cin
	Vacation, hotels
	Other leisure
	Entertainment
	Books, newspapers, television fees
	Mobil phones
	Telephone
Taxable benefits	NaN

Division	Group
Food	Bread and cereals
	Meat

	Fish and seafood
	Milk, cheese and eggs
...	
Housing	Rental for housing (incl. garage)
	Maintenance
	Household's insurance
	Water supply and other services
	Electricity, gas and other fuels
	Interests rate (gross)
	Real estate fee
	Rent for a leasehold site
Furnishings, household equipment and maintenanc...	Furniture and carpets
	Textiles
	Household appliance
Health and hospital services	NaN
Transport	Purchase of motorcars
	Purchase of other vehicles
	Operation of motorcars
	Interests & vehicle tax, motorcars
	Operation of other vehicles
	Interests & vehicle tax, other vehi
	Local travel, transportation servic
Leisure and culture	Weekend cottage
	Radio and tv
	Sport and hobby
	Watch, camera, photographic and ci
	Vacation, hotels
	Other leisure
	Entertainment
	Books, newspapers, television fees
	Mobil phones
	Telephone
Taxable benefits	NaN

Division	Group
Food	Bread and cereals

	Meat
	Fish and seafood
	Milk, cheese and eggs
...	
Housing	Rental for housing (incl. garage)
	Maintenance
	Household's insurance
	Water supply and other services
	Electricity, gas and other fuels
	Interests rate (gross)
	Real estate fee
	Rent for a leasehold site
Furnishings, household equipment and maintenanc...	Furniture and carpets
	Textiles
	Household appliance
Health and hospital services	NaN
Transport	Purchase of motorcars
	Purchase of other vehicles
	Operation of motorcars
	Interests & vehicle tax, motorcars
	Operation of other vehicles
	Interests & vehicle tax, other vehi
	Local travel, transportation servic
Leisure and culture	Weekend cottage

	Radio and tv
	Sport and hobby
	Watch, camera, photographic and ci
	Vacation, hotels
	Other leisure
	Entertainment
	Books, newspapers, television fees
	Mobil phones
	Telephone
Taxable benefits	NaN

Division	Group
Food	Bread and cereals
	Meat
	Fish and seafood
	Milk, cheese and eggs
...	
Housing	Rental for housing (incl. garage)
	Maintenance
	Household's insurance
	Water supply and other services

	Electricity, gas and other fuels
	Interests rate (gross)
	Real estate fee
	Rent for a leasehold site
Furnishings, household equipment and maintenanc...	Furniture and carpets
	Textiles
	Household appliance
Health and hospital services	NaN
Transport	Purchase of motorcars
	Purchase of other vehicles
	Operation of motorcars
	Interests & vehicle tax, motorcars
	Operation of other vehicles
	Interests & vehicle tax, other vehi
	Local travel, transportation servic
Leisure and culture	Weekend cottage
	Radio and tv
	Sport and hobby
	Watch, camera, photographic and cin
	Vacation, hotels
	Other leisure
	Entertainment
	Books, newspapers, television fees
	Mobil phones
	Telephone
Taxable benefits	NaN

Division	Group
Food	Bread and cereals
	Meat

	Fish and seafood
	Milk, cheese and eggs
...	
Housing	Rental for housing (incl. garage)
	Maintenance
	Household's insurance
	Water supply and other services
	Electricity, gas and other fuels
	Interests rate (gross)
	Real estate fee
	Rent for a leasehold site
Furnishings, household equipment and maintenanc...	Furniture and carpets
	Textiles
	Household appliance
Health and hospital services	NaN
Transport	Purchase of motorcars
	Purchase of other vehicles
	Operation of motorcars
	Interests & vehicle tax, motorcars
	Operation of other vehicles
	Interests & vehicle tax, other vehi
	Local travel, transportation servic
Leisure and culture	Weekend cottage
	Radio and tv
	Sport and hobby
	Watch, camera, photographic and ci
	Vacation, hotels
	Other leisure
	Entertainment
	Books, newspapers, television fees
	Mobil phones
	Telephone
Taxable benefits	NaN

Division	Group
Food	Bread and cereals

	Meat
	Fish and seafood
	Milk, cheese and eggs
...	
Housing	Rental for housing (incl. garage)
	Maintenance
	Household's insurance
	Water supply and other services
	Electricity, gas and other fuels
	Interests rate (gross)
	Real estate fee
	Rent for a leasehold site
Furnishings, household equipment and maintenanc...	Furniture and carpets
	Textiles
	Household appliance
Health and hospital services	NaN
Transport	Purchase of motorcars
	Purchase of other vehicles
	Operation of motorcars
	Interests & vehicle tax, motorcars
	Operation of other vehicles
	Interests & vehicle tax, other vehi
	Local travel, transportation servic
Leisure and culture	Weekend cottage

	Radio and tv
	Sport and hobby
	Watch, camera, photographic and ci
	Vacation, hotels
	Other leisure
	Entertainment
	Books, newspapers, television fees
	Mobil phones
	Telephone
Taxable benefits	NaN

Division	Group
Food	Bread and cereals
	Meat
	Fish and seafood
	Milk, cheese and eggs
...	
Housing	Rental for housing (incl. garage)
	Maintenance
	Household's insurance
	Water supply and other services

	Electricity, gas and other fuels
	Interests rate (gross)
	Real estate fee
	Rent for a leasehold site
Furnishings, household equipment and maintenanc...	Furniture and carpets
	Textiles
	Household appliance
Health and hospital services	NaN
Transport	Purchase of motorcars
	Purchase of other vehicles
	Operation of motorcars
	Interests & vehicle tax, motorcars
	Operation of other vehicles
	Interests & vehicle tax, other vehi
	Local travel, transportation servic
Leisure and culture	Weekend cottage
	Radio and tv
	Sport and hobby
	Watch, camera, photographic and cin
	Vacation, hotels
	Other leisure
	Entertainment
	Books, newspapers, television fees
	Mobil phones
	Telephone
Taxable benefits	NaN

Division	Group
Food	Bread and cereals
	Meat

	Fish and seafood
	Milk, cheese and eggs
...	
Housing	Rental for housing (incl. garage)
	Maintenance
	Household's insurance
	Water supply and other services
	Electricity, gas and other fuels
	Interests rate (gross)
	Real estate fee
	Rent for a leasehold site
Furnishings, household equipment and maintenanc...	Furniture and carpets
	Textiles
	Household appliance
Health and hospital services	NaN
Transport	Purchase of motorcars
	Purchase of other vehicles
	Operation of motorcars
	Interests & vehicle tax, motorcars
	Operation of other vehicles
	Interests & vehicle tax, other vehi
	Local travel, transportation servic
Leisure and culture	Weekend cottage
	Radio and tv
	Sport and hobby
	Watch, camera, photographic and ci
	Vacation, hotels
	Other leisure
	Entertainment
	Books, newspapers, television fees
	Mobil phones
	Telephone
Taxable benefits	NaN

Division	Group
Food	Bread and cereals

	Meat
	Fish and seafood
	Milk, cheese and eggs
...	
Housing	Rental for housing (incl. garage)
	Maintenance
	Household's insurance
	Water supply and other services
	Electricity, gas and other fuels
	Interests rate (gross)
	Real estate fee
	Rent for a leasehold site
Furnishings, household equipment and maintenanc...	Furniture and carpets
	Textiles
	Household appliance
Health and hospital services	NaN
Transport	Purchase of motorcars
	Purchase of other vehicles
	Operation of motorcars
	Interests & vehicle tax, motorcars
	Operation of other vehicles
	Interests & vehicle tax, other vehi
	Local travel, transportation servic
Leisure and culture	Weekend cottage

	Radio and tv
	Sport and hobby
	Watch, camera, photographic and ci
	Vacation, hotels
	Other leisure
	Entertainment
	Books, newspapers, television fees
	Mobil phones
	Telephone
Taxable benefits	NaN

Division	Group
Food	Bread and cereals
	Meat
	Fish and seafood
	Milk, cheese and eggs
...	
Housing	Rental for housing (incl. garage)
	Maintenance
	Household's insurance
	Water supply and other services

	Electricity, gas and other fuels
	Interests rate (gross)
	Real estate fee
	Rent for a leasehold site
Furnishings, household equipment and maintenanc...	Furniture and carpets
	Textiles
	Household appliance
Health and hospital services	NaN
Transport	Purchase of motorcars
	Purchase of other vehicles
	Operation of motorcars
	Interests & vehicle tax, motorcars
	Operation of other vehicles
	Interests & vehicle tax, other vehi
	Local travel, transportation servic
Leisure and culture	Weekend cottage
	Radio and tv
	Sport and hobby
	Watch, camera, photographic and cin
	Vacation, hotels
	Other leisure
	Entertainment
	Books, newspapers, television fees
	Mobil phones
	Telephone
Taxable benefits	NaN

Division	Group
Food	Bread and cereals
	Meat

	Fish and seafood
	Milk, cheese and eggs
...	
Housing	Rental for housing (incl. garage)
	Maintenance
	Household's insurance
	Water supply and other services
	Electricity, gas and other fuels
	Interests rate (gross)
	Real estate fee
	Rent for a leasehold site
Furnishings, household equipment and maintenanc...	Furniture and carpets
	Textiles
	Household appliance
Health and hospital services	NaN
Transport	Purchase of motorcars
	Purchase of other vehicles
	Operation of motorcars
	Interests & vehicle tax, motorcars
	Operation of other vehicles
	Interests & vehicle tax, other vehi
	Local travel, transportation servic
Leisure and culture	Weekend cottage
	Radio and tv
	Sport and hobby
	Watch, camera, photographic and ci
	Vacation, hotels
	Other leisure
	Entertainment
	Books, newspapers, television fees
	Mobil phones
	Telephone
Taxable benefits	NaN

Division	Group
Food	Bread and cereals

	Meat
	Fish and seafood
	Milk, cheese and eggs
...	
Housing	Rental for housing (incl. garage)
	Maintenance
	Household's insurance
	Water supply and other services
	Electricity, gas and other fuels
	Interests rate (gross)
	Real estate fee
	Rent for a leasehold site
Furnishings, household equipment and maintenanc...	Furniture and carpets
	Textiles
	Household appliance
Health and hospital services	NaN
Transport	Purchase of motorcars
	Purchase of other vehicles
	Operation of motorcars
	Interests & vehicle tax, motorcars
	Operation of other vehicles
	Interests & vehicle tax, other vehi
	Local travel, transportation servic
Leisure and culture	Weekend cottage

	Radio and tv
	Sport and hobby
	Watch, camera, photographic and ci
	Vacation, hotels
	Other leisure
	Entertainment
	Books, newspapers, television fees
	Mobil phones
	Telephone
Taxable benefits	NaN

Division	Group
Food	Bread and cereals
	Meat
	Fish and seafood
	Milk, cheese and eggs
...	
Housing	Rental for housing (incl. garage)
	Maintenance
	Household's insurance
	Water supply and other services

	Electricity, gas and other fuels
	Interests rate (gross)
	Real estate fee
	Rent for a leasehold site
Furnishings, household equipment and maintenanc...	Furniture and carpets
	Textiles
	Household appliance
Health and hospital services	NaN
Transport	Purchase of motorcars
	Purchase of other vehicles
	Operation of motorcars
	Interests & vehicle tax, motorcars
	Operation of other vehicles
	Interests & vehicle tax, other vehi
	Local travel, transportation servic
Leisure and culture	Weekend cottage
	Radio and tv
	Sport and hobby
	Watch, camera, photographic and cin
	Vacation, hotels
	Other leisure
	Entertainment
	Books, newspapers, television fees
	Mobil phones
	Telephone
Taxable benefits	NaN

Division	Group
Food	Bread and cereals
	Meat

	Fish and seafood
	Milk, cheese and eggs
...	
Housing	Rental for housing (incl. garage)
	Maintenance
	Household's insurance
	Water supply and other services
	Electricity, gas and other fuels
	Interests rate (gross)
	Real estate fee
	Rent for a leasehold site
Furnishings, household equipment and maintenanc...	Furniture and carpets
	Textiles
	Household appliance
Health and hospital services	NaN
Transport	Purchase of motorcars
	Purchase of other vehicles
	Operation of motorcars
	Interests & vehicle tax, motorcars
	Operation of other vehicles
	Interests & vehicle tax, other vehi
	Local travel, transportation servic
Leisure and culture	Weekend cottage
	Radio and tv
	Sport and hobby
	Watch, camera, photographic and ci
	Vacation, hotels
	Other leisure
	Entertainment
	Books, newspapers, television fees
	Mobil phones
	Telephone
Taxable benefits	NaN

Division	Group
Food	Bread and cereals

	Meat
	Fish and seafood
	Milk, cheese and eggs
...	
Housing	Rental for housing (incl. garage)
	Maintenance
	Household's insurance
	Water supply and other services
	Electricity, gas and other fuels
	Interests rate (gross)
	Real estate fee
	Rent for a leasehold site
Furnishings, household equipment and maintenanc...	Furniture and carpets
	Textiles
	Household appliance
Health and hospital services	NaN
Transport	Purchase of motorcars
	Purchase of other vehicles
	Operation of motorcars
	Interests & vehicle tax, motorcars
	Operation of other vehicles
	Interests & vehicle tax, other vehi
	Local travel, transportation servic
Leisure and culture	Weekend cottage

	Radio and tv
	Sport and hobby
	Watch, camera, photographic and ci
	Vacation, hotels
	Other leisure
	Entertainment
	Books, newspapers, television fees
	Mobil phones
	Telephone
Taxable benefits	NaN

Division	Group
Food	Bread and cereals
	Meat
	Fish and seafood
	Milk, cheese and eggs
...	
Housing	Rental for housing (incl. garage)
	Maintenance
	Household's insurance
	Water supply and other services

	Electricity, gas and other fuels
	Interests rate (gross)
	Real estate fee
	Rent for a leasehold site
Furnishings, household equipment and maintenanc...	Furniture and carpets
	Textiles
	Household appliance
Health and hospital services	NaN
Transport	Purchase of motorcars
	Purchase of other vehicles
	Operation of motorcars
	Interests & vehicle tax, motorcars
	Operation of other vehicles
	Interests & vehicle tax, other vehi
	Local travel, transportation servic
Leisure and culture	Weekend cottage
	Radio and tv
	Sport and hobby
	Watch, camera, photographic and cin
	Vacation, hotels
	Other leisure
	Entertainment
	Books, newspapers, television fees
	Mobil phones
	Telephone
Taxable benefits	NaN

Division	Group
Food	Bread and cereals
	Meat

	Fish and seafood
	Milk, cheese and eggs
...	
Housing	Rental for housing (incl. garage)
	Maintenance
	Household's insurance
	Water supply and other services
	Electricity, gas and other fuels
	Interests rate (gross)
	Real estate fee
	Rent for a leasehold site
Furnishings, household equipment and maintenanc...	Furniture and carpets
	Textiles
	Household appliance
Health and hospital services	NaN
Transport	Purchase of motorcars
	Purchase of other vehicles
	Operation of motorcars
	Interests & vehicle tax, motorcars
	Operation of other vehicles
	Interests & vehicle tax, other vehi
	Local travel, transportation servic
Leisure and culture	Weekend cottage
	Radio and tv
	Sport and hobby
	Watch, camera, photographic and ci
	Vacation, hotels
	Other leisure
	Entertainment
	Books, newspapers, television fees
	Mobil phones
	Telephone
Taxable benefits	NaN

Division	Group
Food	Bread and cereals

	Meat
	Fish and seafood
	Milk, cheese and eggs
...	
Housing	Rental for housing (incl. garage)
	Maintenance
	Household's insurance
	Water supply and other services
	Electricity, gas and other fuels
	Interests rate (gross)
	Real estate fee
	Rent for a leasehold site
Furnishings, household equipment and maintenanc...	Furniture and carpets
	Textiles
	Household appliance
Health and hospital services	NaN
Transport	Purchase of motorcars
	Purchase of other vehicles
	Operation of motorcars
	Interests & vehicle tax, motorcars
	Operation of other vehicles
	Interests & vehicle tax, other vehi
	Local travel, transportation servic
Leisure and culture	Weekend cottage

	Radio and tv
	Sport and hobby
	Watch, camera, photographic and ci
	Vacation, hotels
	Other leisure
	Entertainment
	Books, newspapers, television fees
	Mobil phones
	Telephone
Taxable benefits	NaN

Division	Group
Food	Bread and cereals
	Meat
	Fish and seafood
	Milk, cheese and eggs
...	
Housing	Rental for housing (incl. garage)
	Maintenance
	Household's insurance
	Water supply and other services

	Electricity, gas and other fuels
	Interests rate (gross)
	Real estate fee
	Rent for a leasehold site
Furnishings, household equipment and maintenanc...	Furniture and carpets
	Textiles
	Household appliance
Health and hospital services	NaN
Transport	Purchase of motorcars
	Purchase of other vehicles
	Operation of motorcars
	Interests & vehicle tax, motorcars
	Operation of other vehicles
	Interests & vehicle tax, other vehi
	Local travel, transportation servic
Leisure and culture	Weekend cottage
	Radio and tv
	Sport and hobby
	Watch, camera, photographic and cin
	Vacation, hotels
	Other leisure
	Entertainment
	Books, newspapers, television fees
	Mobil phones
	Telephone
Taxable benefits	NaN

Division	Group
Food	Bread and cereals
	Meat

```
                                                   Fish and seafood
                                                   Milk, cheese and eggs
...
Housing                                            Rental for housing (incl. garage)
                                                   Maintenance
                                                   Household's insurance
                                                   Water supply and other services
                                                   Electricity, gas and other fuels
                                                   Interests rate (gross)
                                                   Real estate fee
                                                   Rent for a leasehold site
Furnishings, household equipment and maintenanc... Furniture and carpets
                                                   Textiles
                                                   Household appliance
Health and hospital services                       NaN
Transport                                          Purchase of motorcars
                                                   Purchase of other vehicles
                                                   Operation of motorcars
                                                   Interests & vehicle tax, motorcars
                                                   Operation of other vehicles
                                                   Interests & vehicle tax, other vehi
                                                   Local travel, transportation servic
Leisure and culture                                Weekend cottage
                                                   Radio and tv
                                                   Sport and hobby
                                                   Watch, camera, photographic and cin
                                                   Vacation, hotels
                                                   Other leisure
                                                   Entertainment
                                                   Books, newspapers, television fees
                                                   Mobil phones
                                                   Telephone
Taxable benefits                                   NaN

[120 rows x 1104 columns]
```

5.4 RO 2.4 Characterisation

The result needs to be characterised in the four environmental footprint in `FD_HBS_chr` For this step the same characterisation factors as in RO1.3 are used, based on Steinmann et al. (2018). The output lists the four environmental footprint of all the HBS categories. It is important to note that the following units apply: >Carbon footprint: t CO2 eq.
>Land footprint: 1000m2
>Material footprint: t
>Water footprint: m3

In furhter steps the units are adjusted to common units as: kg CO2 eq., km2, t, m3

In [68]:
```
# Multiplication of the transposed characterisation-file with the final HBS data.

characterisation_HBS = pd.read_csv('C:/Users/tobia/Dropbox/KTH Sustainable Technology/
characterisation_HBS


FD_HBS_chr = FD_HBS.dot(characterisation_HBS.T)


# Use the groupby() and sum() function to sum up specific columns.
FD_HBS_chr.groupby(level=0, sort=False).sum().iloc[:,0].sum()
FD_HBS_chr.groupby(level=1,sort=False).sum()
```

Out[68]:
```
                                                   Carbon Footprint [t CO2-eq.]  \
Group
Bread and cereals                                                       0.666702
Meat                                                                    0.346929
Fish and seafood                                                        0.202018
Milk, cheese and eggs                                                   0.647422
Oils and fats                                                           0.074179
Fruit and berries                                                       0.136238
Vegetables                                                              0.180385
Sugar and sweets                                                        0.176665
Sauces, spices                                                          0.041248
Salt and herbs                                                          0.019062
Baby food, dietary preparations, baking powder ...                      0.014941
Snacks                                                                  0.019578
Other food                                                              0.066461
Fruit- and vegetable juices                                             0.022891
Soft drinks                                                             0.045589
Mineral or spring waters                                                0.009205
Coffee                                                                  0.103983
Tea                                                                     0.023633
Cacao                                                                   0.006302
Personal hygiene                                                        0.243586
Other consumer goods                                                    0.151433
Childcare                                                               0.009380
Trade fee, other insurance (excluding vehicle, ...                      0.031314
```

Other services	0.078921
Clothing	0.903561
Footwear	0.177994
Rental for housing (incl. garage)	0.317876
Maintenance	0.060601
Household's insurance	0.018062
Water supply and other services	2.452854
Electricity, gas and other fuels	0.549027
Interests rate (gross)	0.156971
Real estate fee	0.022827
Rent for a leasehold site	0.001067
Furniture and carpets	0.412804
Textiles	0.100240
Household appliance	0.299209
Purchase of motorcars	0.511821
Purchase of other vehicles	0.121512
Operation of motorcars	2.871254
Interests & vehicle tax, motorcars	0.022668
Operation of other vehicles	0.009481
Interests & vehicle tax, other vehicles	0.000916
Local travel, transportation services	0.078652
Weekend cottage	0.032961
Radio and tv	0.171559
Sport and hobby	0.091115
Watch, camera, photographic and cinematographic...	0.073625
Vacation, hotels	0.910232
Other leisure	0.086199
Entertainment	0.014777
Books, newspapers, television fees	0.139205
Mobil phones	0.023075
Telephone	0.028103

	Land footprint [1000 m2] \
Group	
Bread and cereals	9.023251
Meat	1.922364
Fish and seafood	0.553452
Milk, cheese and eggs	2.698919
Oils and fats	0.451474
Fruit and berries	0.772888
Vegetables	1.023336
Sugar and sweets	0.860486
Sauces, spices	0.278231
Salt and herbs	0.087676
Baby food, dietary preparations, baking powder ...	0.068719
Snacks	0.090045
Other food	0.305680
Fruit- and vegetable juices	0.113200

Soft drinks	0.162030
Mineral or spring waters	0.032718
Coffee	0.244520
Tea	0.055573
Cacao	0.014819
Personal hygiene	1.100777
Other consumer goods	0.957708
Childcare	0.025585
Trade fee, other insurance (excluding vehicle, ...	0.034506
Other services	0.264075
Clothing	0.953593
Footwear	0.313688
Rental for housing (incl. garage)	1.191964
Maintenance	0.007743
Household's insurance	0.019136
Water supply and other services	1.191788
Electricity, gas and other fuels	2.188985
Interests rate (gross)	0.172975
Real estate fee	0.085597
Rent for a leasehold site	0.004000
Furniture and carpets	1.003318
Textiles	0.099846
Household appliance	0.179422
Purchase of motorcars	0.326976
Purchase of other vehicles	0.081638
Operation of motorcars	0.332023
Interests & vehicle tax, motorcars	0.017477
Operation of other vehicles	0.001022
Interests & vehicle tax, other vehicles	0.000706
Local travel, transportation services	0.043888
Weekend cottage	0.123596
Radio and tv	0.107599
Sport and hobby	0.221629
Watch, camera, photographic and cinematographic...	0.045655
Vacation, hotels	0.439690
Other leisure	0.286013
Entertainment	0.049031
Books, newspapers, television fees	0.513964
Mobil phones	0.018781
Telephone	0.022873

	Material footprint [t] \
Group	
Bread and cereals	2.690926
Meat	1.080250
Fish and seafood	0.436280
Milk, cheese and eggs	0.705638
Oils and fats	0.164503

Fruit and berries	0.884052
Vegetables	1.170522
Sugar and sweets	2.030034
Sauces, spices	0.116773
Salt and herbs	0.054475
Baby food, dietary preparations, baking powder ...	0.042697
Snacks	0.055947
Other food	0.189927
Fruit- and vegetable juices	0.161197
Soft drinks	0.123039
Mineral or spring waters	0.024844
Coffee	0.098385
Tea	0.022360
Cacao	0.005963
Personal hygiene	0.766790
Other consumer goods	0.479467
Childcare	0.020196
Trade fee, other insurance (excluding vehicle, ...	0.042927
Other services	0.133513
Clothing	1.640493
Footwear	0.336986
Rental for housing (incl. garage)	1.306911
Maintenance	0.018511
Household's insurance	0.025827
Water supply and other services	1.480983
Electricity, gas and other fuels	2.602336
Interests rate (gross)	0.215187
Real estate fee	0.093852
Rent for a leasehold site	0.004386
Furniture and carpets	0.927478
Textiles	0.180872
Household appliance	0.635007
Purchase of motorcars	0.860398
Purchase of other vehicles	0.208104
Operation of motorcars	4.070710
Interests & vehicle tax, motorcars	0.018876
Operation of other vehicles	0.002688
Interests & vehicle tax, other vehicles	0.000763
Local travel, transportation services	0.083199
Weekend cottage	0.135515
Radio and tv	0.281848
Sport and hobby	0.189654
Watch, camera, photographic and cinematographic...	0.102445
Vacation, hotels	0.498117
Other leisure	0.200460
Entertainment	0.034365
Books, newspapers, television fees	0.231373
Mobil phones	0.024135

Telephone	0.029394

Group	Water footprint [m3]
Bread and cereals	58.043857
Meat	11.349231
Fish and seafood	13.964317
Milk, cheese and eggs	8.721652
Oils and fats	5.659828
Fruit and berries	39.599885
Vegetables	52.431903
Sugar and sweets	26.326749
Sauces, spices	4.492206
Salt and herbs	1.367520
Baby food, dietary preparations, baking powder ...	1.071840
Snacks	1.404480
Other food	4.767838
Fruit- and vegetable juices	4.572997
Soft drinks	2.770242
Mineral or spring waters	0.559376
Coffee	53.198698
Tea	12.090613
Cacao	3.224164
Personal hygiene	5.997158
Other consumer goods	3.454137
Childcare	0.152792
Trade fee, other insurance (excluding vehicle, ...	0.353084
Other services	0.854403
Clothing	26.017243
Footwear	4.548753
Rental for housing (incl. garage)	3.329552
Maintenance	0.151483
Household's insurance	0.133917
Water supply and other services	1.044707
Electricity, gas and other fuels	12.756870
Interests rate (gross)	1.769959
Real estate fee	0.239102
Rent for a leasehold site	0.011173
Furniture and carpets	3.558070
Textiles	2.912513
Household appliance	2.956165
Purchase of motorcars	4.379888
Purchase of other vehicles	1.100185
Operation of motorcars	7.442482
Interests & vehicle tax, motorcars	0.160335
Operation of other vehicles	0.021228
Interests & vehicle tax, other vehicles	0.006478
Local travel, transportation services	0.405089

```
Weekend cottage                                                    0.345246
Radio and tv                                                       1.862428
Sport and hobby                                                    1.370745
Watch, camera, photographic and cinematographic...                 0.682933
Vacation, hotels                                                   5.488573
Other leisure                                                      1.697941
Entertainment                                                      0.291076
Books, newspapers, television fees                                 2.137634
Mobil phones                                                       0.218677
Telephone                                                          0.266326
```

In [69]:
```
with pd.option_context('display.max_rows', None, 'display.max_columns', None):
    display(FD_HBS_chr)
```

```
Division                                                   Group
Food                                                       Bread and cereals
                                                           Meat
                                                           Fish and seafood
                                                           Milk, cheese and eggs
```

	Oils and fats
	Fruit and berries
	Vegetables
	Sugar and sweets
	Sauces, spices
	Salt and herbs
	Baby food, dietary preparations, baking pow
	Snacks
	Other food
Non-alcoholic beverages	Fruit- and vegetable juices
	Soft drinks
	Mineral or spring waters
	Coffee
	Tea
	Cacao
Meals out	NaN
Alcoholic beverages (with light beer)	NaN
Tobacco	NaN

Consumer goods	Personal hygiene
	Other consumer goods
Household services	Childcare
	Trade fee, other insurance (excluding vehicl
	Other services
Clothing and footwear	Clothing
	Footwear
Housing	Rental for housing (incl. garage)
	Maintenance
	Household's insurance
	Water supply and other services
	Electricity, gas and other fuels
	Interests rate (gross)
	Real estate fee
	Rent for a leasehold site
Furnishings, household equipment and maintenanc...	Furniture and carpets
	Textiles
	Household appliance
Health and hospital services	NaN
Transport	Purchase of motorcars
	Purchase of other vehicles
	Operation of motorcars
	Interests & vehicle tax, motorcars
	Operation of other vehicles
	Interests & vehicle tax, other vehicles
	Local travel, transportation services
Leisure and culture	Weekend cottage
	Radio and tv
	Sport and hobby
	Watch, camera, photographic and cinematograp
	Vacation, hotels
	Other leisure
	Entertainment
	Books, newspapers, television fees
	Mobil phones
	Telephone
Taxable benefits	NaN

Division	Group
Food	Bread and cereals

Meat

Fish and seafood

Milk, cheese and eggs

Oils and fats

Fruit and berries

Vegetables

	Sugar and sweets
	Sauces, spices
	Salt and herbs
	Baby food, dietary preparations, baking pow
	Snacks
	Other food
Non-alcoholic beverages	Fruit- and vegetable juices
	Soft drinks
	Mineral or spring waters
	Coffee
	Tea
	Cacao
Meals out	NaN
Alcoholic beverages (with light beer)	NaN
Tobacco	NaN
Consumer goods	Personal hygiene
	Other consumer goods
Household services	Childcare
	Trade fee, other insurance (excluding vehic
	Other services
Clothing and footwear	Clothing
	Footwear
Housing	Rental for housing (incl. garage)
	Maintenance
	Household's insurance
	Water supply and other services
	Electricity, gas and other fuels
	Interests rate (gross)
	Real estate fee
	Rent for a leasehold site
Furnishings, household equipment and maintenanc...	Furniture and carpets
	Textiles
	Household appliance

Health and hospital services	NaN
Transport	Purchase of motorcars
	Purchase of other vehicles
	Operation of motorcars
	Interests & vehicle tax, motorcars
	Operation of other vehicles
	Interests & vehicle tax, other vehicles
	Local travel, transportation services
Leisure and culture	Weekend cottage
	Radio and tv
	Sport and hobby
	Watch, camera, photographic and cinematograp
	Vacation, hotels
	Other leisure
	Entertainment
	Books, newspapers, television fees
	Mobil phones
	Telephone
Taxable benefits	NaN

Division	Group
Food	Bread and cereals
	Meat
	Fish and seafood

	Milk, cheese and eggs
	Oils and fats
	Fruit and berries
	Vegetables
	Sugar and sweets
	Sauces, spices
	Salt and herbs
	Baby food, dietary preparations, baking powd
	Snacks
	Other food
Non-alcoholic beverages	Fruit- and vegetable juices
	Soft drinks
	Mineral or spring waters

	Coffee
	Tea
	Cacao
Meals out	NaN
Alcoholic beverages (with light beer)	NaN
Tobacco	NaN
Consumer goods	Personal hygiene
	Other consumer goods
Household services	Childcare
	Trade fee, other insurance (excluding vehicl
	Other services
Clothing and footwear	Clothing
	Footwear
Housing	Rental for housing (incl. garage)
	Maintenance
	Household's insurance
	Water supply and other services
	Electricity, gas and other fuels
	Interests rate (gross)
	Real estate fee
	Rent for a leasehold site
Furnishings, household equipment and maintenanc...	Furniture and carpets
	Textiles
	Household appliance
Health and hospital services	NaN
Transport	Purchase of motorcars
	Purchase of other vehicles
	Operation of motorcars
	Interests & vehicle tax, motorcars
	Operation of other vehicles
	Interests & vehicle tax, other vehicles
	Local travel, transportation services
Leisure and culture	Weekend cottage
	Radio and tv
	Sport and hobby
	Watch, camera, photographic and cinematograp
	Vacation, hotels
	Other leisure
	Entertainment
	Books, newspapers, television fees
	Mobil phones
	Telephone
Taxable benefits	NaN

Division	Group
Food	Bread and cereals
	Meat
	Fish and seafood
	Milk, cheese and eggs
	Oils and fats
	Fruit and berries

	Vegetables
	Sugar and sweets
	Sauces, spices
	Salt and herbs
	Baby food, dietary preparations, baking powd
	Snacks
	Other food
Non-alcoholic beverages	Fruit- and vegetable juices
	Soft drinks
	Mineral or spring waters
	Coffee
	Tea
	Cacao
Meals out	NaN
Alcoholic beverages (with light beer)	NaN
Tobacco	NaN
Consumer goods	Personal hygiene
	Other consumer goods
Household services	Childcare
	Trade fee, other insurance (excluding vehicl
	Other services
Clothing and footwear	Clothing
	Footwear
Housing	Rental for housing (incl. garage)
	Maintenance
	Household's insurance
	Water supply and other services
	Electricity, gas and other fuels

```
                                                    Interests rate (gross)
                                                    Real estate fee
                                                    Rent for a leasehold site
Furnishings, household equipment and maintenanc... Furniture and carpets
                                                    Textiles
                                                    Household appliance
Health and hospital services                        NaN
Transport                                           Purchase of motorcars
                                                    Purchase of other vehicles
                                                    Operation of motorcars
                                                    Interests & vehicle tax, motorcars
                                                    Operation of other vehicles
                                                    Interests & vehicle tax, other vehicles
                                                    Local travel, transportation services
Leisure and culture                                 Weekend cottage
                                                    Radio and tv
                                                    Sport and hobby
                                                    Watch, camera, photographic and cinematograp
                                                    Vacation, hotels
                                                    Other leisure
                                                    Entertainment
                                                    Books, newspapers, television fees
                                                    Mobil phones
                                                    Telephone
Taxable benefits                                    NaN
```

5.5 RO 2.5 Visualisation of the Results

In a next step the results of the calculations `FD_HBS_chr` are visualised in different ways. First in a Treemap, and afterwards in a Treemap with tree subplots.

In [70]:
```
# Here is the raw data shown, with the groupby() & sum(), the different levels (e.g.

data_test=FD_HBS_chr.groupby(level=0,sort=False).sum()
data_test
```

Out[70]:
```
                                                    Carbon Footprint [t CO2-eq.]  \
Division
Food                                                                    2.591827
Non-alcoholic beverages                                                 0.211603
Meals out                                                               0.281836
Alcoholic beverages (with light beer)                                   0.152785
Tobacco                                                                 0.087985
Consumer goods                                                          0.395019
Household services                                                      0.119614
Clothing and footwear                                                   1.081555
Housing                                                                 3.579284
```

Furnishings, household equipment and maintenanc...	0.812254
Health and hospital services	0.045774
Transport	3.616303
Leisure and culture	1.570850
Taxable benefits	0.021638

	Land footprint [1000 m2] \
Division	
Food	18.136520
Non-alcoholic beverages	0.622860
Meals out	0.786523
Alcoholic beverages (with light beer)	0.543026
Tobacco	0.110244
Consumer goods	2.058485
Household services	0.324166
Clothing and footwear	1.267281
Housing	4.862189
Furnishings, household equipment and maintenanc...	1.282585
Health and hospital services	0.121809
Transport	0.803731
Leisure and culture	1.828833
Taxable benefits	0.016683

	Material footprint [t] \
Division	
Food	9.622024
Non-alcoholic beverages	0.435789
Meals out	0.478641
Alcoholic beverages (with light beer)	0.412350
Tobacco	0.085844
Consumer goods	1.246257
Household services	0.196635
Clothing and footwear	1.977479
Housing	5.747994
Furnishings, household equipment and maintenanc...	1.743356
Health and hospital services	0.098450
Transport	5.244738
Leisure and culture	1.727306
Taxable benefits	0.018018

	Water footprint [m3]
Division	
Food	229.201305
Non-alcoholic beverages	76.416089
Meals out	8.966053
Alcoholic beverages (with light beer)	9.284145
Tobacco	15.419551
Consumer goods	9.451295

```
Household services                                                  1.360278
Clothing and footwear                                              30.565996
Housing                                                            19.436762
Furnishings, household equipment and maintenanc...                  9.426748
Health and hospital services                                        0.962595
Transport                                                          13.515684
Leisure and culture                                                14.361578
Taxable benefits                                                    0.153047
```

In [71]:
```
data_test.sum()
```

Out[71]:
```
Carbon Footprint [t CO2-eq.]     14.568327
Land footprint [1000 m2]         32.764935
Material footprint [t]           29.034882
Water footprint [m3]            438.521127
dtype: float64
```

In [75]:
```
data_test.sum()
```

Out[75]:
```
Carbon Footprint [t CO2-eq.]     14.568327
Land footprint [1000 m2]         32.764935
Material footprint [t]           29.034882
Water footprint [m3]            438.521127
dtype: float64
```